UM-2 provenance

Where every ingredient, interaction, field equation and number of the universe model UM-2 comes from, how it was computed, and how it was verified

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Provenance, version 4, 2026-10-18 Versions 3 and 2: 2026-10-18; version 1: 2026-10-17

Repository once-ere/Dirac_claude, folder universe_model_2

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1 How to read this document

What it contains. This document records the provenance of UM-2, a model of the Universe built on the author's 4+4-dimensional Einstein-Lovelock theory and his dirac16complex field, coupled to gravity and to every field of the Standard Model without a Higgs particle. It states, for every ingredient, where it comes from (the author's orders and notebooks, the Revision record of this repository, standard physics and measured values, or the choices and inventions of UM-2), writes it out as an equation, names the computation that produced or used it and the checks that verify it. The sections are: the author's orders, quoted (section 2); who did what (section 3); the ingredients taken from the author (section 4) and from the Revision record (section 5); the action of UM-2 term by term, each of the 18 interaction terms IN-1 to IN-18 written out with its origin, and for every invented term the reason and its consistency with QCD and the electroweak model (section 6); the measured values and standard physics used, with their sources (section 7); the origin of every one of the 136 field equations FE-1 to FE-136 in one table (section 8); the computations, their outputs and their 287 checks (section 9); the residuals of the approximate solution in its field equations and in concert, and the precision fixes of version 4 (section 10); the verification record (section 11); the history of the corrections (section 12); what is not in the repository (section 13); the reproduction (section 14); and the history of UM-2 in the repository (section 15).

Status labels. Every ingredient and result carries one of these labels:

- **AUTHOR**: taken from the author's Mathematica notebooks or from the author's written orders;
- **RECORD**: taken from the Revision record of this repository (earlier work carried out under the author's orders, with its own verification);
- **STANDARD**: textbook physics or a measured value; a measured value is marked *verified* (read from the source on 2026-10-17 or 2026-10-18) or *from memory* (quoted from memory of the standard tables, to be

checked against the current editions);
- **ASSUMED**: a modelling choice made for UM-2;
- **INVENTED**: a new phenomenological interaction or ingredient of UM-2;
- **DERIVED**: computed and verified in UM-2; the verification is named with the result;
- **ESTIMATE**: an order-of-magnitude evaluation, not a precision computation;
- **OPEN**: not resolved in this version.

Equations and symbols. Every display equation is followed by the complete list of its symbols: every quantity, field, coupling, constant, operator, index (with the values it takes) and special mark, with its meaning, its unit, its value where it has one, and its status. Typography needs no definition (digits, $+$, $-$, $=$, brackets, fractions, roots, $\cdot$, $\ldots$, $\leq$, $\approx$, $\sim$, $\times$, the standard functions sin, cos, tan, cot, sinh, cosh, tanh, exp, ln, log, det, min, max, lim and the signs of sums, products and integrals; their index letters and integration variables are symbols and are listed). The tags IN-n and FE-n are one numbering, shared by all documents of UM-2, of the 18 interaction terms of the action and of the 136 field equations; an interaction term or a field equation that this document uses is written out here with its tag and its symbols. A symbol has one meaning throughout the document, in the displays, in the inline formulas and in the text. Euler's number is written e, the imaginary unit i and the differential d. Names of the computations' checks are quoted literally in typewriter type, in the spelling of the code.

Units and conventions. $\hbar = c = 1$. H is the author's constant (an inverse length); in the numerical work on the block $H = 1$ unless a physical unit is given, and its physical value is $H = 14.296$ TeV. The coordinates are always the author's $X = \{x_0, x_1, \ldots, x_7\}$: x_0 the hidden space-like direction, listed first, periodic with the period $\pi/(6H)$; x_1, x_2, x_3 ordinary space; x_4 the time; x_5, x_6, x_7 the three extra time-like directions. W^+ carries the electric charge $+e$ and W^- the charge $-e$. UM-2 contains no Higgs particle.

The numbers. Every number is taken from the outputs of the computations of UM-2 (the scripts in `universe_model_2/code/`, which write `universe_model_2/outputs/` and `universe_model_2/reports/`) and is quoted with the precision of the output; section 9 names the output that holds it. These files are named for reproduction only: every equation, definition and argument that the reader needs is written in this document.

Versions. Version 1 (2026-10-17) had one generation of quarks and leptons. Version 2 (2026-10-18) has three generations and three neutrino flavours with their mixing and flavour conversion, ordered by the author on 2026-10-18, and removes the chiral tip Z portal of version 1 (corrections, item 21; its independent review, item 22). Version 3 (2026-10-18) answers the author's order to find and document an accurate approximate solution of the coupled field equations with Kohn-Sham theory for dirac16complex coupled to gravity and to every Standard-Model field: the portal I13, the benchmark strengths of I11 and I12, the coupled solution in environments, the thermal history and the dark-matter relic, the magnitudes of the terms of the metric series, σ_8 and the Majorana alternative (item 23; its independent review, item 24). **Version 4** (2026-10-18) answers the author's order of 2026-10-18, 15:42, to organise and present the solution better: one notation in which every symbol has one meaning, a registry of every symbol with its meaning and unit, an inventory of the 136 field equations and 18 interaction terms, every equation followed by its symbols, the residuals of the approximate solution in each of its field equations and in concert (52 new checks, 287 in all), and five precision fixes of the computations (item 25). The physics and the claims of version 3 are unchanged; every changed number is stated in item 25 and in section 10.

2 The orders

2.1 The order of 2026-10-17

The model answers the author's order of 2026-10-17. Its key parts, in the author's words: "Create a new model of the Universe that includes [a] gravity (Einstein-Lovelock gravity); [b] the dirac16complex quantized spinor field (which is hypothesied to represent dark matter); the dirac16complex is coupled to gravity using a non-trivial canonical spin connection; ... interactions of gravity and dirac16complex with SOME of the following Standard Model particles: [c] photons; [d] neutrinos; [e] electrons; [f] quarks (only up and down, for the protons and neutrons); [g] the 3 weak interaction gauge bosons, the W and Z 'particles'. However do NOT include the Higgs particle. You must invent/create new phenomenological interactions ... consistent with qcd and electro-weak models. You want to be able to model: CMB, CNB, BAO, so your model for the metric tensor ... must be very general, but perhaps expressible as a power series of terms of decreasing magnitude. Put each and every interaction in equation form, and explain and analyze it in documentation and provenance files; these should be both markdown and latex file types; compile all latex to pdf. NEVER refer the reader to some other file! PUT EVERY EQUATION

you create and/or use IN THESE FILES. CRITICALLY IMPORTANT: Mass will be given to all non-massless particles, (except the massless particles: graviton, photon and perhaps the quanta of the dirac16complex field), by appropriate wave function dependence on the space coordinate x0 ... periodic in x0 with period pi/(6H). Localization with respect to x0 implies, using Heisenberg ... Delta x0 * Delta (p)x0 >= hbar/2 ... [2] push your new, improved, model ... along with provenance (both .md and .pdf files). [3] Compute the coupled field equations and push to repo; create concomitant documention (both .md and .pdf files) and push. [4] Compute an accurate approximate solution to the coupled field equations (use Kohn-Sham theory for the dirac16complex field, which couples to all other fields) and push ... Check and verify this repo."

Further orders of the same day: "analyze the x0 mass mechanism, and ... analyze the validity of associating dark matter (and dark energy) with dirac16complex"; "the W+ and W- bosons carry positive and negative electric charges. Do not forget this"; the coordinates are always the author's $\{x_0, x_1, \ldots, x_7\}$ of the notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb`, never relabelled; every verified result is pushed to the repository as soon as it is obtained; no single calculation runs longer than 7 minutes and 3 seconds (423 s) (this limit was lifted on 2026-10-18).

2.2 The orders of 2026-10-18: three neutrino flavours

On 2026-10-18 the author stated that neutrino flavour and flavour conversion are critically important in this project and asked that the paper of Gogilashvili and Tamborra on neutrino flavour conversion in core-collapse supernovae (his file "Neutrino Flavor Conversion.pdf", arXiv:2605.16504; not committed, it is a third-party paper) be read, understood and remembered (paraphrase). He then ordered (08:02), in his words: "correct universe model 2 to include three neutrino flavors with mixing and flavor conversion. this is your first and only priority!! ... do correct universe model 2 to include three neutrino flavors with mixing and flavor conversion NOW." Two ways were put to him: (A) three full generations of quarks and leptons, anomaly free generation by generation; (B) three lepton generations with the single quark generation of version 1, whose gauge anomalies would have to be cancelled by an added Wess-Zumino term. His answer (08:24): "NOT (B). USE (A)!!!". Version 2 implements (A). A flavour is defined by the charged lepton produced with the neutrino in a charged-current process, so three neutrino flavours require the muon and the tau; with them, anomaly freedom requires the second and third quark generations (a single quark generation would leave the mixed $SU(2)^2U(1)_Y$ anomaly coefficient -1 and the $U(1)_Y^3$ coefficient $3/2$; DERIVED).

2.3 The order of 2026-10-18 for version 3

After the three-flavour correction the author ordered (09:48), in his words (excerpts): "find and document (with provenance) an accurate approximate solution for your new model universe, i.e., of the coupled field equations, with Kohn-Sham theory for the dirac16complex field coupled to gravity and to all of the Standard-Model fields; coupled fields include gravity (Einstein-Lovelock gravity, which couples to all fields), the dirac16complex quantized fermion spinor field (which couples to all other fields), the three-flavour version of left-handed neutrinos, the usual left-handed electrons, the usual singlet right-handed electron, three quark generations, gluons, photons, the 3 weak interaction gauge bosons, the W^(+), W^(-), and Z 'particles'. However do NOT include the Higgs particle. You must invent/create and document new phenomenological interactions for these standard model particles that should be consistent with qcd and electro-weak models. You want to be able to model: CMB ..., CNB ..., BAO ..., so your model for the metric tensor of the gravitation field must be very general, but perhaps expressible as a power series of terms of decreasing magnitude. CRITICALLY IMPORTANT: Mass will be given to all non-massless particles ... by appropriate wave function dependence on the space coordinate x0 ..., which must be periodic in x0 with period pi/(6H). ... Put each and every interaction in equation form, and explain and analyze it in documentation and provenance files; these should be both markdown and latex file types; compile all latex to pdf. NEVER refer the reader to some other file! PUT EVERY EQUATION you create and/or use IN THESE FILES. In your documentations and provenances i want you to analyze the x0 mass mechanism, and to analyze the validity of associating dark matter (and dark energy) with dirac16complex. [2] push your new, improved, model of the Universe to the repo, along with provenance (both .md and .pdf files). [3] Compute the coupled field equations and push to repo ... [4] Compute an accurate approximate solution to the coupled field equations (use Kohn-Sham theory for the dirac16complex field, which couples to all other fields) and push to repo ... Check and verify this repo. DO NOT TAKE SHORTCUTS. ... [5] Continue and Complete and push all stages that are still open". Version 3 keeps the three generations of version 2 (with right-handed neutrinos for the Dirac masses of the x_0 mechanism; the Majorana alternative without them is analysed) and adds the items listed in section 1.

2.4 The order of 2026-10-18, 15:42, for version 4

The author's order of 2026-10-18 at 15:42, in his words:

> "STOP or PAUSE everything. We need to organize and present your solution better. ... precisely, but approximately, solve the coupled field equations and find and document (with provenance) an accurate approximate solution for your new model universe, i.e., of the coupled field equations, with Kohn-Sham theory for the dirac16complex field coupled to gravity and to all of the Standard-Model fields; coupled fields include gravity (Einstein-Lovelock gravity, which couples to all fields), the dirac16complex quantized fermion spinor field (which couples to all other fields), the three-flavour version of left-handed neutrinos, the usual left-handed electrons, the usual singlet right-handed electron, three quark generations, gluons, photons, the 3 weak interaction gauge bosons, the Wˆ(+), Wˆ(-), and Z 'particles'. However do NOT include the Higgs particle. You must invent/create and document new phenomenological interactions for these standard model particles that should be consistent with qcd and electro-weak models. For each and every field equation that you use, you must clearly state, write down, document the field equation itself, define each and every symbol that appears in the field equation EVEN if you have done this before. This a command not a request. I do NOT want students struggling to find definitions, when they should be finding (numerical or analytical) solutions. Analyze your approximate solution (usually in concert with other field equation) in a documentation presentation (markdown and latex, compiled to pdf versions, ALWAYS). You want to be able to model: CMB, CNB, BAO, so your model for the metric tensor of the gravitation field must be very general, but perhaps expressible as a power series of terms of decreasing magnitude. CRITICALLY IMPORTANT: Mass will be given to all non-massless particles (except the massless particles: graviton, photon and perhaps the quanta of the dirac16complex field), by appropriate wave function dependence on the space coordinate x0 (either explicit, derived, or assumed), which must be periodic in x0 with period pi/(6H). Localization with respect to x0 implies, using Heisenberg Uncertainty Principle, Delta x0 * Delta (p)x0 >= hbar/2 (here (p)x0 is the x0 th component of momentum), and you can think of momentum as being proportional to mass, and you can think of kinetic energy as being proportional to momentum-squared divided by mass, hence giving rise to an (effective) mass for all particles whose wave function is periodic in space coordinate x0. This is true, except for the graviton, photon and perhaps the quanta of the dirac16complex field. Put each and every interaction in equation form, and explain and analyze it in documentation and provenance files; these should be both markdown and latex file types; compile all latex to pdf. NEVER refer the reader to some other file! PUT EVERY EQUATION you create and/or use IN THESE FILES. In your documentations and provenances i want you to analyze the x0 mass mechanism, and to analyze the validity of associating dark matter (and dark energy) with dirac16complex."

Two minutes later (15:44): "do not forget to push results as soon as they are completed and verified".

Version 4 answers the order with six binding rules for all documents of UM-2: (S1) every display equation is followed by the complete list of its symbols, generated from one registry, and every field equation or interaction term that a document uses is written out there with its tag and its symbols; (S2) one symbol, one meaning, across all documents (the notation of version 4 resolves every symbol that version 3 used for several things); (S3) the registry of the symbols and the inventory of the field equations and interaction terms; (S4) every document starts with "How to read this document", and a solution is presented as the field equation, the approximation, the solution, its check in its own equation and in concert with the other field equations, and the status labels; (S5) the reader is never sent to another file; (S6) the status labels and the conflicts with data stay as in version 3, and every change of a claim is stated here. How each rule was implemented, and what was computed anew for it, is item 25 of the corrections and section 10.

3 Who did what

- The 4+4 framework, the coordinates, the primordial metric of the notebooks, the Einstein-Lovelock action with its weights and its cosmological term, the dirac16complex field and its real 16×16 gamma matrices of the Clifford algebra of signature $(4, 4)$, the mass parameter, the idea that masses come from the x_0 dependence of wave functions periodic with the period $\pi/(6H)$, the dark-matter hypothesis, the demand for three neutrino flavours with mixing and flavour conversion and the presentation rules of version 4 are the AUTHOR's

(Patrick L. Nash).
- The corrected primordial metric, the regular tip cutoff, the Dirac-type Lagrangian with the canonical spin connection, the 2×2 block reduction of the Kohn-Sham problem and its conventions are RECORD items: produced earlier in this repository under the author's orders and verified there.
- Everything labelled ASSUMED, INVENTED or DERIVED below, all the code of UM-2 and all its documents are by Claude (Anthropic), working in this repository on 2026-10-17 (version 1) and 2026-10-18 (versions 2, 3 and 4).

4 Ingredients from the author (AUTHOR)

4.1 Coordinates and signature

The author's coordinates (notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb`, `X = {x0[], x1[], ..., x7[]}`) are

$$X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}, \qquad z = 6Hx_0, \qquad 0 < z < \pi$$

Symbols in this equation.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

with x_0 a hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time and x_5, x_6, x_7 three extra time-like directions; the interval $0 < z < \pi$ is one period of x_0, of length $\pi/(6H)$. The orthonormal frame metric, with the frame directions aligned with $x_0, \ldots, x_7$ in this order, is

$$\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$$

Symbols in this equation.

- η_{ab}: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ (η^{ab} the same matrix). Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.

4.2 The primordial metric of the notebooks

The notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb` (the author's local notebook) gives the primordial metric

$$g_{\rm nb} = \mathrm{diag}\big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\\ -1,\ -\mathrm{e}^{-2a_4}(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}(\sin^2 z)^{1/6}\big)$$

Symbols in this equation.

- $g_{\rm nb}$: The metric of the author's notebook (before the author's order of 2026-10-14 added the factor b_4 to the extra-time components), written as the diagonal matrix of its components in the order $x_0, x_1, \ldots, x_7$; it is quoted for comparison only, the model uses the corrected metric $\bar{g}_{MN}$. Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.

with the vielbein $\mathrm{diag}\big(\cot z$, three times $\mathrm{e}^{a_4}(\sin^2 z)^{1/12}$, 1, three times $\mathrm{e}^{-a_4}(\sin^2 z)^{1/12}\big)$ and $a_4 = a_4(x_4)$. By the author's orders of 2026-10-14 a second function $b_4(x_4) > 0$ multiplies the extra-time components ("b4 is a function of x4; b4 is NEVER 1"), and $(\sin z)^{1/3}$ means $(\sin^2 z)^{1/6}$, the positive branch, for every real z. The metric that UM-2 uses, the corrected metric of the Revision record (FE-31), is written out in section 5.

4.3 The Einstein-Lovelock equations, the weights and the cosmological term

The notebook writes the vacuum Einstein-Lovelock equations (after Lovelock and Rund) as

$$0 = -10\,\delta^M_N + \frac{w_{(1)}}{H^2}\,\mathcal{E}_{(1)}{}^M{}_N + \frac{w_{(2)}}{H^4}\,\mathcal{E}_{(2)}{}^M{}_N + \frac{w_{(3)}}{H^6}\,\mathcal{E}_{(3)}{}^M{}_N, \qquad w_{(1)} = 1, \quad w_{(2)} = \frac{1}{32}, \quad w_{(3)} = \frac{1}{1152}$$

Symbols in this equation.

- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^M{}_N - \frac{1}{2}\delta^M_N R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

where the notebook's tensor "Lovelock$p$" is the Lovelock tensor $\mathcal{E}_{(p)}{}^M{}_N$ of order p. In the notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb` (In[133] to In[136]) the author sets $w_{(1)} = 1$, $w_{(2)} = 1/2^5$ and solves the vacuum equations on the history $a_4 = Hx_4$, $b_4 = 1$ of the notebook metric for $w_{(3)}$ and for the cosmological number (In[134]); In[136] factors $1152 = 2^7\,3^2$; the Revision specification records the solution $w_{(3)} = 1/1152$ and

the number 10. Multiplying by H^2 and writing the equations as the Einstein-Lovelock equations with a source gives the couplings and the cosmological constant of UM-2,

$$\alpha_{(p)} = w_{(p)}\, H^{2-2p}, \qquad \Lambda = -10H^2$$

Symbols in this equation.

- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.

which enter the action term IN-2 (section 6). With these couplings and the source of all matter, the notebook's equations become the field equations of gravity of UM-2, with the Lovelock tensors written with the generalised Kronecker delta:

$$\begin{aligned} &\sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^{M}{}_{N} + \Lambda\, \delta^{M}_{N} = \kappa\, \mathcal{T}^{M}{}_{N}, \\ &{\mathcal{E}_{(p)}}^{M}{}_{N} = -\frac{1}{2^{p+1}}\, \delta^{M M_1 N_1 \cdots M_p N_p}_{N P_1 Q_1 \cdots P_p Q_p}\, R^{P_1 Q_1}{}_{M_1 N_1} \cdots R^{P_p Q_p}{}_{M_p N_p} \end{aligned} \tag{FE-1}$$

Symbols in (FE-1).

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order p, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length^{2p}. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.
- $\mathcal{T}^{M}{}_{N}$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane (a, b). Unit: 1/length^2. Status: STANDARD.

That this normalisation reproduces the record's field equations (which were checked against the notebook) is verified exactly by `um2_geometry.py` (checks `record_equation constraint_x4`, `record_equation space_x1_eq_x2_eq_x3`, `record_equation hidden_and_extraTimes_x0_eq_x5_eq_x6_eq_x7`). UM-2 keeps the author's weights and Λ unchanged; an earlier model of this repository (UM-1) had changed $w_{(3)}$ and Λ.

4.4 The gamma matrices of the Clifford algebra of signature (4, 4)

Both notebooks of the author named above build real 16×16 matrices in three steps. First, two real 4×4 matrices for each $h = 1, 2, 3$ (the notebook's arrays `s` and `t`, built as `Qa - Qb` and `Qa + Qb` from the Levi-Civita array `Qa` and a second array `Qb` of Kronecker deltas):

$$(\mathsf{s}_h)_{kq} = \epsilon^{hkq4}_{\mathrm{LC}} - \delta_{k4}\delta_{qh} + \delta_{kh}\delta_{q4}, \qquad (\mathsf{t}_h)_{kq} = \epsilon^{hkq4}_{\mathrm{LC}} + \delta_{k4}\delta_{qh} - \delta_{kh}\delta_{q4}$$

Symbols in this equation.

- s_h: The author's real antisymmetric 4×4 matrices s_h ($h = 1, 2, 3$; in the notebook $s_h = Q_a - Q_b$), the building blocks of the matrices Z_1, Z_2, Z_3. Unit: dimensionless. Status: AUTHOR.
- t_h: The author's real antisymmetric 4×4 matrices t_h ($h = 1, 2, 3$; in the notebook $t_h = Q_a + Q_b$), the building blocks of the matrices Z_6, Z_5, Z_4. Unit: dimensionless. Status: AUTHOR.
- h: The label $h = 1, 2, 3$ of the author's real 4×4 matrices s_h, t_h and of the 8×8 matrices Z_h and Z_{7-h} built from them. Status: AUTHOR.
- k, q: The row and column indices $k, q = 1, 2, 3, 4$ of the author's 4×4 matrices s_h and t_h. Status: AUTHOR.
- $\epsilon^{hkq4}_{\mathrm{LC}}$: The Levi-Civita symbol of four indices: $+1$ if $(h, k, q, 4)$ is an even permutation of $(1, 2, 3, 4)$, -1 if it is an odd permutation, and 0 if two indices are equal. Unit: dimensionless. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.

Second, eight real 8×8 matrices and the matrix that pairs them (the notebook's arrays `zeta` and `tau`, which are equal, and `sigma`; the identity $\mathbf{1}$ is 8×8 for Z_0 and 4×4 inside S):

$$\mathsf{Z}_0 = \mathbf{1}, \qquad \mathsf{Z}_h = \begin{pmatrix} 0 & \mathsf{s}_h \\ \mathsf{s}_h & 0 \end{pmatrix}, \qquad \mathsf{Z}_{7-h} = \begin{pmatrix} 0 & \mathsf{t}_h \\ -\mathsf{t}_h & 0 \end{pmatrix}, \qquad \mathsf{Z}_7 = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3\mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6, \qquad \mathsf{S} = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}$$

Symbols in this equation.

- Z_a: The author's real 8×8 matrices Z_a ($a = 0, \ldots, 7$; the notebook's $\zeta_A = \tau_A$): Z_0 is the identity, $\mathsf{Z}_1, \mathsf{Z}_2, \mathsf{Z}_3$ and $\mathsf{Z}_6, \mathsf{Z}_5, \mathsf{Z}_4$ are built from s_h and t_h, and $\mathsf{Z}_7 = \mathsf{Z}_1 \cdots \mathsf{Z}_6$; they are the lower-left 8×8 blocks of the gamma matrices γ^a. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- h: The label $h = 1, 2, 3$ of the author's real 4×4 matrices s_h, t_h and of the 8×8 matrices Z_h and Z_{7-h} built from them. Status: AUTHOR.
- s_h: The author's real antisymmetric 4×4 matrices s_h ($h = 1, 2, 3$; in the notebook $s_h = Q_a - Q_b$), the building blocks of the matrices Z_1, Z_2, Z_3. Unit: dimensionless. Status: AUTHOR.
- t_h: The author's real antisymmetric 4×4 matrices t_h ($h = 1, 2, 3$; in the notebook $t_h = Q_a + Q_b$), the building blocks of the matrices Z_6, Z_5, Z_4. Unit: dimensionless. Status: AUTHOR.
- S: The author's real 8×8 matrix with the 4×4 identity in its two off-diagonal blocks and zeros elsewhere (the notebook's σ); $\mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7$. Unit: dimensionless. Status: AUTHOR.

Third, the gamma matrices along the frame directions $a = 0, \ldots, 7$ (the notebook's `T16`):

$$\bar{\mathsf{Z}}_a = \mathsf{S}\,\mathsf{Z}_a^T\,\mathsf{S}, \qquad \gamma^a = \begin{pmatrix} 0 & \bar{\mathsf{Z}}_a \\ \mathsf{Z}_a & 0 \end{pmatrix}$$

Symbols in this equation.

- $\bar{\mathsf{Z}}_a$: The partner matrices $\bar{\mathsf{Z}}_a = \mathsf{S}\,\mathsf{Z}_a^T\,\mathsf{S}$ of the author ($a = 0, \ldots, 7$; the notebook's $\bar{\tau}_A$), the upper-right 8×8 blocks of the gamma matrices γ^a. Unit: dimensionless. Status: AUTHOR.
- S: The author's real 8×8 matrix with the 4×4 identity in its two off-diagonal blocks and zeros elsewhere (the notebook's σ); $\mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7$. Unit: dimensionless. Status: AUTHOR.
- Z_a: The author's real 8×8 matrices Z_a ($a = 0, \ldots, 7$; the notebook's $\zeta_A = \tau_A$): Z_0 is the identity, $\mathsf{Z}_1, \mathsf{Z}_2, \mathsf{Z}_3$ and $\mathsf{Z}_6, \mathsf{Z}_5, \mathsf{Z}_4$ are built from s_h and t_h, and $\mathsf{Z}_7 = \mathsf{Z}_1 \cdots \mathsf{Z}_6$; they are the lower-left 8×8 blocks of the gamma matrices γ^a. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.

They satisfy the Clifford algebra and the notebook identities (notebook Out[322] to Out[324]; the identity $\mathbf{1}$ is 16×16 in the first relation and 8×8 in the second):

$$\gamma^a\gamma^b + \gamma^b\gamma^a = 2\eta^{ab}\,\mathbf{1}, \qquad \tfrac{1}{2}\big(\mathsf{Z}_a\bar{\mathsf{Z}}_b + \mathsf{Z}_b\bar{\mathsf{Z}}_a\big) = \eta_{ab}\,\mathbf{1}, \qquad \mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7$$

Symbols in this equation.

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ (η^{ab} the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- Z_a: The author's real 8×8 matrices Z_a ($a = 0, \ldots, 7$; the notebook's $\zeta_A = \tau_A$): Z_0 is the identity, $\mathsf{Z}_1, \mathsf{Z}_2, \mathsf{Z}_3$ and $\mathsf{Z}_6, \mathsf{Z}_5, \mathsf{Z}_4$ are built from s_h and t_h, and $\mathsf{Z}_7 = \mathsf{Z}_1 \cdots \mathsf{Z}_6$; they are the lower-left 8×8 blocks of the gamma matrices γ^a. Unit: dimensionless. Status: AUTHOR.
- $\bar{\mathsf{Z}}_a$: The partner matrices $\bar{\mathsf{Z}}_a = \mathsf{S}\,\mathsf{Z}_a^T\,\mathsf{S}$ of the author ($a = 0, \ldots, 7$; the notebook's $\bar{\tau}_A$), the upper-right 8×8 blocks of the gamma matrices γ^a. Unit: dimensionless. Status: AUTHOR.
- S: The author's real 8×8 matrix with the 4×4 identity in its two off-diagonal blocks and zeros elsewhere (the notebook's σ); $\mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7$. Unit: dimensionless. Status: AUTHOR.

The derived matrices that UM-2 uses (the matrix of the Dirac adjoint, the chirality of eight dimensions, the author's operator that separates the two halves of dirac16complex, and the generators of frame rotations) are, with the author's matrices,

$$C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3} = \begin{pmatrix} -\mathsf{S} & 0 \\ 0 & \mathsf{S} \end{pmatrix}, \qquad \Gamma_8 = \gamma^{x_0}\gamma^{x_1}\cdots\gamma^{x_7} = \begin{pmatrix} -\mathbf{1} & 0 \\ 0 & \mathbf{1} \end{pmatrix}, \qquad \mathcal{B} = -\mathrm{i}C\gamma^{x_4}, \qquad \mathcal{S}^{ab} = \tfrac{1}{4}\big[\gamma^a, \gamma^b\big]$$

Symbols in this equation.

- C: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar{\Psi} = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.

- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- S: The author's real 8×8 matrix with the 4×4 identity in its two off-diagonal blocks and zeros elsewhere (the notebook's σ); $\mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7$. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality of eight dimensions, $\Gamma_8 = \gamma^{x_0}\gamma^{x_1}\cdots\gamma^{x_7}$, equal to $\mathrm{diag}(-\mathbf{1}, \mathbf{1})$ for the author's matrices; its eigenspaces (eigenvalues -1 and $+1$) are the two inequivalent 8-dimensional irreducible representations of Spin(4,4). Unit: dimensionless. Status: DERIVED.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues ∓ 1 define the halves Ψ_A and Ψ_B, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{S}^{ab}$: The generators of frame rotations on spinors, $\mathcal{S}^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$ (the commutator). Unit: dimensionless. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.

All of these relations are verified exactly with the rebuilt matrices by `um2_algebra.py` (31 checks, among them `tau_relations_of_the_notebook`, `notebook_identities_sigma`, `clifford_relations`, `identical_to_record_gammas`, `C_is_sigma16`, `chirality_diag`); the matrices enter the action through IN-6 (section 6).

4.5 The mass parameter, the x_0 mass idea and the dark-matter hypothesis

The mass of dirac16complex is written $m_\Psi = 3H\mathcal{M}$ in the author's notebooks (the action term IN-5). The orders require every non-massless particle to get its mass from the dependence of its wave function on x_0, periodic with the period $\pi/(6H)$, with the uncertainty relation of x_0 and its momentum as the physical reason, and state the hypothesis that the quanta of dirac16complex represent dark matter (and ask for the validity of a dark-energy association to be analysed).

5 Ingredients from the Revision record (RECORD)

5.1 The corrected primordial metric

The author's order of 2026-10-16: "a diagonal form for the metric of the primordial gravitational field is only self-consistent if all off-diagonal terms of all source terms in the Einstein field equations ARE ZERO. This is not the case here. I task you to: [1] find the simplest modification for the current metric of the primordial gravitational field that is probably self-consistent with the Einstein field equations". The metric of 2026-10-14 (the notebook metric with b_4) has an off-diagonal Lovelock component $\mathcal{E}_{(p)}{}^{x_4}{}_{x_0} \propto \cot z\, \dot{b}_4/b_4$ at every Lovelock order p. The record examined the family of diagonal metrics with $g_{x_4x_4} = -1$ and

$$g_{x_0x_0} = b_4^{2\mathfrak{p}}\, \mathrm{e}^{2\mathfrak{q}a_4} \cot^2 z, \qquad g_{x_1x_1} = g_{x_2x_2} = g_{x_3x_3} = \mathrm{e}^{2a_4}(\sin^2 z)^{3\mathfrak{a}},$$
$$g_{x_5x_5} = g_{x_6x_6} = g_{x_7x_7} = -\mathrm{e}^{-2a_4} b_4\, (\sin^2 z)^{3\mathfrak{b}}$$

Symbols in this equation.

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.

- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\mathfrak{p}$, $\mathfrak{q}$, $\mathfrak{a}$, $\mathfrak{b}$: The four exponents of the family of diagonal primordial metrics examined in the Revision record on 2026-10-16: $\mathfrak{p}$ and $\mathfrak{q}$ are the powers of b_4 and of e^{a_4} in the hidden component $g_{x_0x_0} = b_4^{2\mathfrak{p}}\mathrm{e}^{2\mathfrak{q}a_4}\cot^2 z$, $\mathfrak{a}$ and $\mathfrak{b}$ the warp powers of ordinary 3-space and of the extra times, $(\sin^2 z)^{3\mathfrak{a}}$ and $(\sin^2 z)^{3\mathfrak{b}}$. The record's corrected metric has $\mathfrak{p} = \frac{1}{2}$, $\mathfrak{q} = -1$, $\mathfrak{a} = 0$, $\mathfrak{b} = \frac{1}{6}$. Unit: dimensionless. Value: $\mathfrak{p} = \frac{1}{2}$, $\mathfrak{q} = -1$, $\mathfrak{a} = 0$, $\mathfrak{b} = \frac{1}{6}$ for the metric used by UM-2. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.

whose mixed curvature (3-space with the hidden direction, and the extra times with the hidden direction) vanishes for every $a_4(x_4)$ and $b_4(x_4)$ only if

$$\mathfrak{a}\Big((\mathfrak{q}-1)\,\dot{a}_4 + \mathfrak{p}\,\frac{\dot{b}_4}{b_4}\Big) = 0, \qquad \mathfrak{b}\Big((\mathfrak{q}+1)\,\dot{a}_4 + \big(\mathfrak{p}-\tfrac{1}{2}\big)\frac{\dot{b}_4}{b_4}\Big) = 0$$

Symbols in this equation.

- $\mathfrak{p}$, $\mathfrak{q}$, $\mathfrak{a}$, $\mathfrak{b}$: The four exponents of the family of diagonal primordial metrics examined in the Revision record on 2026-10-16: $\mathfrak{p}$ and $\mathfrak{q}$ are the powers of b_4 and of e^{a_4} in the hidden component $g_{x_0x_0} = b_4^{2\mathfrak{p}}\mathrm{e}^{2\mathfrak{q}a_4}\cot^2 z$, $\mathfrak{a}$ and $\mathfrak{b}$ the warp powers of ordinary 3-space and of the extra times, $(\sin^2 z)^{3\mathfrak{a}}$ and $(\sin^2 z)^{3\mathfrak{b}}$. The record's corrected metric has $\mathfrak{p} = \frac{1}{2}$, $\mathfrak{q} = -1$, $\mathfrak{a} = 0$, $\mathfrak{b} = \frac{1}{6}$. Unit: dimensionless. Value: $\mathfrak{p} = \frac{1}{2}$, $\mathfrak{q} = -1$, $\mathfrak{a} = 0$, $\mathfrak{b} = \frac{1}{6}$ for the metric used by UM-2. Status: RECORD.
- $\dot{a}_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\dot{b}_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.

Keeping the warp of the extra times ($\mathfrak{b} = 1/6$) forces $\mathfrak{a} = 0$, $\mathfrak{q} = -1$, $\mathfrak{p} = 1/2$: the record's current metric, the order-0 metric of UM-2,

$$\begin{aligned}\bar{g} = \mathrm{diag}\Big(&\mathrm{e}^{-2a_4}b_4\cot^2 z,\ \mathrm{e}^{2a_4},\ \mathrm{e}^{2a_4},\ \mathrm{e}^{2a_4},\ -1,\\ &-\mathrm{e}^{-2a_4}b_4(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4(\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0\end{aligned} \tag{FE-31}$$

Symbols in (FE-31).

- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

Every off-diagonal component of $\mathcal{E}_{(1)}$, $\mathcal{E}_{(2)}$, $\mathcal{E}_{(3)}$ vanishes identically for all $a_4(x_4)$, $b_4(x_4) > 0$ and every real z (record: sympy 14/14, Wolfram 13/13; re-derived in UM-2 by `um2_geometry.py`, checks `y_chart_pair_diagonal`, `author_chart_pair_diagonal`, `charts_agree`). The alternative K of the record (hidden component $\mathrm{e}^{2a_4}\cot^2 z$, 3-space components $\mathrm{e}^{2a_4}(\sin^2 z)^{1/6}$, extra-time components $-\mathrm{e}^{-2a_4}b_4$) is also diagonal at every order; the record leaves the choice to the author, and UM-2 uses the metric FE-31 (status AUTHOR in the inventory: the author's coordinates and functions, corrected by the record under his order).

5.2 The hidden coordinate, the fold and the regular tip

On every interval with $\sin z \neq 0$ the record introduces the hidden coordinate y and the block scale f:

$$
\begin{aligned}
&y = \frac{\ln|\sin z|}{6H}, \qquad \mathrm{d}y = \cot z\,\mathrm{d}x_0, \qquad \xi = \frac{y}{L} \in [-1, 0],\\
&\bar{g}_{MN}\,\mathrm{d}X^M \mathrm{d}X^N = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 + f^2\Big[\mathrm{d}y^2 - \mathrm{e}^{2Hy}\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big],\\
&f = \mathrm{e}^{-a_4} b_4^{1/2}, \qquad K = \frac{1}{f^2}, \qquad \mathcal{H} = \dot{a}_4, \qquad \mathcal{H}_f = \frac{\dot{f}}{f} = \frac{\dot{b}_4}{2b_4} - \dot{a}_4, \qquad \dot{K} = -2K\mathcal{H}_f,\\
&\sqrt{|\bar{g}|} = \mathrm{e}^{3a_4} f^4 |\cot z| (\sin^2 z)^{1/4} = \mathrm{e}^{3a_4} f^4\, \mathrm{e}^{3Hy}\Big|\frac{\mathrm{d}y}{\mathrm{d}x_0}\Big|
\end{aligned}
\tag{FE-32}
$$

Symbols in (FE-32).

- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \leq y \leq 0$. Unit: length. Status: DERIVED.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\dot{a}_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and x_0 deflate). Unit: GeV. Status: DERIVED.
- $\dot{f}$: The time derivative $\mathrm{d}f/\mathrm{d}x_4$ of the block scale. Unit: GeV. Status: DERIVED.
- $\dot{b}_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\dot{K}$: The time derivative $\mathrm{d}K/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.

The fold is $z = \pi/2$ ($y = 0$); the tips are $z \to 0^+$ and $z \to \pi^-$ ($y \to -\infty$). The record replaces each tip by a regular cutoff at $y = -L$ with $HL = 3$, a chosen depth (the extended levels depend on it like those of a box of proper length fL). UM-2 keeps $HL = 3$ (ASSUMED).

5.3 The dirac16complex Lagrangian and the canonical spin connection

The specification of the record (under the author's orders) fixes the Dirac-type Lagrangian of dirac16complex with the contact self-interaction and the canonical (vielbein-postulate, torsion-free) spin connection: the action term IN-5 with the connection IN-7, written out in section 6. The record chose the Dirac-type form because the notebook's real Majorana-type Lagrangian is a total derivative for anticommuting real fields (it has no field equations), and it uses the canonical connection rather than the notebook's contraction of the mixed connection coefficients, which lacks a metric factor. The field equation of the record is $\gamma^M D_M \Psi = (m_\Psi + \lambda \bar{\Psi}\Psi)\Psi$, the part of the dirac16complex field equation FE-12 without the portals.

5.4 Canonical quantisation and the Krein metric

With x_4 as the evolution time, the canonical anticommutator of Ψ and $\Psi^\dagger$ on a slice \$x_4 = \$ const is the matrix $\mathcal{B}$ times the Dirac delta function of the seven remaining coordinates divided by $\sqrt{|g|}$; $\mathcal{B}$ has the eigenvalues +1 (8 times) and −1 (8 times), so the inner product is indefinite (a Krein space). The record works in the good sector (no extra-time momentum), counts the positive-energy branch of every block as particles, and evaluates the expectation values of bilinears with the Krein metric $\mathcal{B}$ inserted between the spinors of the modes (`um2_algebra.py`, checks `B_hermitian_involution`, `B_signature_8_8`, `B_gamma4_gamma_a_hermitian`).

5.5 The Kohn-Sham reduction

For the corrected metric the record reduces the 16-component field equation, in the eight blocks of the commuting operators $\gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$, $\gamma^{x_2}\gamma^{x_3}$, $\gamma^{x_5}\gamma^{x_6}$ (eigenvalues ς, $\mathrm{i}\varsigma_2$ and $\mathrm{i}\varsigma_3$, each sign ±1), to 2×2 block Hamiltonians; UM-2 re-derived the reduction exactly for all eight blocks and general a_4, b_4 (`um2_fields.py`, check `dirac_reduction_exact`; `um2_algebra.py`, checks `block_basis_unitary` to `block_form B gammaˆ(x0) = j s2 sigma3`):

$$
\begin{aligned}
&\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\mathcal{V}\chi(y,x_4)\,:\\
&\gamma^M D_M\Psi - \big(M(y) - \mathrm{i}v(y)\gamma^{x_4}\big)\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\gamma^{x_4}\mathcal{V}\big[\partial_{x_4}\chi + \mathrm{i}h\chi\big],\\
&h = \varsigma\Big[-\frac{\mathrm{i}}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + \mathrm{e}^{-a_4}k\,\sigma_3\Big] + v(y), \qquad \mathrm{i}\,\partial_{x_4}\chi = h\chi;\\
&\text{in each block: } \gamma^{x_0} = \sigma_3, \qquad \mathcal{B} = \varsigma\varsigma_2, \qquad C = \varsigma_2\sigma_2, \qquad \Gamma_4 = -\varsigma\varsigma_2\sigma_3
\end{aligned}
\tag{FE-56}
$$

Symbols in (FE-56).

- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{V}$: The constant 16×2 matrix of the reduction ansatz that places the two-component spinor χ of one block into the 16 components of the eight-dimensional spinor. Unit: dimensionless. Status: DERIVED.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^0 \chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_\mathrm{eff}$ for dirac16complex. Unit: GeV. Status: DERIVED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status:

STANDARD.

- h: The block Hamiltonian: the 2×2 differential operator of one block that generates the time evolution of χ, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies ε. Unit: GeV. Status: DERIVED.
- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- y: The label y in ∂_y: the derivative with respect to the hidden coordinate y. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues ∓ 1 define the halves Ψ_A and Ψ_B, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- ς_2: The second sign that labels the blocks (± 1). Unit: dimensionless. Status: DERIVED.
- C: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar{\Psi} = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

The record treats the contact interaction in the Hartree-plus-exchange (local) approximation; its mean fields are the first terms of M_{eff} and v in FE-64, to which UM-2 adds the mean fields of the portals I11, I12 and I13 (DERIVED):

$$M_{\mathrm{eff}}(y) = m_\Psi + \frac{15}{16}\,\lambda\, n^{\mathrm{sc}}(y) + \frac{\langle \mathcal{F} \rangle}{\Lambda_\star^3}, \qquad v(y) = -\frac{\lambda}{16}\, n(y) + \frac{c_{\mathrm{V}}}{\Lambda_\star^6} \sum_{i=1}^{3} \sum_{\psi} q_\psi\, n_{\psi i}(y),$$
$$v_{\mathrm{b}}(y) = \frac{c_{\mathrm{A}}}{\Lambda_\star^6} \sum_{i=1}^{3} \sum_{\psi} q_\psi^B\, n_{\psi i}(y) \tag{FE-64}$$

Symbols in (FE-64).

- $M_{\mathrm{eff}}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: m_Ψ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- $n^{\mathrm{sc}},\ n_o^{\mathrm{sc}}$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and n_o^{sc} the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_{\mathrm{G}} G^A_{MN} G^{AMN} + c_{\mathrm{W}} W^I_{MN} W^{IMN} + c_{\mathrm{B}} B_{MN} B^{MN}$; $\langle \mathcal{F} \rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $n,\ n_o,\ n_{\psi i},\ n_f,\ n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- c_{V}: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38 \times 10^{-17}$ GeV^{-2}. Status: INVENTED.

- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- q_ψ, q_f: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- $v_{\mathrm{b}}(y)$: The mean field of I13 (baryon densities), acting on the lower component of χ only. Unit: GeV. Status: DERIVED.
- c_{A}: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278 \times 10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- q^B_ψ, q^B_f: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.

The densities, the momentum shells, the normalisation volume of a cell and the occupations, and the Kohn-Sham energy, are the record's (both parities at the fold are filled, each with the weight $\frac{1}{2}$; the record's mass is $m_\Psi = 3H\mathcal{M}$ with $\mathcal{M} = 1/3$, i.e. $m_\Psi = H$, which UM-2 keeps, ASSUMED):

$$
\begin{aligned}
&n = \sum_o \tfrac{1}{2}\, d_o\, \phi_o\, n_o, \qquad n^{\mathrm{sc}} = \sum_o \tfrac{1}{2}\, d_o\, \phi_o\, n^{\mathrm{sc}}_o, \qquad n_o = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \chi_o, \\
&n^{\mathrm{sc}}_o = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \varsigma\sigma_2 \chi_o, \qquad n^{\mathrm{t}}_o = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\, \chi_o^\dagger \varsigma\sigma_3 \chi_o, \qquad d_o = 4\, r_3\big(|\vec{n}_o|^2\big), \qquad \vec{k}_o = k_{\mathrm{lat}}\, \vec{n}_o, \\
&\mathrm{Vol}_7 = \Big(\frac{2\pi}{k_{\mathrm{lat}}}\Big)^3 v_{\mathrm{t}}\, f^4, \qquad \phi_o = \frac{1}{\mathrm{e}^{(\varepsilon_o - \mu_{\mathrm{chem}})/T} + 1} \quad \text{(Mermin; or 0 and 1 for the aufbau)}
\end{aligned}
\tag{FE-65}
$$

Symbols in (FE-65).

- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- d_o, d_s: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- ϕ_o: The occupation of the Kohn-Sham orbital o (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- n^{sc}, n^{sc}_o: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and n^{sc}_o the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- Vol_7: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - \mathrm{e}^{-3HL})/(3H)$). Unit: GeV^{-6}. Value: 34.27 in units $H = 1$, $v_{\mathrm{t}} = 1$, $f = f_*$. Status: RECORD.
- χ, χ_o, χ_1, χ_2: The two-component spinor of one block of the reduction (a function of y and x_4), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; χ_o the Kohn-Sham orbital o, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: GeV$^{1/2}$ (normalised orbital). Status: DERIVED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

- ς: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each ± 1, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor χ ($\sigma_3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- n_o^{t}: The density of the orbital o along the momentum term ($\chi_o^\dagger \varsigma \sigma_3 \chi_o$ times the weight $\mathrm{e}^{-3Hy}/\mathrm{Vol}_7$); it carries the 3-space pressure. Unit: GeV^7. Status: RECORD.
- r_3: The number of representations of an integer as an ordered sum of three squares of integers (signs counted). Unit: dimensionless. Status: STANDARD.
- $\vec{n}_o$: The integer lattice vector of the momentum shell of the orbital o ($\vec{n} \in \mathbb{Z}^3$). Unit: dimensionless. Status: RECORD.
- $\vec{k},\ \vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- k_{lat}: The spacing of the momentum lattice of the Kohn-Sham cell. Unit: GeV. Value: $0.25\,H$. Status: RECORD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- v_{t}: The coordinate volume of the compact extra times x_5, x_6, x_7. Unit: GeV^{-3}. Value: not fixed; $v_{\mathrm{t}} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\varepsilon,\ \varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- μ_{chem}: The chemical potential of the Mermin occupations. Unit: GeV. Status: RECORD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

$$E_{\mathrm{KS}} = \sum_o d_o\, \phi_o\, \varepsilon_o - 2\,\mathrm{Vol}_7 \int_{-L}^{0} \mathrm{e}^{3Hy}\, e_{\mathrm{int}}\, \mathrm{d}y, \qquad e_{\mathrm{int}} = \frac{15}{32}\,\lambda\left(n^{\mathrm{sc}}\right)^2 - \frac{1}{32}\,\lambda\, n^2 \tag{FE-66}$$

Symbols in (FE-66).

- E_{KS}: The Kohn-Sham energy of a cell. Unit: GeV. Status: RECORD.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- $d_o,\ d_s$: A degeneracy: $d_o = 4 r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- ϕ_o: The occupation of the Kohn-Sham orbital o (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- $\varepsilon,\ \varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- Vol_7: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - \mathrm{e}^{-3HL})/(3H)$). Unit: GeV^{-6}. Value: 34.27 in units $H = 1$, $v_{\mathrm{t}} = 1$, $f = f_*$. Status: RECORD.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- e_{int}: The interaction energy density of the contact term in the Kohn-Sham energy and energy-momentum tensor. Unit: GeV^8. Status: RECORD.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \leq 0.001$). Status: RECORD.
- n^{sc}, n_o^{sc}: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and n_o^{sc} the contribution of the orbital o. Unit: GeV^7. Status: RECORD.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.

Two reference states of the record (at $f = 1$, $a_4 = 0$: 8 particles in a cell with $\lambda = 157.7\, H^{-6}$, and 592 particles with $\lambda = 2.131\, H^{-6}$) serve as validation targets of the independent UM-2 solver: both are reproduced to 10^{-13} in the energy, the highest occupied and the lowest unoccupied level and to 10^{-9} in the integrated hidden pressure (`um2_ks_jobs.py validate`, checks `validate_N8_lamp1_a00`, `validate_N592_lamp1_a00`). The proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - \mathrm{e}^{-3HL})/(3H)$.

5.6 Earlier work used only for comparison

The first universe model of this repository (UM-1, folder `Revision/universe_model`) is not an input of UM-2. UM-2 differs from it in keeping the author's weights and Λ, in the electroweak sector, in the number of generations and in the dark-matter candidate. Version 1 of UM-2 (2026-10-17, one generation) is superseded; its differences are item 21 of the corrections.

6 The action of UM-2 term by term, and the origin of each interaction

The action of UM-2 consists of the 18 interaction terms IN-1 to IN-18 below, each written out with its symbols and followed by its origin. Varying the action with respect to every field gives the field equations FE-1 to FE-30; the other field equations are their exact reductions on the static block and the equations of the four-dimensional effective theory (section 8). The terms labelled INVENTED are the new phenomenological interactions that the author's orders require; for each the reason for its form and its consistency with QCD and the electroweak model are stated.

6.1 IN-1: the action

$$S = S_{\mathrm{EL}} + S_{\mathrm{stab}} + S_{\Psi} + S_{\mathrm{gauge}} + S_{\mathrm{f}} + S_{\mathrm{tip}} + S_{\mathrm{I11}} + S_{\mathrm{I12}} + S_{\mathrm{I13}} \tag{IN-1}$$

Symbols in (IN-1).

- S: The action of UM-2 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar = 1$). Status: ASSUMED.
- S_{EL}: The Einstein-Lovelock action of gravity. Unit: dimensionless. Status: AUTHOR.
- S_{stab}: The action of the block stabiliser (a vacuum energy of the block). Unit: dimensionless. Status: ASSUMED.
- S_{Ψ}: The action of the dirac16complex field (kinetic term with the canonical spin connection, mass term and contact self-interaction). Unit: dimensionless. Status: RECORD.
- S_{gauge}: The action of the gauge fields of SU(3), SU(2) and $\mathrm{U}(1)_Y$ in eight dimensions. Unit: dimensionless. Status: ASSUMED.
- S_{f}: The action of the Standard-Model fermions of three generations (kinetic and bulk-mass terms). Unit: dimensionless. Status: ASSUMED.
- S_{tip}: The action on the tip $y = -L$: the sigma field and the tip Yukawa couplings. Unit: dimensionless. Status: INVENTED.
- S_{I11}: The action of the portal I11 (dirac16complex vector current times the $B - L$ current of the fermions). Unit: dimensionless. Status: INVENTED.

- S_{I12}: The action of the portal I12 (scalar density of dirac16complex times the gauge kinetic invariants). Unit: dimensionless. Status: INVENTED.
- S_{I13}: The action of the portal I13 (the x_0-chiral vector current of dirac16complex times the baryon-number current). Unit: dimensionless. Status: INVENTED.

Origin (ASSUMED). The choice of the terms is the model: the author's gravity (IN-2, IN-3); a stabiliser of the block (IN-4); the record's dirac16complex field with its gamma matrices and its coupling to gravity (IN-5 to IN-7); the gauge fields and the three generations of the Standard Model in eight dimensions (IN-8 to IN-12); the tip sigma field in place of a Higgs field (IN-13 to IN-15); and three portals between dirac16complex and the Standard Model (IN-16 to IN-18). There is no Higgs field: the orders exclude the Higgs particle. The chiral tip Z portal I9 of version 1 is no longer part of the action (its coefficients are zero; the end of this section).

6.2 IN-2: Einstein-Lovelock gravity with the author's weights

$$S_{\mathrm{EL}} = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\Big), \qquad \kappa = 8\pi G_8,$$
$$\alpha_{(p)} = w_{(p)}\,H^{2-2p}, \qquad w_{(1)}=1, \quad w_{(2)}=\frac{1}{32}, \quad w_{(3)}=\frac{1}{1152}, \qquad \Lambda = -10H^2 \tag{IN-2}$$

Symbols in (IN-2).

- S_{EL}: The Einstein-Lovelock action of gravity. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X=\{x_0,\dots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\dots,x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1,\dots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p=1,2,3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)}=w_{(p)}H^{2-2p}$ ($\alpha_{(1)}=1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p of the metric (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)}=R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda=-10H^2$. Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa=8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)}=1$, $w_{(2)}=1/32$, $w_{(3)}=1/1152$. Unit: dimensionless. Value: $(1,1/32,1/1152)$. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

Origin (AUTHOR). The author's Einstein-Lovelock action (section 4): the weights $w_{(p)}$ and the cosmological constant $\Lambda=-10H^2$ are the author's, unchanged. Its metric variation is the left side of FE-1, verified by four routes (pairing formula, Ricci tensor, explicit Gauss-Bonnet tensor, minisuperspace variation with a lapse), by the Bianchi identities and against the record's equations (`um2_geometry.py`, 25 checks).

6.3 IN-3: the Lovelock densities

$$\mathcal{L}_{(p)} = \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p},$$
$$\mathcal{L}_{(1)} = R, \qquad \mathcal{L}_{(2)} = R^2 - 4R_{MN}R^{MN} + R_{MNPQ}R^{MNPQ} \tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p of the metric (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: 1/length^{2p}. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, δ_{l0}, δ^{I3}); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane (a, b). Unit: 1/length^2. Status: STANDARD.

Origin (STANDARD). The Lovelock densities (Lovelock 1971), built with the generalised Kronecker delta (the definition of the author's file `Generalized_Kronecker_Delta`, which is not committed). For the pair-diagonal curvature of the order-0 metric they reduce to the pairing formula FE-34 (DERIVED; `um2_geometry.py`, checks `einstein_from_ricci_equals_pairing_k1`, `gauss_bonnet_tensor_equals_pairing_k2`).

6.4 IN-4: the block stabiliser

$$S_{\rm stab} = -\int \mathrm{d}^8X\,\sqrt{|g|}\,\rho_{\rm stab}(f), \qquad \kappa\rho_{\rm stab}(f) = \kappa\rho_{\rm stab}(f_*)\Big(\frac{f_*}{f}\Big)^8, \qquad f = \mathrm{e}^{-a_4}\,b_4^{1/2} \tag{IN-4}$$

Symbols in (IN-4).

- $S_{\rm stab}$: The action of the block stabiliser (a vacuum energy of the block). Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\rho_{\rm stab}$: The energy density of the block stabiliser, a function of the block scale, $\rho_{\rm stab} \propto f^{-8}$; its value at the static block is $\rho_{\rm stab}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_{\rm stab}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\rm N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV^{-2}. Status: AUTHOR.

- f_*: The block scale of the stabilised (static) block, $f_* = K_*^{-1/2}$. Unit: dimensionless. Value: 0.215556. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.

Origin (ASSUMED; its microscopic origin is OPEN). Introduced in version 1 because the author's weights and Λ admit no static block with attractive four-dimensional gravity without a source: the only sourceless static block with $\mathcal{H} = 0$ has $K = 1/(4w_{(2)}) = 8$, where the coefficient of the four-dimensional Einstein-Hilbert term is $\mathcal{G} = -1$ (repulsive), and it would need $\Lambda = -24H^2 \neq -10H^2$ (`um2_background.py`, checks `static_vacuum_general`, `newton_factor_at_vacuum`). The stabiliser is conserved ($\rho_{\mathrm{stab}} \propto f^{-8}$), four-dimensionally Lorentz invariant ($p_3 = -\rho_{\mathrm{stab}}$) and isotropic in the block ($p_0 = p_5$); its normalisation is fixed by the static block (FE-43). The family in which the hidden pressure is w_{stab} times the energy density, $p_0 = p_5 = w_{\mathrm{stab}}\rho_{\mathrm{stab}}$ with $\rho_{\mathrm{stab}} \propto f^{-4(1+w_{\mathrm{stab}})}$, was also computed at $w_{\mathrm{stab}} = 1/4, 1/2, 3/4, 1, 5/4, 3/2, 2, 3$: every member has a linearly stable block with attractive gravity (check `block_stable_family`); UM-2 uses $w_{\mathrm{stab}} = 1$.

6.5 IN-5: the dirac16complex field

$$S_\Psi = \int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda}{2}\big(\bar\Psi\Psi\big)^2\Big],$$
$$m_\Psi = 3H\mathcal{M} = H \qquad (\mathcal{M} = 1/3) \tag{IN-5}$$

Symbols in (IN-5).

- S_Ψ: The action of the dirac16complex field (kinetic term with the canonical spin connection, mass term and contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- λ: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV^{-6}. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}} \leq 0.001$). Status: RECORD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296

TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\mathcal{M}$: The record's dimensionless mass parameter of dirac16complex, $\mathcal{M} = 1/3$ ($m_\Psi = 3H\mathcal{M}$). Unit: dimensionless. Value: 1/3. Status: RECORD.

Origin (RECORD). The record's Dirac-type Lagrangian (section 5) with the author's form of the mass, $m_\Psi = 3H\mathcal{M}$, and the record's value $\mathcal{M} = 1/3$, which UM-2 keeps (ASSUMED). The contact self-coupling λ is the record's; UM-2 does not fix it: the Kohn-Sham machinery is tested with $\lambda = 0$ and $\lambda = \pm 7 \times 10^{-5} H^{-6}$, and the fold-band radiation of the self-interaction channel (FE-135, FE-136) requires $|\lambda| \leq 1.4 \times 10^{-10} H^{-6}$ for the coordinate volume $v_{\mathrm{t}} = H^{-3}$ of the extra times (ASSUMED bound; corrections, item 24).

6.6 IN-6: the Clifford algebra, the Dirac adjoint and the two halves of dirac16complex

$$\begin{aligned}
&\gamma^a\gamma^b + \gamma^b\gamma^a = 2\eta^{ab}\,\mathbf{1}, \qquad \eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1), \qquad \gamma^M = \mathbf{e}_a{}^M \gamma^a, \\
&\bar\Psi = \Psi^\dagger C, \qquad C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}, \qquad \Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}, \qquad \mathcal{B} = -\mathrm{i}C\gamma^{x_4}, \\
&\Psi = \Psi_A + \Psi_B, \qquad \mathcal{B}\Psi_A = -\Psi_A, \qquad \mathcal{B}\Psi_B = +\Psi_B
\end{aligned} \tag{IN-6}$$

Symbols in (IN-6).

- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, ... the matrices along x_0, x_4, ...), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ (η^{ab} the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}^a{}_M$, $\mathbf{e}_a{}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- C: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar\Psi = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- Γ_4: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues ∓ 1 define the halves Ψ_A and Ψ_B, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- Ψ_A, Ψ_B: The two halves of dirac16complex, $\Psi = \Psi_A + \Psi_B$, the eigenspaces of $\mathcal{B}$ with the eigenvalues -1 and $+1$; each half carries one dark-matter species. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

Origin (AUTHOR). The author's real 16×16 matrices of section 4 and the matrices C, Γ_4, $\mathcal{B}$ derived from them. The Clifford algebra, the reality, $C^T = C$ and $C^2 = \mathbf{1}$, the antisymmetry of $C\gamma^a$, the irreducibility of the 16-dimensional representation of Pin(4,4), the two inequivalent halves of Spin(4,4) and the block forms are verified exactly (`um2_algebra.py`, 31 checks). The halves Ψ_A and Ψ_B are the eigenspaces of $\mathcal{B}$; $\mathcal{B}$ commutes with every interaction of the good sector, so the numbers of quanta of the two halves are conserved separately (computed with the author's matrices: `um2_analysis.py`, check `B5_stability_and_neutrality`; `um2_electroweak.py`, checks `B_lorentz_invariant`, `Gamma4_equals_minus_B_gamma0`).

6.7 IN-7: the canonical spin connection

$$
\begin{aligned}
&D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M, \qquad \Omega_M = \tfrac{1}{2}\,\omega_{Mab}\,\mathcal{S}^{ab}, \qquad \mathcal{S}^{ab} = \tfrac{1}{4}\big[\gamma^a,\gamma^b\big],\\
&\omega_M{}^a{}_b = \mathbf{e}^a{}_N\big(\partial_M \mathbf{e}_b{}^N + \Gamma^N{}_{MP}\,\mathbf{e}_b{}^P\big), \qquad \omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b, \qquad g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\,\mathbf{e}^b{}_N,\\
&\Gamma^N{}_{MP} = \tfrac{1}{2}\,g^{NQ}\big(\partial_M g_{QP} + \partial_P g_{QM} - \partial_Q g_{MP}\big)
\end{aligned}
\tag{IN-7}
$$

Symbols in (IN-7).

- $D_M,\ D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Psi,\ \bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- Ω_M: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- $\omega_M{}^a{}_b$: The coefficients of the spin connection of the vielbein (the Levi-Civita connection in the frame basis); $\omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b$. Unit: 1/length. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with η_{ab}; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- b: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- $\mathcal{S}^{ab}$: The generators of frame rotations on spinors, $\mathcal{S}^{ab} = \frac{1}{4}[\gamma^a,\gamma^b]$ (the commutator). Unit: dimensionless. Status: STANDARD.
- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}^a{}_M,\ \mathbf{e}_a{}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^N{}_{MP}$: The Christoffel symbols (the Levi-Civita connection) of the metric, $\Gamma^N{}_{MP} = \frac{1}{2}g^{NQ}(\partial_M g_{QP} + \partial_P g_{QM} - \partial_Q g_{MP})$. Unit: 1/length. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ (η^{ab} the same matrix). Unit: dimensionless. Status: AUTHOR.
- c: A frame index running over $0, \ldots, 7$ (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

Origin (STANDARD; chosen by the record). The canonical (vielbein-postulate, torsion-free) spin connection, the standard coupling of a spinor to gravity, chosen by the record instead of the notebook's contraction of the mixed connection coefficients (section 5); the order of 2026-10-17 asks for a non-trivial canonical spin connection. On the order-0 metric it makes γ^N covariantly constant, gives FE-13 and vanishes only in flat 4+4 space (`um2_geometry.py`, checks `gamma_covariantly_constant`, `gamma_mu_Omega_mu`, `nontrivial_unless_flat`).

6.8 IN-8: the gauge fields in eight dimensions

$$S_{\text{gauge}} = -\frac{1}{4}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\frac{1}{\hat g_3^2}\,G^A_{MN}G^{AMN} + \frac{1}{\hat g_2^2}\,W^I_{MN}W^{IMN} + \frac{1}{\hat g_1^2}\,B_{MN}B^{MN}\Big) \tag{IN-8}$$

Symbols in (IN-8).

- S_{gauge}: The action of the gauge fields of SU(3), SU(2) and U(1)$_Y$ in eight dimensions. Unit: dimensionless. Status: ASSUMED.
- d: The differential: dX is an infinitesimal change of X, dX/dY the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\hat g_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat g_3^2$); $g_3^2 = \hat g_3^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat g_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat g_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\hat g_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat g_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- B_M, B_{MN}: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

Origin (ASSUMED structure; STANDARD form). The gauge group of the Standard Model, $SU(3)_c \times SU(2)_L \times U(1)_Y$, in eight dimensions with the standard Yang-Mills action in the geometric normalisation (the couplings $1/\hat g_i^2$ in front of the kinetic terms). The gauge structure is that of QCD and of the electroweak model; the zero modes of the gauge fields are flat in x_0 and are the four-dimensional gauge fields with the couplings $g_i^2 = \hat g_i^2/V_\perp$ (FE-49). The gluons and the photon stay exactly massless (`um2_spectra.py`, check `massless_zero_mode`).

6.9 IN-9: the field strengths

$$\begin{aligned}
&G_M = G^A_M\,t^A, \qquad G_{MN} = \partial_M G_N - \partial_N G_M - \mathrm{i}\big[G_M, G_N\big], \qquad G^A_{MN} = 2\,\mathrm{tr}\big(t^A G_{MN}\big),\\
&W_M = W^I_M\,\frac{\tau^I}{2}, \qquad W_{MN} = \partial_M W_N - \partial_N W_M - \mathrm{i}\big[W_M, W_N\big], \qquad W^I_{MN} = \mathrm{tr}\big(\tau^I W_{MN}\big),\\
&B_{MN} = \partial_M B_N - \partial_N B_M
\end{aligned} \tag{IN-9}$$

Symbols in (IN-9).

- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- t^A: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

Origin (STANDARD). The field strengths of the three gauge groups.

6.10 IN-10: the physical electroweak fields

$$W^\pm_M = \frac{W^1_M \mp \mathrm{i} W^2_M}{\sqrt{2}\,\hat{g}_2}, \qquad Z_M = \frac{W^3_M - B_M}{\hat{g}_Z}, \qquad A_M = \frac{\hat{g}_1^2\, W^3_M + \hat{g}_2^2\, B_M}{\hat{g}_1 \hat{g}_2 \hat{g}_Z},$$
$$\hat{g}_Z = \sqrt{\hat{g}_2^2 + \hat{g}_1^2}, \qquad \cos\theta_\mathrm{W} = \frac{\hat{g}_2}{\hat{g}_Z}, \qquad \sin\theta_\mathrm{W} = \frac{\hat{g}_1}{\hat{g}_Z} \tag{IN-10}$$

Symbols in (IN-10).

- $W^\pm_M$, W^+, W^-: The charged weak bosons, canonically normalised: $W^\pm_M = (W^1_M \mp \mathrm{i} W^2_M)/(\sqrt{2}\,\hat{g}_2)$; W^+ carries the electric charge $+e$ and W^- the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV^3. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- Z_M: The Z boson field, canonically normalised, $Z_M = (W^3_M - B_M)/\hat{g}_Z$. Unit: GeV^3. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- $\hat{g}_Z$: The eight-dimensional coupling of the Z, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: GeV^{-2}. Status: STANDARD.
- A_M: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W^3_M + \hat{g}_2^2 B_M)/(\hat{g}_1 \hat{g}_2 \hat{g}_Z)$. Unit: GeV^3. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.

- θ_{W}: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured W and Z masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.

Origin (STANDARD). The physical electroweak fields of the Standard Model, canonically normalised. W^+ carries the electric charge $+e$ and W^- the charge $-e$; W^- is the antiparticle of W^+ (DERIVED: `um2_electroweak.py`, check `W_charges`; the author's order "the W+ and W- bosons carry positive and negative electric charges. Do not forget this"). The angle of the couplings is fixed by the measured W and Z masses with the measured electroweak parameter (FE-55).

6.11 IN-11: the Standard-Model fermions of three generations

$$S_{\mathrm{f}} = \sum_{i=1}^{3}\sum_{\psi}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\tfrac{1}{2}\big(\bar{\psi}_i\gamma^M D_M\psi_i - (D_M\bar{\psi}_i)\gamma^M\psi_i\big) - \mu_{\psi i}\,\bar{\psi}_i\psi_i\Big],$$
$$\psi \in \{Q,\ u_R,\ d_R,\ \ell,\ e_R,\ \nu_R\}, \qquad \mu_{\psi i} = \mu_Q\ (\psi = Q), \qquad \mu_{\psi i} = \mu_\ell\ (\psi = \ell) \tag{IN-11}$$

Symbols in (IN-11).

- S_{f}: The action of the Standard-Model fermions of three generations (kinetic and bulk-mass terms). Unit: dimensionless. Status: ASSUMED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- ψ_i, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet ψ of the generation i; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation c of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

- e_R, e_{Ri}: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: GeV$^{7/2}$. Status: ASSUMED.
- ν_R, ν_{Ri}: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: GeV$^{7/2}$. Status: ASSUMED.
- μ_Q: The common bulk mass of the three quark doublets. Unit: GeV. Value: $5.2304\,H$ ($c_Q = 3.38230$). Status: DERIVED.
- μ_ℓ: The common bulk mass of the three lepton doublets. Unit: GeV. Value: $9.2055\,H$ ($c_\ell = 5.95288$). Status: DERIVED.

Origin (ASSUMED). Three generations of quarks and leptons (the author's choice (A) of 2026-10-18), each multiplet a 16-component spinor of the same Clifford algebra times its gauge indices, with bulk masses; one doublet bulk mass per sector, μ_Q and μ_ℓ (ASSUMED), and right-handed neutrinos ν_{Ri} for the Dirac masses of the neutrinos in the x_0 mechanism (ASSUMED; the Majorana alternative without them is analysed). The hypercharges are those of the Standard Model ($\frac{1}{6}$, $\frac{2}{3}$, $-\frac{1}{3}$, $-\frac{1}{2}$, -1, 0 for Q, u_R, d_R, ℓ, e_R, ν_R), and every generation is anomaly free (DERIVED: `um2_electroweak.py`, check `anomaly_cancellation`). The neutrinos of the lepton doublets are the weak partners of the charged leptons (the flavours ν_e, ν_μ, ν_τ), not the mass states ν_1, ν_2, ν_3.

6.12 IN-12: the covariant derivative of a fermion and the electric charge

$$D_M = \partial_M + \Omega_M - \mathrm{i}G^A_M\, t^A - \mathrm{i}W^I_M\,\frac{\tau^I}{2} - \mathrm{i}\,Y B_M, \qquad Q_{\mathrm{em}} = T_3 + Y \tag{IN-12}$$

Symbols in (IN-12).

- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- Ω_M: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- t^A: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- Y: The hypercharge (the generator of U(1)$_Y$); Y_ψ the hypercharge of the multiplet ψ (Q: $\frac{1}{6}$, u_R: $\frac{2}{3}$, d_R: $-\frac{1}{3}$, ℓ: $-\frac{1}{2}$, e_R: -1, ν_R: 0). Unit: dimensionless. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- Q_{em}: The electric charge in units of the elementary charge, $Q_{\mathrm{em}} = T_3 + Y$; Q^f_{em} that of the fermion f. Unit: dimensionless. Status: STANDARD.
- T_3: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); T^f_3 that of the fermion f. Unit: dimensionless. Status: STANDARD.

Origin (STANDARD). The covariant derivative of the Standard Model with the spin connection; the electric charge $Q_{\rm em} = T_3 + Y$ in every generation (`um2_electroweak.py`, checks `electric_charges`, `photon_coupling`, `Z_coupling`).

6.13 IN-13: the tip sigma field and the tip Yukawa couplings (no Higgs particle)

$$
\begin{aligned}
S_{\rm tip} &= \int_{y=-L} \mathrm{d}^7 X_{\rm tip} \sqrt{|g_{\rm tip}|}\, \Big[-\frac{v_\Sigma^2}{4}\, \mathrm{tr}\big((D_\mu \Sigma)^\dagger D^\mu \Sigma\big) - \mathcal{L}_{\rm Y} \Big], \\
\mathcal{L}_{\rm Y} &= \sum_{i,j=1}^{3} \Big[\bar{\ell}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_\nu)_{ij}\, \nu_{Rj} \\ (\mathcal{Y}_e)_{ij}\, e_{Rj} \end{pmatrix} + \bar{Q}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_u)_{ij}\, u_{Rj} \\ (\mathcal{Y}_d)_{ij}\, d_{Rj} \end{pmatrix} \Big] + \text{h.c.}
\end{aligned}
\tag{IN-13}
$$

Symbols in (IN-13).

- $S_{\rm tip}$: The action on the tip $y = -L$: the sigma field and the tip Yukawa couplings. Unit: dimensionless. Status: INVENTED.
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X_{\rm tip}$: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\rm tip}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- $g_{\rm tip}$: The metric induced on the seven-dimensional tip $y = -L$ (the metric g_{MN} restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\rm tip}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\rm W}$. Unit: GeV$^{5/2}$. Value: fixed through $\beta_{\rm W} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathcal{L}_{\rm Y}$: The tip Yukawa Lagrangian: the couplings of the doublets to the singlets through the sigma field and the tip matrices $\mathcal{Y}$ (plus the Hermitian conjugate, h.c.). Unit: GeV^7. Status: INVENTED.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- ℓ_i, $\bar{\ell}_i$: The lepton doublet of the generation i (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex 3×3 tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices i, j); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- ν_R, ν_{Ri}: The right-handed neutrino (a gauge singlet): ν_R the multiplet, ν_{Ri} that of the generation i; its zero modes give Dirac neutrino masses. Unit: GeV$^{7/2}$. Status: ASSUMED.
- e_R, e_{Ri}: The right-handed charged-lepton singlet: e_R the multiplet, e_{Ri} that of the generation i (e_R, μ_R, τ_R). Unit: GeV$^{7/2}$. Status: ASSUMED.
- Q_i, $\bar{Q}_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.

- u_R, u_{Ri}, $\bar{u}_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar{d}_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.

Origin (INVENTED for this geometry). *Reason.* The orders exclude the Higgs particle, but W^+, W^- and Z are massive and the quarks and leptons need their masses and mixing. UM-2 breaks the electroweak symmetry without a Higgs particle by a unitary sigma field Σ ($\Sigma^\dagger\Sigma = \mathbf{1}$: three Goldstone modes, which become the longitudinal W^+, W^- and Z, and no physical scalar) that lives only on the tip $y = -L$ of the x_0 cell. The concepts of symmetry breaking by a non-linear sigma field and of fermion localisation in an extra dimension are STANDARD; their realisation in this geometry is INVENTED. The tip Yukawa couplings $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$ couple the doublets and the singlets at the tip; on the zero modes they become the tip mixing conditions FE-30, whose tip matrices are the PMNS and the CKM matrices (ASSUMED values). *Consistency with the electroweak model and QCD.* Every tip term is invariant under $SU(2)_L \times U(1)_Y$ for arbitrary complex tip matrices (DERIVED with random group elements, deviation 1.1×10^{-14}: `um2_fields.py`, check `tip_terms_gauge_invariant`) and is a colour singlet; the custodial symmetry of the sigma-field term gives $\rho_{\mathrm{EW}} = 1$ at leading order (`um2_spectra.py`, check `custodial_leading_order`); the photon receives no tip term (IN-15). *Conflicts with data* (inherent in the specified content, stated in every version): there is no Higgs boson, while one is observed at 125 GeV; without it the scattering of longitudinal W^+ and W^- violates perturbative unitarity near 1.2 TeV, far below the Kaluza-Klein scale of 121 TeV.

6.14 IN-14: the covariant derivative of the sigma field and its gauge transformation

$$D_\mu\Sigma = \partial_\mu\Sigma - \mathrm{i}W^I_\mu \frac{\tau^I}{2}\,\Sigma + \mathrm{i}B_\mu\,\Sigma\,\frac{\tau^3}{2}, \qquad \Sigma^\dagger\Sigma = \mathbf{1}, \qquad \Sigma \to U_L\,\Sigma\,U_Y^\dagger, \qquad U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2} \tag{IN-14}$$

Symbols in (IN-14).

- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M\tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- B_M, B_{MN}: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- U_L: An SU(2) gauge transformation (a unitary 2×2 matrix acting on doublets from the left). Unit: dimensionless. Status: STANDARD.
- U_Y: A U(1)$_Y$ gauge transformation of the sigma field, $U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2}$. Unit: dimensionless. Status: STANDARD.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- ϑ: The angle (gauge parameter) of a $\mathrm{U}(1)_Y$ transformation. Unit: dimensionless. Status: STANDARD.

Origin (INVENTED). The covariant derivative of the sigma field with the gauge transformations of the electroweak model (U_L an $SU(2)_L$ matrix, U_Y the hypercharge rotation by the angle ϑ); with it every tip term is gauge invariant (DERIVED, as for IN-13; the hypercharge phase of a fermion is $\psi \to \mathrm{e}^{\mathrm{i}Y\vartheta}\psi$, corrections, item 1).

6.15 IN-15: the tip mass term of W^+, W^- and Z

$$-\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big)\Big|_{\Sigma=\mathbf{1}} = -\frac{v_\Sigma^2}{8}\Big[2\hat{g}_2^2\,W_\mu^+W^{-\mu} + \hat{g}_Z^2\,Z_\mu Z^\mu\Big] \tag{IN-15}$$

Symbols in (IN-15).

- v_Σ: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter β_{W}. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV^{-2}. Status: ASSUMED.
- $W_M^\pm$, W^+, W^-: The charged weak bosons, canonically normalised: $W_M^\pm = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; W^+ carries the electric charge $+e$ and W^- the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV^3. Status: STANDARD.
- $\hat{g}_Z$: The eight-dimensional coupling of the Z, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: GeV^{-2}. Status: STANDARD.
- Z_M: The Z boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: GeV^3. Status: STANDARD.

Origin (DERIVED). The sigma-field term of IN-13 in the unitary gauge: W^+ and W^- share one mass (particle and antiparticle), and the photon, the combination without a tip term, stays massless. On the static block the term becomes the Robin condition FE-51 of the boson profiles (`um2_fields.py`, check `gauge_boundary_conditions`; `um2_spectra.py`, checks `rho_fit`, `mZ_reproduced`). Its sign is that of the signature $(+,+,+,-)$ of $x_1,\ldots,x_4$ (corrections, item 12).

6.16 IN-16: portal I11, the vector contact portal to the $B-L$ currents

$$S_{\mathrm{I11}} = \int \mathrm{d}^8X\,\sqrt{|g|}\,\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\left(\mathrm{i}\bar{\Psi}\gamma^M\Psi\right)J_M^{B-L}, \qquad J_M^{B-L} = \sum_{i=1}^{3}\sum_{\psi} q_\psi\left(\mathrm{i}\bar{\psi}_i\gamma_M\psi_i\right),$$
$$q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}, \qquad q_\ell = q_{e_R} = q_{\nu_R} = -1 \tag{IN-16}$$

Symbols in (IN-16).

- S_{I11}: The action of the portal I11 (dirac16complex vector current times the $B-L$ current of the fermions). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- c_V: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ GeV^{-2}. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- J^{B-L}_M: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: GeV^7. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- ψ: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over ψ runs over the six. Status: STANDARD.
- q_ψ, q_f: The $B-L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- ψ_i, $\bar\psi_i$: A Standard-Model fermion field of eight dimensions: ψ_i is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation i, ψ_f one fermion component f (in four-dimensional formulas its zero mode); $\bar\psi = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

Origin (INVENTED; version 1, with its charges fixed in version 3). *Reason.* A contact coupling of the conserved vector current of the gauge-singlet field Ψ to a gauge-invariant conserved current of all quarks and leptons, so that dirac16complex feels every Standard-Model fermion. *Consistency with QCD and the electroweak model.* Both currents are singlets of $SU(3)_c \times SU(2)_L \times U(1)_Y$; the $B-L$ charge is flavour universal and conserved by every interaction of the Standard Model (STANDARD; with the right-handed neutrinos it is free of anomalies), so the portal breaks no gauge symmetry and leaves the Standard-Model interactions unchanged; the currents are the Hermitian currents $\mathrm{i}\bar\Psi\gamma^M\Psi$ and $\mathrm{i}\bar\psi\gamma_M\psi$ (corrections, item 13). Its effects on the Standard Model are mean fields of relative size 10^{-17} at most in the fermion equations FE-14 to FE-19. *Strength.* On the zero modes it is the contact interaction FE-70 with $G_\mathrm{V} = c_\mathrm{V}H/(2f^4v_\mathrm{t}\Lambda_\star^6)$, set to the benchmark $G_\mathrm{V} = 4.38 \times 10^{-17}$ GeV^{-2}, at which the freeze-in of the fold bands from the highest temperature $T_\mathrm{max} = 10$ TeV adds $\Delta N_\mathrm{eff} = 0.001$ (both ASSUMED; FE-129). Version 2 had left the charges general and set $c_\mathrm{V} = 0$.

6.17 IN-17: portal I12, the gauge-kinetic scalar portal

$$S_\mathrm{I12} = -\int \mathrm{d}^8X\,\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\,\mathcal{F}, \qquad \mathcal{F} = c_\mathrm{G}\,G^A_{MN}G^{AMN} + c_\mathrm{W}\,W^I_{MN}W^{IMN} + c_\mathrm{B}\,B_{MN}B^{MN} \tag{IN-17}$$

Symbols in (IN-17).

- S_I12: The action of the portal I12 (scalar density of dirac16complex times the gauge kinetic invariants). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_\mathrm{G} G^A_{MN} G^{AMN} + c_\mathrm{W} W^I_{MN} W^{IMN} + c_\mathrm{B} B_{MN} B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV^4. Status: INVENTED.
- c_G: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{G}/\Lambda_\star^3 = c_{\mathrm{I}12}/\Lambda_\star^3$. Status: INVENTED.
- G^A_M, G_M, G_{MN}, G^A_{MN}: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components G^A_M ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength G_{MN} (matrix) and its components G^A_{MN}; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in x_0). Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- c_W: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{W}/\Lambda_\star^3 = c_{\mathrm{I}12}/\Lambda_\star^3$. Status: INVENTED.
- W^I_M, W_M, W_{MN}, W^I_{MN}: The SU(2) gauge field (geometric normalisation): the components W^I_M ($I = 1, 2, 3$; W^3 the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength W_{MN} and its components W^I_{MN}; with Greek indices the components along the observable directions. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_B: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{B}/\Lambda_\star^3 = c_{\mathrm{I}12}/\Lambda_\star^3$. Status: INVENTED.
- B_M, B_{MN}: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: ASSUMED.

Origin (INVENTED; version 1). *Reason.* The coupling of the scalar density $\bar\Psi\Psi$ to the gauge-invariant kinetic densities of the gluons and of the electroweak fields: the gauge-singlet field couples to the gauge bosons of all three groups, its density makes the gauge couplings field dependent (FE-21), and the dark matter can annihilate into gauge bosons. *Consistency with QCD and the electroweak model.* $G^A_{MN}G^{AMN}$, $W^I_{MN}W^{IMN}$ and $B_{MN}B^{MN}$ are gauge invariant (the first is the operator of the QCD trace anomaly), so the portal preserves $SU(3)_c \times SU(2)_L \times U(1)_Y$ and keeps the photon and the gluons exactly massless (`um2_coupled.py`, check `C5_WZ_and_photon`); the shifts of the couplings are at most 1.7×10^{-16} (FE-79). The power of the scale is $\Lambda_\star^3$ in the geometric normalisation (corrections, item 10). *Strength.* The unitarity-safe benchmark $G_{\mathrm{I}12} = \pi/(2M^3_\mathrm{DM})$ with the universal coefficients $c_\mathrm{G} = c_\mathrm{W} = c_\mathrm{B} = c_{\mathrm{I}12}$ (ASSUMED; FE-127, FE-128). Version 2 found that I12 alone cannot give the relic abundance within perturbative unitarity, because the operator annihilates a pair only from the state ${}^3P_0$ (corrections, item 22); this led to I13.

6.18 IN-18: portal I13, the x_0-chiral vector portal to the baryon-number current

$$S_{\mathrm{I}13} = \int \mathrm{d}^8X\,\sqrt{|g|}\,\frac{c_\mathrm{A}}{\Lambda_\star^6}\sum_{M\neq x_0}\left(\mathrm{i}\bar\Psi\gamma^M P_b\Psi\right)J^B_M, \qquad P_b = \frac{1-\gamma^{x_0}}{2}, \qquad P_a = \frac{1+\gamma^{x_0}}{2},$$
$$J^B_M = \frac{1}{3}\sum_{i=1}^{3}\left(\mathrm{i}\bar Q_i\gamma_M Q_i + \mathrm{i}\bar u_{Ri}\gamma_M u_{Ri} + \mathrm{i}\bar d_{Ri}\gamma_M d_{Ri}\right) \tag{IN-18}$$

Symbols in (IN-18).

- S_{I13}: The action of the portal I13 (the x_0-chiral vector current of dirac16complex times the baryon-number current). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, ${g^M}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- c_{A}: The coefficient of the x_0-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278 \times 10^{-9}$ GeV^{-2} (dark-matter relic). Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^μ: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = {\mathbf{e}_a}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^μ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- P_a, P_b: The projectors on the two x_0-chiralities, $P_a = (1 + \gamma^{x_0})/2$ and $P_b = (1 - \gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- J^B_M: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: GeV^7. Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- Q_i, $\bar Q_i$: The quark doublet of the generation i (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- u_R, u_{Ri}, $\bar u_{Ri}$: The up-type quark singlet: u_R the multiplet, u_{Ri} that of the generation i (u_R, c_R, t_R), $\bar u_{Ri}$ its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- d_R, d_{Ri}, $\bar d_{Ri}$: The down-type quark singlet: d_R the multiplet, d_{Ri} that of the generation i (d_R, s_R, b_R), $\bar d_{Ri}$ its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.

Origin (INVENTED; version 3). *Reason.* An s-wave annihilation of the dark matter into quarks, which gives the relic abundance within partial-wave unitarity (FE-126 to FE-128). The x_0-chiral current $\mathrm{i}\bar\Psi\gamma^M P_b\Psi$ was chosen because it is Hermitian for the seven directions $M \neq x_0$ (for $M = x_0$ the Hermitian conjugate of $\mathrm{i}C\gamma^{x_0}P_b$ is $\mathrm{i}C\gamma^{x_0}P_a$, so the sum leaves that direction out; corrections, item 24), is a four-dimensional vector, is diagonal in the halves on the zero modes and vanishes on the fold bands (to 7×10^{-14}), so it cannot produce dark radiation; the baryon-number current was chosen because the $B - L$ current would thermalise the right-handed neutrinos and add $\Delta N_{\mathrm{eff}} = 0.144$ (FE-131; corrections, item 23). *Consistency with QCD and the electroweak model.* The baryon current is a colour singlet, flavour universal and invariant under $SU(2)_L \times U(1)_Y$, and it is conserved in perturbation theory (STANDARD; its electroweak anomaly acts only through non-perturbative processes); the algebra of the bilinear with the author's matrices is exact (`um2_coupled.py`, check `C8_I13_algebra`). *Strength.* $G_{\mathrm{A}} = c_{\mathrm{A}}H/(2f^4 v_{\mathrm{t}}\Lambda_\star^6) = 1.278 \times 10^{-9}$ GeV^{-2}, DERIVED from $\Omega_{\mathrm{DM}}h^2_{100} = 0.120$; its physics accuracy is about 1 % (section 10).

6.19 Removed and rejected operators

Removed in version 2 (corrections, item 21): the chiral tip Z portal I9 of version 1,

$$S_{\mathrm{I9}} = \int_{y=-L} \mathrm{d}^7 X_{\mathrm{tip}} \sqrt{|g_{\mathrm{tip}}|}\, \big(c_{\mathrm{Z}} + \tilde{c}_{\mathrm{Z}}\, \varsigma_{\mathrm{t}}\big) \big(\mathrm{i}\bar{\Psi}\gamma^{\mu} P_L \Psi\big)\, \frac{\mathrm{i} v_{\Sigma}^2}{2}\, \mathrm{tr}\big(\Sigma^{\dagger} D_{\mu}\Sigma\, \tau^3\big), \qquad c_{\mathrm{Z}} = \tilde{c}_{\mathrm{Z}} = 0$$

Symbols in this equation.

- S_{I9}: The action of the chiral tip Z portal I9 of version 1, which coupled the four-dimensional left-handed current of dirac16complex to the Z part of the current of the tip sigma field; removed from UM-2 in version 2 (its coefficients are set to zero). Unit: dimensionless (an action, $\hbar = 1$). Status: INVENTED (removed).
- y: The hidden coordinate of the y chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of x_0 is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- X_{tip}: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\mathrm{tip}}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- g_{tip}: The metric induced on the seven-dimensional tip $y = -L$ (the metric g_{MN} restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\mathrm{tip}}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- c_{Z}, $\tilde{c}_{\mathrm{Z}}$: The coefficients of the removed portal I9: c_{Z} of its part that is even under the fold reflection, $\tilde{c}_{\mathrm{Z}}$ of its oriented (odd) part; UM-2 sets both to zero. Unit: GeV^{-6}. Value: 0. Status: ASSUMED.
- ς_{t}: The sign ($+1$ or -1) that distinguishes the two sides of a tip (the tips at $z \to 0^+$ and $z \to \pi^-$ of neighbouring periods), used only in the oriented part of the removed portal I9. Unit: dimensionless. Status: INVENTED (removed).
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^{\dagger} C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- γ^a, γ^M, γ^{μ}: The Dirac matrices: γ^a with a frame index are the author's real 16×16 matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate (γ^{x_0}, γ^{x_4}, … the matrices along x_0, x_4, …), $\gamma_M = g_{MN}\gamma^N$; γ^{μ} the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- P_L, P_R: The projectors on the four-dimensional chiralities, $P_L = (1+\Gamma_4)/2$ (left-handed) and $P_R = (1-\Gamma_4)/2$ (right-handed). Unit: dimensionless. Status: STANDARD.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- v_{Σ}: The scale of the tip sigma field (its kinetic term is $-\frac{v_{\Sigma}^2}{4}\mathrm{tr}((D_{\mu}\Sigma)^{\dagger} D^{\mu}\Sigma)$); on the static block it fixes the Robin parameter β_{W}. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- Σ: The tip sigma field: a unitary 2×2 matrix field on the tip ($\Sigma^{\dagger}\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal W^+, W^- and Z, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_{\mu}\Sigma$ with the W and B fields. Unit: 1/length. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.

With three neutrinos it has no role left: its even part is bounded by the invisible width of the Z, and its oriented part would thermalise the fold band of Ψ_A (item 21); setting both coefficients to zero is a modelling choice

(ASSUMED; item 22). Considered and rejected: a dipole portal $\bar{\Psi}\mathcal{S}^{MN}\Psi B_{MN}$ (it would thermalise the sterile fold bands through the photon: too much dark radiation) and any operator coupling the neutral Ψ to a single W^+ or W^- (none is gauge invariant).

6.20 Parameter choices (ASSUMED)

$HL = 3$ and $m_\Psi = H$ (from the record); the tip mixing $\mathcal{K} = 1$ of one generation, i.e. unitary tip matrices of three generations with $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\rm CKM}$; the doublet localisation c_L of each sector fixed by its heaviest member ($c_R = -c_L$ for the tau and for the top quark); the lightest neutrino mass $m_1 = 1$ meV and the normal ordering; the Kohn-Sham test value of the self-coupling, $\lambda_{\rm test} = 7 \times 10^{-5} H^{-6}$ (mean fields below 5 % of m_Ψ in the densest test state; it is only the value with which the Kohn-Sham machinery is tested); the coordinate volume $v_{\rm t}$ of the extra times (it enters only $V_\perp$ and the conversion of $c_{\rm V}$, $c_{\rm A}$ and λ into $G_{\rm V}$, $G_{\rm A}$ and G_λ). Version 3: the I11 strength $G_{\rm V} = 4.38 \times 10^{-17}$ GeV^{-2} and the highest temperature $T_{\rm max} = 10$ TeV of the hot Universe; the I12 strength $G_{\rm I12} = \pi/(2M_{\rm DM}^3)$ with universal coefficients (chosen to respect the $^3P_0$ unitarity bound up to $E_{\rm cm} = 2.4 M_{\rm DM}$); the bound $G_\lambda = \lambda H \tilde{I}_\lambda/(2f^4 v_{\rm t}) \leq 2.3 \times 10^{-15}$ GeV^{-2} on the self-interaction channel (from $\Delta N_{\rm eff} \leq 0.001$; item 24). The I13 strength is not a choice: it is DERIVED from the relic density.

7 Standard physics and measured values (STANDARD)

Every measured value enters UM-2 in one of three roles, never as a derived result: as the target of a fit (the $W^\pm$ and Z masses with $\rho_{\rm EW,0}$, the twelve fermion masses, the two neutrino splittings with the chosen m_1, and $\Omega_{\rm DM}h^2_{100}$ for the freeze-out requirement), as a direct input (the angles and phases of the PMNS and CKM matrices, $G_{\rm F}$, the $\overline{\rm MS}$ weak angle in the Z widths, α_3, $\alpha_{\rm em}$, the CMB temperature, the Planck 2018 parameters of the cosmology, the effective couplings and kernel factors of the decoupling equations, the matter densities and the supernova parameters), or as a comparison (the Z widths, the number of light neutrinos, $N_{\rm eff}$, the BAO, the explosion statistics, the supernova dark-energy fits, the LZ limit).

input	value used	source	status
$W^\pm$ mass	$m_{\rm W} = 80.3625 \pm 0.0077$ GeV	PDG 2026, OUR EVALUATION (pdgLive S043M)	verified
Z mass	$m_{\rm Z} = 91.1879 \pm 0.0020$ GeV	PDG 2026, OUR AVERAGE (pdgLive S044M)	verified
weak angle ($\overline{\rm MS}$, at $m_{\rm Z}$)	$\sin^2\theta_{\overline{\rm MS}} = 0.23122 \pm 0.00004$	PDG electroweak review	from memory
electroweak parameter	$\rho_{\rm EW,0} - 1 = 0.00038 \pm 0.00020$	PDG electroweak review (global fit)	from memory
charged-lepton masses	electron $m_{\rm e} = 0.51099895000$ MeV, muon 105.6583755 MeV, tau 1776.93 ± 0.09 MeV	PDG 2025 summary tables	verified
light-quark masses ($\overline{\rm MS}$, 2 GeV)	$m_{\rm u} = 2.16 \pm 0.07$ MeV, $m_{\rm d} = 4.70 \pm 0.07$ MeV, $m_{\rm s} = 93.5 \pm 0.8$ MeV	PDG 2025 summary tables	verified
heavy-quark masses	$m_{\rm c}(m_{\rm c}) = 1.2730 \pm 0.0046$ GeV, $m_{\rm b}(m_{\rm b}) = 4.183 \pm 0.007$ GeV, $m_{\rm t} = 172.56 \pm 0.31$ GeV (direct)	PDG 2025 summary tables	verified
neutrino oscillation parameters (normal ordering)	$\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta_{\rm CP} = 212°$, $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.513 \times 10^{-3}$ eV^2; the inverted ordering is disfavoured by $\Delta\chi^2_{\rm fit} = 6.1$; largest possible $J_{\rm CP}$: 0.0333 ± 0.0007	NuFIT 6.0, arXiv:2410.05380 (JHEP 12 (2024) 216), Table 1, IC24 with SK atmospheric data	verified
lightest neutrino mass	$m_1 = 1.000$ meV	chosen	ASSUMED

input	value used	source	status
sum of the neutrino masses	$\sum_i m_i < 0.064$ eV (95 %, ΛCDM)	DESI DR2 Results II, arXiv:2503.14738	verified
CKM parameters	$s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta_{\mathrm{CP}} = 1.147$ rad; $J_{\mathrm{CP}} = 3.12 \times 10^{-5}$	PDG 2024 review "CKM quark-mixing matrix", global fit	verified
strong coupling	$\alpha_3(m_{\mathrm{Z}}) = 0.1180$	PDG world average	from memory
Fermi constant, electroweak scale	$G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2}; 246.22 GeV	PDG	from memory
fine-structure constant	$1/\alpha_{\mathrm{em},0} = 137.035999$	CODATA	from memory
number of light neutrinos	$N_\nu = 2.9963 \pm 0.0074$	PDG 2025 (Janot and Jadach, Phys. Lett. B803 (2020) 135319)	verified
(superseded, version 1)	$N_\nu = 2.9840 \pm 0.0082$	LEP/SLD, Phys. Rept. 427 (2006) 257	verified
Z width and branching fractions	2495.5 ± 2.3 MeV; invisible 20.000 %, hadrons 69.911 %, one charged-lepton pair 3.3658 %	PDG 2025 Z listing	verified
nucleon masses	$m_{\mathrm{p}} = 938.272$ MeV, $m_{\mathrm{n}} = 939.565$ MeV	PDG	from memory
$\hbar c$, Boltzmann and Avogadro constants	$1.973269804 \times 10^{-16}$ GeV m, $8.617333262 \times 10^{-5}$ eV/K, $6.02214076 \times 10^{23}$	CODATA, SI	from memory
CMB temperature today	$T_\gamma = 2.7255$ K	Fixsen 2009	from memory
N_{eff} in the Standard Model	3.0432 ± 0.0002 (three flavours, oscillations, NLO)	Cielo, Escudero, Mangano, Pisanti, arXiv:2306.05460	verified
N_{eff} by the method of temperatures	3.000, 3.011, 3.010, 3.039, 3.049, 3.048 (Table 1); couplings $\tilde{g}_L^{\nu_e e} = 0.727$, $\tilde{g}_R^{\nu_e e} = 0.233$, $\tilde{g}_L^{\nu_\mu e} = -0.273$, $\tilde{g}_R^{\nu_\mu e} = 0.233$; $f_{\mathrm{a}} = 0.884$, $f_{\mathrm{s}} = 0.829$	Escudero, arXiv:2001.04466	verified (2026-10-18)
finite-temperature QED pressure	the second- and third-order pressures of the electron-photon plasma (FE-113, FE-114)	Bennett et al., arXiv:1911.04504	from memory; massless limits reproduced to 10^{-16}
N_{eff} measured	2.99 ± 0.17 (Planck 2018 + BAO); 2.86 ± 0.13 (ACT DR6)	Planck 2018 VI, A&A 641 (2020) A6; Calabrese et al., arXiv:2503.14454	verified
Planck 2018 base parameters	$\Omega_{\mathrm{b}}h^2_{100} = 0.02237$, $\Omega_{\mathrm{DM}}h^2_{100} = 0.1200$, $100\,\theta_{\mathrm{MC}} = 1.04092$ (the angular acoustic scale of CosmoMC), optical depth $\tau_{\mathrm{reio}} = 0.0544$, $\ln(10^{10}A_{\mathrm{s}}) = 3.044$, $n_{\mathrm{s}} = 0.9649$	Planck 2018 results VI, Table 2	from memory
BAO	DESI DR2: 13 measurements of $d_{\mathrm{V}}/r_{\mathrm{d}}$, $d_{\mathrm{M}}/r_{\mathrm{d}}$, $d_{\mathrm{H}}/r_{\mathrm{d}}$ (table below)	DESI collaboration, arXiv:2503.14738; the values checked against Table 2 of arXiv:2603.26861	values verified (2026-10-17); correlations from memory
dark-matter density	$\Omega_{\mathrm{DM}}h^2_{100} = 0.120$	Planck 2018	from memory
Planck mass (freeze-out)	$M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV		from memory

input	value used	source	status
reduced Planck mass (radion decay)	$\bar{M}_{\rm Pl} = (8\pi G_{\rm N})^{-1/2} = 2.435 \times 10^{18}$ GeV		from memory
angle of the gauge couplings	$\cos^2\theta_{\rm W} = (m_{\rm W}^2/m_{\rm Z}^2)/\rho_{\rm EW,0} = 0.776368$ (used for $\beta_{\rm Z} = \beta_{\rm W}/\cos^2\theta_{\rm W}$ and in $\rho_{\rm EW}$); the on-shell value $m_{\rm W}^2/m_{\rm Z}^2 = 0.776663$ for comparison	computed from the $W^{\pm}$ and Z masses and $\rho_{\rm EW,0}$ above	computed
solar electron density	$n_{\rm e} = 245$ times the Avogadro constant times exp(−10.54 times the radius in solar radii) per cm^3 outside the core, at most 100 times the Avogadro constant per cm^3 (exponential fit)	after Bahcall	ASSUMED profile
matter of the Earth	densities 2.6 (T2K), 2.84 (NOvA), 2.848 (DUNE), 2.8 g/cm^3 (resonance); electron fraction 0.5	representative constant densities	ASSUMED
supernova envelope and neutrinospheres	density 10^4 times the cube of (10^9 cm over the radius) g/cm^3, electron fraction 0.5; luminosities 6, 6 and 4×10^{52} erg/s and mean energies 11, 14 and 15.5 MeV (ν_e, $\bar{\nu}_e$, the other flavours); $n_{\nu_e} = 10^{32}$ to 10^{33} cm^{-3}	representative values	ASSUMED
supernova explosion statistics with flavour conversion	failed fractions 25.6 % (no conversion), 50.8, 77.4, 93.3, 96.4 % (conversion above the densities 10^9 to 10^{13} g/cm^3); Salpeter-weighted 27.0, 47.6, 61.9, 73.9, 88.3 %	Gogilashvili and Tamborra, arXiv:2605.16504 (the author's file)	verified (read 2026-10-18)
fast flavour instability criterion	instability if and only if the electron-lepton-number distribution has a crossing	Morinaga, Phys. Rev. D 105 (2022) L101301	from memory
direct-detection limit	LZ: 2.2×10^{-48} cm^2 at 40 GeV, scaled linearly with the mass	LZ 2024	from memory; the extrapolation is an ESTIMATE
nucleon gluon content, low-scale strong coupling	$f_{\rm TG} = 0.92$, $\alpha_3 = 0.3$	standard	from memory; ESTIMATE
dark-energy density	$\rho_{\rm de} \approx 2.5 \times 10^{-47}$ GeV^4		from memory
gluon condensate (version 3)	0.012 GeV^4 for α_3/π times the squared field strength of the canonically normalised gluon field	Shifman, Vainshtein, Zakharov	from memory
QCD interaction measure (version 3)	$\Theta_{\rm QCD}/T^4$ peaks at about 4 near 200 MeV (lattice)	lattice QCD (HotQCD and Wuppertal-Budapest)	from memory; the shape used is an ESTIMATE
nucleon sigma terms (version 3)	$\sigma_{\pi N} = 45$ MeV, $\sigma_{\rm s} = 40$ MeV	standard	from memory; ESTIMATE
electromagnetic coupling at $m_{\rm Z}$ (version 3)	$\alpha_{\rm em} = 1/127.95$	PDG	from memory
baryon-to-photon ratio (version 3)	$\eta_{\rm b} = 6.1 \times 10^{-10}$	Planck 2018, BBN	from memory

input	value used	source	status
neutron-proton mass difference (version 3)	$m_{\mathrm{n}} - m_{\mathrm{p}} = 1.293$ MeV	PDG	from memory
hadron resonance gas (version 3)	pion (0.138, 3), kaon (0.4956, 4), eta (0.5479, 1), rho (0.7753, 9), omega (0.7827, 3), K-star (0.8917, 12), nucleon (0.9389, 8), eta-prime (0.9578, 1), phi (1.0195, 3), Lambda (1.1157, 4), Sigma (1.1932, 12), Delta (1.232, 32) (mass in GeV, number of states)	PDG	from memory
QCD crossover (version 3)	$T_{\mathrm{c}} = 155$ MeV, $T_{\mathrm{w}} = 15$ MeV (smooth switch)	lattice QCD	from memory; the switch is ASSUMED
σ_8 (version 3)	0.811	Planck 2018	from memory
tensor-to-scalar ratio (version 3)	$r_{\mathrm{ts}} < 0.036$ (95 %)	BICEP/Keck 2021	from memory
astrophysical bodies (version 3)	$G_{\mathrm{N}} = 6.67430 \times 10^{-11}$ m^{3}kg^{-1}s^{-2}; Sun 1.98892×10^{30} kg, 6.957×10^{8} m; Earth 5.9722×10^{24} kg, 6.371×10^{6} m; white dwarf 0.6 solar masses, 8700 km; neutron star 1.4 solar masses, 12 km, core density 0.3 fm^{-3} with 10 % protons; Earth mean density 5.51 g/cm^3; solar core 150 g/cm^3; white dwarf 10^6 g/cm^3	standard	from memory; representative
supernova dark-energy fits	constant $w_{\mathrm{de}} = -0.764$; $(w_{\mathrm{de},0}, w_{\mathrm{de},a}) = (-0.861, -0.60)$	the author's private file (not in the repository)	AUTHOR

The DESI DR2 BAO measurements used (redshift; the three distance ratios; the correlation coefficient r_{MH} of the measured $d_{\mathrm{M}}/r_{\mathrm{d}}$ and $d_{\mathrm{H}}/r_{\mathrm{d}}$):

tracer	z_{red}	$d_{\mathrm{V}}/r_{\mathrm{d}}$	$d_{\mathrm{M}}/r_{\mathrm{d}}$	$d_{\mathrm{H}}/r_{\mathrm{d}}$	r_{MH}
BGS	0.295	7.942 ± 0.075			
LRG1	0.510		13.588 ± 0.167	21.863 ± 0.425	-0.459
LRG2	0.706		17.351 ± 0.177	19.455 ± 0.330	-0.404
LRG3+ELG1	0.934		21.576 ± 0.152	17.641 ± 0.193	-0.416
ELG2	1.321		27.601 ± 0.318	14.176 ± 0.221	-0.434
QSO	1.484		30.512 ± 0.760	12.817 ± 0.516	-0.500
Lyman-alpha	2.330		38.988 ± 0.531	8.632 ± 0.101	-0.431

The goodness of fit of a model is

$$\chi^2_{\mathrm{BAO}} = \sum_{\mathrm{tracers}} \left(\mathbf{d} - \mathbf{d}^{\mathrm{th}}\right)^T \mathbf{C}^{-1} \left(\mathbf{d} - \mathbf{d}^{\mathrm{th}}\right)$$

Symbols in this equation.

- χ^2_{BAO}: The chi-squared of the model distances against the 13 DESI DR2 BAO measurements (the pairs of the anisotropic tracers with their correlations, and one point for BGS). Unit: dimensionless. Value: 31.49 for UM-2 (32.55 for ΛCDM). Status: DERIVED.

- $\mathbf{d}$, $\mathbf{d}^{\mathrm{th}}$: The measured pair $(d_{\mathrm{M}}/r_{\mathrm{d}}, d_{\mathrm{H}}/r_{\mathrm{d}})$ of an anisotropic BAO tracer, and the pair $\mathbf{d}^{\mathrm{th}}$ computed in the model. Unit: dimensionless. Status: STANDARD.
- ^{T}: The transpose: a row written with ^{T} is a column. Status: STANDARD.
- $\mathbf{C}$: The covariance matrix of the measured pair of one BAO tracer, built from the published errors and the published correlation coefficient of the pair. Unit: dimensionless. Status: STANDARD.

where, for an anisotropic tracer, $\mathbf{d}$ is the measured pair of $d_{\mathrm{M}}/r_{\mathrm{d}}$ and $d_{\mathrm{H}}/r_{\mathrm{d}}$, $\mathbf{d}^{\mathrm{th}}$ the computed pair and $\mathbf{C}$ the 2×2 covariance matrix with the squared published errors on the diagonal and r_{MH} times the product of the errors off the diagonal; for the isotropic tracer BGS, $\mathbf{d}$ is the single ratio $d_{\mathrm{V}}/r_{\mathrm{d}}$ and $\mathbf{C}$ its squared error. The distances of a flat model with the Hubble rate $\mathcal{H}$ as a function of the redshift are

$$d_{\mathrm{M}}(z_{\mathrm{red}}) = \int_0^{z_{\mathrm{red}}} \frac{\mathrm{d}\tilde{z}_{\mathrm{red}}}{\mathcal{H}(\tilde{z}_{\mathrm{red}})}, \qquad d_{\mathrm{H}}(z_{\mathrm{red}}) = \frac{1}{\mathcal{H}(z_{\mathrm{red}})}, \qquad d_{\mathrm{V}} = \left(z_{\mathrm{red}}\, d_{\mathrm{M}}^2\, d_{\mathrm{H}}\right)^{1/3}, \qquad 1 + z_{\mathrm{red}} = \mathrm{e}^{-a_4}$$

Symbols in this equation.

- d_{M}, d_{H}, d_{V}: The distances of the BAO analysis at the redshift z_{red} (flat space, $c = 1$): the comoving transverse distance d_{M}, the Hubble distance $d_{\mathrm{H}} = 1/\mathcal{H}$ at that redshift, and the volume-averaged distance $d_{\mathrm{V}} = (z_{\mathrm{red}} d_{\mathrm{M}}^2 d_{\mathrm{H}})^{1/3}$ (written with a lower-case d because D_M is the covariant derivative). Unit: length (Mpc). Status: STANDARD.
- z_{red}, $\tilde{z}_{\mathrm{red}}$: The cosmological redshift, $1 + z_{\mathrm{red}} = \mathrm{e}^{-a_4}$ with $a_4 = 0$ today (not $z = 6Hx_0$, the angle variable of the hidden direction); $\tilde{z}_{\mathrm{red}}$ is the redshift as the variable of an integral. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.

with the sound horizon r_{d} at the drag epoch from CAMB (`um2_cosmology.py`, check `bao_desi`).

Standard theory used: the Standard-Model gauge theory $SU(3)_c \times SU(2)_L \times U(1)_Y$ with three generations; QCD confinement of the quarks into hadrons (protons and neutrons); the PMNS and CKM matrices in the standard parametrisation (FE-61); the flavour evolution of neutrinos in matter (Wolfenstein, Mikheyev and Smirnov; FE-103 to FE-108), with antineutrinos obtained by complex conjugation of the mixing matrix and a change of sign of the matter potential; the tree-level partial widths of the Z (section 10); the radiative decay $\mu \to e\gamma$ for Dirac neutrinos; the freeze-out of a relic (Kolb and Turner) and its numerical Boltzmann solution with the relativistic thermal average (Gondolo and Gelmini); the trace-anomaly nucleon matrix element; the Einstein-Boltzmann equations of linear cosmological perturbations in the Newtonian gauge (Ma and Bertschinger), solved by CAMB; the linearised equations of a dense neutrino gas (fast flavour conversion); neutrino decoupling by the method of temperatures (Escudero); Higgsless electroweak symmetry breaking by a non-linear sigma field and fermion localisation in an extra dimension (concepts only, no numbers).

8 The field equations and their origin

The 136 field equations of UM-2 are the variations of the action of section 6 (FE-1 to FE-30), their exact reductions on the order-0 metric and the static block (FE-31 to FE-86), and the equations of the four-dimensional effective theory: the metric series and the linear perturbations (FE-87 to FE-99), the neutrinos (FE-100 to FE-120) and the thermal history (FE-121 to FE-136). The table gives, for each, its title and status (as in the inventory of the field equations), how it was obtained and solved, and the checks that verify it. The checks are named by the report and their number in the catalogue of section 9: alg (`um2_algebra.py`), geo (`um2_geometry.py`), bg (`um2_background.py`), fld (`um2_fields.py`), mass (`um2_spectra.py`), flav (`um2_flavour.py`), nu (`um2_neutrino.py`), ksv, kss, ks8, ks592, ksdm, ksp, ksfb (`um2_ks_jobs.py` validate, spectrum, ground8, ground592, darkmatter, portals, feedback), ew (`um2_electroweak.py`), prim (`um2_primordial.py`), cos (`um2_cosmology.py`), ana (`um2_analysis.py`), th

(`um2_thermal.py`), cpl (`um2_coupled.py`), ser (`um2_series.py`) and res (`um2_residuals.py`, version 4: res 1 to 26 are the residuals of the solution in its equations, res 27 to 52 the links in concert). The field equations that this document uses are written out where they are used (FE-1, FE-31, FE-32, FE-56, FE-64 to FE-66 in sections 4 and 5; FE-2, FE-47, FE-55, FE-67, FE-75, FE-76, FE-79, FE-102, FE-115, FE-120 to FE-123 and FE-131 in section 10).

FE	field equation (status)	how it was obtained and solved	verified by
FE-1	The Einstein-Lovelock equations (AUTHOR)	metric variation of IN-2 and of the matter action: the author's notebook equations with a source (section 4); solved at order 0 through FE-38 to FE-40 and at order ϵ through FE-49	geo 4-12, 16-18; res 1
FE-2	The energy-momentum tensor: definition and components (STANDARD)	definition (for spinors the variation with respect to the vielbein)	ks8 2; ks592 2
FE-3	The total source of gravity (DERIVED)	sum of the energy-momentum tensors of the terms of IN-1	cpl 14; res 9, 11
FE-4	The energy-momentum tensor of the stabiliser (DERIVED)	metric variation of IN-4 with $\rho_{\mathrm{stab}} \propto f^{-8}$	bg 10, 13, 15; res 1
FE-5	The energy-momentum tensor of dirac16complex (DERIVED)	symmetrised vielbein variation of IN-5; for Kohn-Sham states it is FE-67	ks8 2, 3; ks592 2-4; ksdm 3; ana 20
FE-6	The energy-momentum tensor of the Standard-Model fermions (DERIVED)	vielbein variation of IN-11; used through ideal and degenerate gases with the x_0 masses	th 1; cpl 14
FE-7	The energy-momentum tensor of the gauge fields (STANDARD)	metric variation of IN-8	bg 19; ks592 4
FE-8	The energy-momentum tensor of the tip (DERIVED)	metric variation of IN-13 at $y = -L$; its effect is part of the rest energies of $W^{\pm}$, Z and the fermions	(no separate check)
FE-9	The energy-momentum tensors of the portals I11 and I13 (DERIVED)	metric variation of IN-16 and IN-18 (the frame currents do not depend on the vielbein)	cpl 13
FE-10	The energy-momentum tensor of the portal I12 (DERIVED)	metric variation of IN-17	cpl 15; res 12
FE-11	Conservation of the source (DERIVED)	diffeomorphism invariance: the Bianchi identities of the Lovelock tensors	geo 6-8; ks8 2; ks592 2
FE-12	The dirac16complex field equation with the canonical spin connection and the portals (DERIVED)	variation of IN-5 and IN-16 to IN-18 with respect to $\bar{\Psi}$; reduced exactly to FE-56	fld 1; cpl 6; ksp 5
FE-13	The canonical spin connection on the order-0 metric (DERIVED)	IN-7 evaluated on FE-31 (sympy, both halves of a period)	geo 22-25
FE-14	The Dirac equation of the quark doublets Q_i (DERIVED)	variation of IN-11, IN-13, IN-16, IN-18 with respect to $\bar{Q}_i$; solved by FE-56 to FE-60	fld 1, 8, 12, 13; mass 11-13; flav 3

FE	field equation (status)	how it was obtained and solved	verified by
FE-15	The Dirac equation of the up-type quark singlets u_{Ri} (right-handed up, charm and top quarks) (DERIVED)	as FE-14, with respect to $\bar{u}_{Ri}$	fld 9, 12, 13; mass 11-13; flav 3
FE-16	The Dirac equation of the down-type quark singlets d_{Ri} (right-handed down, strange and bottom quarks) (DERIVED)	as FE-14, with respect to $\bar{d}_{Ri}$; the tip matrix of the sector is the CKM matrix	fld 9, 12, 13; mass 11-13; flav 3
FE-17	The Dirac equation of the lepton doublets (left-handed neutrinos and charged leptons of three flavours) (DERIVED)	variation of IN-11, IN-13, IN-16 with respect to $\bar{\ell}_i$ (no I13 term)	fld 8, 12, 13; mass 11-13; flav 3, 7
FE-18	The Dirac equation of the right-handed charged leptons e_{Ri} (electron, muon and tau) (DERIVED)	as FE-17, with respect to $\bar{e}_{Ri}$	fld 9, 12, 13; mass 11-13
FE-19	The Dirac equation of the right-handed neutrinos ν_{Ri} (DERIVED)	as FE-17, with respect to $\bar{\nu}_{Ri}$ (gauge singlets)	mass 11, 13; flav 3; cpl 10
FE-20	The fermion currents of the gauge fields (STANDARD)	variation of IN-11 with respect to the gauge fields	ew 1-4
FE-21	The field-dependent gauge couplings produced by I12 (DERIVED)	IN-8 and IN-17 combined	cpl 15, 16; res 12
FE-22	The Yang-Mills equation of the gluons (DERIVED)	variation of IN-8, IN-11, IN-17 with respect to the gluon field	fld 14; mass 1, 3
FE-23	The Yang-Mills equation of the SU(2) field W^I (matrix form) with the tip current (DERIVED)	variation of IN-8, IN-11, IN-13, IN-17 with respect to the SU(2) field, in the unitary gauge	fld 14, 15; mass 6-8
FE-24	The Yang-Mills equation of the hypercharge field B (DERIVED)	as FE-23, with respect to the hypercharge field	fld 14, 15
FE-25	The W^+, W^-, Z and photon field equations to linear order in the fields (I12 neglected) (DERIVED)	FE-23 and FE-24 at linear order with IN-10	ew 2-4; mass 3, 7, 8
FE-26	The tip sigma field equation (the Noether identity of the gauge symmetry) (STANDARD)	variation of IN-13 with respect to Σ (the Noether identity of the gauge symmetry)	fld 16
FE-27	The boundary conditions of gravity and the gauge fields at the fold and the tip (DERIVED)	boundary terms of the variation at the fold and at the tip	fld 15
FE-28	The boundary conditions of dirac16complex (DERIVED)	the record's fold and tip conditions; self-adjoint	fld 5, 7; kss 1

FE	field equation (status)	how it was obtained and solved	verified by
FE-29	The boundary conditions of the Standard-Model fermions (INVENTED)	selection projector chosen so that each doublet has one left-handed and each singlet one right-handed Weyl zero mode	fld 3-5, 8-10; ew 12
FE-30	The tip mixing conditions of one and of three generations (DERIVED)	IN-13 on the zero modes; self-adjoint for every complex matrix (sympy, 42 real variables); the tip matrices ASSUMED	fld 12, 13; flav 3, 4
FE-31	The order-0 metric in the author's coordinates (AUTHOR)	the record's corrected metric (section 5) in the author's coordinates	geo 1-3
FE-32	The hidden coordinate y, the block scale and the rates of the background (DERIVED)	change of chart to y (section 5)	geo 1-3, 22
FE-33	The sectional curvatures of the order-0 metric (DERIVED)	Riemann tensor of FE-31 in both charts (sympy)	geo 1-3
FE-34	The Lovelock densities and tensors of a pair-diagonal curvature (DERIVED)	pairing formula, coefficients $2^p p!$ and $-2^{p-1}p!$ (sympy)	geo 4-12; bg 4
FE-35	The first Lovelock tensor (the Einstein tensor) of the order-0 metric (DERIVED)	FE-34 with FE-33 (sympy, four routes)	geo 4, 6, 10, 13; bg 1, 4
FE-36	The second Lovelock tensor (Gauss-Bonnet) of the order-0 metric (DERIVED)	as FE-35	geo 5, 7, 11, 14; bg 2, 4
FE-37	The third Lovelock tensor of the order-0 metric (DERIVED)	as FE-35	geo 8, 12, 15; bg 3, 4
FE-38	The background equation: the x_4 component (the constraint) (DERIVED)	FE-1 with FE-35 to FE-37, the author's weights and Λ	geo 16, 19; prim 1; res 1, 3
FE-39	The background equation: the 3-space component (DERIVED)	as FE-38; integrated through the primordial era (DOP853)	geo 17, 20; bg 14; prim 1-5; res 1, 3
FE-40	The background equation: the hidden (block) component (DERIVED)	as FE-38; the x_0 and the extra-time components have the same left side, and the order-0 metric (one block scale) obeys their average with $\bar{p}$	geo 18, 21; bg 1-3, 14; res 1, 3
FE-41	The conservation of the order-0 source (DERIVED)	FE-11 on FE-31	bg 15; prim 1
FE-42	The static block (DERIVED)	FE-38 to FE-40 with $\mathcal{H}_f = 0$, $\dot{\mathcal{H}} = 0$	geo 19-21; bg 5-9; res 1
FE-43	The stabilised block (DERIVED)	closed form (sympy)	bg 10-13, 15; res 1, 48
FE-44	The linearised block and the radion (DERIVED)	Jacobian of FE-39 and FE-40 at the static block	bg 16, 17; prim 2; ana 22; res 2, 47
FE-45	The response of the block to matter to first order (DERIVED)	linearisation with block-averaged sources	bg 18; res 2
FE-46	The coefficients of the first-order block response (DERIVED)	exact algebraic numbers (sympy)	bg 18, 19; res 46

FE	field equation (status)	how it was obtained and solved	verified by
FE-47	The adiabatic block response and the four-dimensional Friedmann acceleration equation (DERIVED)	FE-45 with the second time derivative of ΔK neglected	bg 19; ksfb 1; res 2, 9
FE-48	The four-dimensional reduction of the Lovelock action on the static block (DERIVED)	dimensional reduction on the static block (sympy)	bg 20-22; ksfb 2; ana 21; res 4
FE-49	The four-dimensional effective gravity and the four-dimensional couplings (DERIVED)	the reduced action and the zero modes of the gauge fields	bg 21, 22; ana 23; res 4, 37
FE-50	The x_0 eigenvalue equation of the gauge fields and the graviton (DERIVED)	reduction of FE-22 to FE-25 (and of FE-1 for the transverse-traceless graviton) on the static block	fld 14; mass 1-4; ana 1-7; ser 2; res 5
FE-51	The fold and tip conditions of the boson x_0 profiles (DERIVED)	natural boundary conditions of IN-15	fld 15; mass 6, 7; res 5
FE-52	The boson eigenvalue conditions (DERIVED)	exact solution of FE-50: mpmath roots (30 to 60 digits), finite elements as a check	mass 1-5, 9; flav 5; ana 2; res 5
FE-53	The tower of Kaluza-Klein gravitons (the terms $n \geq 1$ of the metric series) (DERIVED)	exact eigenfunctions of FE-50 with $\beta = 0$	mass 1, 2; ser 2; res 15
FE-54	The potential between two static masses with the Kaluza-Klein gravitons (DERIVED)	exchange of the tower between two static sources	ser 2; res 15
FE-55	The $W^{\pm}$ and Z masses, the electroweak parameter and the Kaluza-Klein scale (DERIVED)	fit of β_{W} by bisection in $\ln \beta_{\mathrm{W}}$ (80 steps)	mass 7, 8, 10; res 5, 13, 29, 30
FE-56	The exact reduction of the eight-dimensional Dirac equation to the block Hamiltonian (DERIVED)	the record's reduction, re-derived exactly for all eight blocks with the author's matrices (sympy)	fld 1, 2; alg 19-29
FE-57	The chiral zero modes (massless four-dimensional fermions) (DERIVED)	exact solutions of FE-56 at $k = 0$	fld 6-11; kss 2
FE-58	The tip values of the doublet and singlet solutions at $k = 0$ (DERIVED)	exact solutions of the doublet and singlet systems	mass 11-13
FE-59	The exact mass condition of one generation and its first-order root (DERIVED)	FE-58 in FE-30; mpmath roots, shooting (DOP853) as a check	mass 11-14; ana 8, 9; res 6
FE-60	The mass condition of three generations with the 3×3 tip matrices (DERIVED)	FE-58 in the three-generation FE-30; mpmath at 50 digits	flav 3, 4, 7; res 6

FE	field equation (status)	how it was obtained and solved	verified by
FE-61	The standard parametrisation of the mixing matrices (the tip matrices of UM-2) (STANDARD)	NuFIT 6.0 and PDG values (section 7)	flav 1, 2, 6; res 33
FE-62	The spectrum of dirac16complex on the static block (DERIVED)	exact solution of FE-56 with $M = m_\Psi$	kss 1-3; ew 13; ana 12; res 7
FE-63	The Kohn-Sham equations of dirac16complex coupled to all fields (DERIVED)	the record's block Hamiltonian with the mean fields FE-64; Galerkin in the free basis, Anderson mixing, Richardson in the basis size	ksv 1, 2; ks8 1; ks592 1; cpl 1, 7; res 8
FE-64	The Kohn-Sham mean fields (DERIVED)	the record's Hartree-plus-exchange fields of the contact term and the portal mean fields of IN-16 to IN-18	ksp 1-4; cpl 1, 7, 13; res 8, 10, 11
FE-65	The Kohn-Sham densities, shells, cell volume and occupations (RECORD)	the record's definitions	ksv 1, 2
FE-66	The Kohn-Sham energy (RECORD)	the record's functional	ks8 3; ks592 3; ksdm 1
FE-67	The energy-momentum tensor of a Kohn-Sham state (DERIVED)	FE-5 for Kohn-Sham states	ksdm 2, 3; ks592 4; ana 19; res 9, 28
FE-68	The energy identity and the y-conservation law of a Kohn-Sham state (DERIVED)	from FE-66, FE-67 and FE-11	ks8 2, 3; ks592 2, 3; res 8
FE-69	The boundary conditions of the Kohn-Sham orbitals (DERIVED)	block form of FE-28 ($\gamma^{x_0} = \sigma_3$ in every block)	alg 20; fld 5; ew 13
FE-70	The portals I11 and I13 on the zero modes: the four-dimensional contact interactions (DERIVED)	reduction of IN-16 and IN-18 with the zero-mode profiles	cpl 1, 3, 5-7, 9; ksp 1-3; res 10, 40
FE-71	The portal I12 on the dark matter (DERIVED)	reduction of IN-17 with the dark-matter orbital	ksp 4; ana 17; res 10, 41
FE-72	The shift of a Kohn-Sham level by the mean fields of an environment (DERIVED)	first order in the mean fields (Hellmann-Feynman)	cpl 1, 2, 13; res 11
FE-73	The level shift of the dark matter in matter of protons, neutrons and electrons (DERIVED)	FE-72 for protons, neutrons and electrons	cpl 13; res 11
FE-74	The thermal gluon condensate from the trace anomaly (STANDARD)	trace anomaly (Shifman, Vainshtein, Zakharov)	cpl 13; res 11
FE-75	The QCD interaction measure (ESTIMATE)	lattice-like shape below 1 GeV, leading order above	cpl 13

FE	field equation (status)	how it was obtained and solved	verified by
FE-76	The thermal condensate of a massive vector boson and the order-α_{G} condensate of a gauge plasma (STANDARD)	free part; Feynman-Hellmann at order α_{G}	cpl 13; res 11, 22
FE-77	The running electroweak couplings without a Higgs particle (STANDARD)	one loop from the $\overline{\mathrm{MS}}$ values at m_{Z}	cpl 13
FE-78	The electroweak condensates of a hot plasma (ESTIMATE)	FE-76 with FE-77; thermal suppression below m_{W}	cpl 13
FE-79	dirac16complex acting on the Standard Model: the coupling shifts and the x_0 problem of $W^{\pm}$ and Z (DERIVED)	FE-21 and FE-50 with the dark-matter scalar density; first-order perturbation theory (version 4)	cpl 15; res 12, 45
FE-80	Gravity in the coupled solution: the block response to every species (DERIVED)	FE-47 with all species, iterated with the masses	cpl 14; res 11
FE-81	The hidden pressure of matter and of the hot plasma (DERIVED)	$\bar{p} = \frac{1}{4}\,\mathrm{d}p/\mathrm{d}\ln f$ at fixed temperature	cpl 14; res 11, 39
FE-82	The block dependence of a gauge coupling and the trace of an electroweak plasma (DERIVED)	$g_{\mathrm{G}}^2 \propto f^{-4}$, $m_{\mathrm{KK}} \propto 1/f$, one loop	cpl 14
FE-83	The block dependence of the QCD scale and of the nucleon mass (ESTIMATE)	one loop; nucleon sigma terms	cpl 12; res 52
FE-84	The block dependence of the QCD scale through the quark thresholds (ESTIMATE)	one-loop matching at the quark thresholds	cpl 12; res 52
FE-85	The block dependence of a fermion mass (DERIVED)	FE-59 with localisations proportional to f	ana 11; cpl 12; res 52
FE-86	The densities of the environments of the coupled solution (ASSUMED)	net baryons in weak equilibrium; dark matter in equilibrium below $x = 40$	cpl 13; res 11
FE-87	The metric as a power series of terms of decreasing magnitude (DERIVED)	magnitudes of the computed terms	ser 1-4; res 16
FE-88	The order-1, $n = 0$ term of the metric in the Newtonian gauge (eight dimensions) (DERIVED)	Newtonian gauge in eight dimensions	ser 1, 3; res 16
FE-89	The four-dimensional perturbed metric in conformal time (STANDARD)	Newtonian gauge (Ma and Bertschinger)	cos 3

FE	field equation (status)	how it was obtained and solved	verified by
FE-90	The linearised Einstein equations for a Fourier mode (STANDARD)	CAMB 2.0.5 (default accuracy, lensing)	cos 1, 3, 4, 8; res 14
FE-91	The perturbations of the dark matter (cold) (STANDARD)	CAMB; cold (a Kohn-Sham quantum at rest has zero pressure)	cos 3; ksdm 2
FE-92	The perturbations of the baryons (STANDARD)	CAMB	cos 3, 4
FE-93	The perturbations of the photons: density and velocity (STANDARD)	CAMB	cos 3
FE-94	The photon temperature hierarchy (STANDARD)	CAMB	cos 3
FE-95	The photon polarisation hierarchy (STANDARD)	CAMB	cos 3
FE-96	The hierarchy of each neutrino mass state (the cosmic neutrino background) (STANDARD)	CAMB, the three mass states separately (masses iterated)	cos 2, 6, 7; res 32, 34
FE-97	The background expansion, the neutrino temperature and the primordial spectrum (STANDARD)	Planck 2018 parameters; $N_{\mathrm{eff}} = 3.0432 + 0.0010$	cos 1, 4, 6; res 35, 38
FE-98	The size of the first-order potentials: rms value, Poisson equation and large scales (STANDARD)	from the CAMB solution	cos 8; ser 1; res 14
FE-99	The amplitude of the matter fluctuations (STANDARD)	from the CAMB matter power spectrum	cos 8; res 50
FE-100	The three neutrino masses (ASSUMED)	m_1 and the ordering chosen, the splittings of NuFIT 6.0; x_0 eigenvalues of FE-59	nu 1, 16; mass 11; res 32
FE-101	The four-fermion interactions at low energy (STANDARD)	exchange of $W^{\pm}$ and Z far below their masses	nu 2-5; ew 6-11
FE-102	The Fermi constant and the neutral-current couplings (STANDARD)	tree level; the measured G_{F}	ew 4, 16, 18, 19; res 13, 30
FE-103	The matter potentials of forward scattering (STANDARD)	forward scattering in an unpolarised medium	nu 3-5, 11
FE-104	The flavour evolution in vacuum and in matter (STANDARD)	exact matrix exponentials; ordered products of layers	nu 2-5; res 17
FE-105	The vacuum oscillation probability, CPT and the T-odd difference (STANDARD)	analytic	nu 2; res 17
FE-106	The MSW resonances of the 1-3 and 1-2 sectors (STANDARD)	two-flavour conditions; the exact three-flavour root as a check	nu 3, 5; res 17

FE	field equation (status)	how it was obtained and solved	verified by
FE-107	The adiabatic survival probability of solar electron neutrinos (STANDARD)	adiabatic formula; numerical evolution as a check	nu 5; res 17
FE-108	The resonance density, the adiabaticity and the Landau-Zener jump probability (STANDARD)	two-flavour conditions	nu 11; res 17
FE-109	The thermal potential of the early Universe (STANDARD)	leading order in T/m_{W}; Brent roots	nu 6; res 18
FE-110	The quantum kinetic equations of a neutrino gas (STANDARD)	linearised (FE-117) and momentum averaged (FE-111)	nu 8, 9, 12-14
FE-111	Neutrino decoupling: the temperature equations (STANDARD)	Escudero's method; LSODA from 20 MeV to 5 keV	nu 8, 9; res 19
FE-112	The energy transfer by collisions (STANDARD)	Escudero's kernels with the radiatively corrected couplings	nu 8, 9
FE-113	The finite-temperature QED pressure of the electron-photon plasma (STANDARD)	Bennett et al.	nu 7
FE-114	The integrals of the QED pressure (STANDARD)	Gauss-Legendre panels	nu 7
FE-115	The effective number of neutrino species and the common temperature (STANDARD)	from FE-111; the precision value 3.0432 is adopted	nu 8, 9; res 19, 35, 49
FE-116	The fast flavour instability: the electron-lepton-number distribution (STANDARD)	definitions; axially symmetric gas	nu 12, 13; res 20
FE-117	The fast flavour instability: the dispersion relation (STANDARD)	vectorised Newton iterations	nu 12, 13; res 20
FE-118	The axisymmetric and azimuthal branches of the dispersion relation (STANDARD)	analytic	nu 13
FE-119	The homogeneous mode and the mode $k_{\mathrm{f}} = \mathcal{I}_1$ (STANDARD)	300-node discretisation; analytic growth	nu 12, 14, 15; res 20
FE-120	The cosmic neutrino background (STANDARD)	adaptive quadrature (to 3×10^{-16} since version 4)	nu 10; res 21, 34
FE-121	The ideal-gas densities of a species of the plasma (STANDARD)	Gauss-Legendre quadrature with the x_0 masses	th 1; res 22, 31, 39
FE-122	The effective degrees of freedom and the Hubble rate of the hot Universe (STANDARD)	sums of FE-121	th 1; res 22, 37

FE	field equation (status)	how it was obtained and solved	verified by
FE-123	The QCD crossover, the neutrino temperature after decoupling and the cooling of the plasma (ASSUMED)	smooth switch; instantaneous neutrino decoupling at 2 MeV	th 1; res 22, 38
FE-124	The Boltzmann equation of one dark-matter species (STANDARD)	implicit Radau on $\ln Y_{\mathrm{DM}}$	th 3, 4; cpl 8; res 23
FE-125	The relativistic thermal average of the annihilation cross-section (STANDARD)	Gondolo and Gelmini; adaptive quadrature	th 2; res 23
FE-126	The relic density (STANDARD)	from FE-124	th 3, 4; cpl 8; res 23, 36
FE-127	The annihilation cross-sections of the dark matter through I12 and I13 (DERIVED)	spin traces and polarisation sums of FE-70 and FE-71	ana 13-16; cpl 8; res 23
FE-128	The partial-wave unitarity bounds (DERIVED)	partial waves ${}^3P_0$ and ${}^3S_1$	th 5-7; ana 16; cpl 8
FE-129	The freeze-in of the fold bands through I11 (DERIVED)	FE-70 with Maxwell-Boltzmann statistics; trapezoid rule	cpl 4; res 24
FE-130	The dark radiation of the fold bands and their departure from equilibrium (DERIVED)	from FE-129	cpl 4; cos 6, 7; res 24, 35
FE-131	The right-handed neutrinos if I13 coupled to $B - L$ (DERIVED)	Radau in the shifted variable $x - 2$ (version 4)	cpl 10; res 25
FE-132	The elastic scattering of right-handed neutrinos on the dark matter (ESTIMATE)	a Weyl ν_R on a heavy Dirac fermion	cpl 10; res 25
FE-133	The extra radiation of species that decouple early (DERIVED)	entropy conservation with the heating factor	cpl 10, 11; cos 5; res 49
FE-134	The production of right-handed neutrinos through I11 (with the baryon current in I13) (DERIVED)	ratio of the I11 and I13 rates	cpl 10; res 25
FE-135	The self-interaction channel of the dark matter into the fold bands (ESTIMATE)	IN-5 on the dark matter and the fold bands; Fierz coefficients not computed	cpl 11; res 10, 26
FE-136	The energy put into the fold bands by the self-interaction channel (DERIVED)	trapezoid rule from $T_{\max}$ to the freeze-out	cpl 11; res 26

9 The computations and their verification (DERIVED)

All computations are Python scripts in the folder `universe_model_2/code`; each writes its numbers to `universe_model_2/outputs` and a report of named checks (name, verdict, detail) to `universe_model_2/reports`, and exits with code 0 only if every check passes. Two modules are shared: `um2_common.py` (the output folders, the report class, reading and writing the JSON files) and `um2_constants.py` (the measured inputs of section 7). In total 287 checks in version 4: the 235 of version 3 (233 before the fixes of its independent review, item 24; 205 in version 2; 179 to 182 in version 1) and the 52 of the new residual analysis; none failed. In the table the prefix `um2_` of every script name is omitted.

script	what it computes	method	output	checks
algebra.py	the author's gamma matrices rebuilt from the notebook formulas; C, Γ_8, $\mathcal{B}$; the irreducibility of the representation of Pin(4,4); the two inequivalent halves of Spin(4,4); the 2×2 block basis and the block forms of every matrix the reductions use	sympy, exact	algebra.json	31
geometry.py	the Riemann tensor of the order-0 metric in two charts; the Lovelock tensors by four routes (pairing formula, Ricci tensor, explicit Gauss-Bonnet tensor, minisuperspace variation with a lapse); the Bianchi identities; the record's equations; the static block; the canonical spin connection	sympy, exact	geometry.json	25
background.py	the background equations in class form; the static block; the stabiliser and its family; the radion; the linear response of the block to matter sources; the dimensional reduction	sympy, exact; eigenvalues numerically	background.json	22
fields.py	the exact x_0 reduction of the eight-dimensional Dirac equation (all blocks, general a_4, b_4, M, v); boundary conditions; zero modes; tip mixing for one and three generations (self-adjoint with an arbitrary complex 3×3 matrix); the gauge x_0 operator; gauge invariance of the tip terms of three generations	sympy and numpy	fields.json	16
spectra.py	the x_0 spectra of the gauge bosons and the graviton; the $W^{\pm}$, Z fit to $\rho_{\mathrm{EW},0}$ with the angle of the couplings fixed by the measured masses (both masses reproduced); the twelve fermion masses as exact x_0 eigenvalues	mpmath root finding; quadratic finite elements with Richardson; shooting	masses.json	14
flavour.py	the PMNS and CKM matrices; the exact tip determinant of three generations and its null vectors; a general complex tip matrix; the flavour-changing couplings of the Kaluza-Klein gauge modes; $\mu \to e\gamma$; lepton universality	mpmath (50 digits), numpy	flavour.json	7

script	what it computes	method	output	checks
neutrino.py	vacuum and matter oscillations, the MSW resonances, long-baseline appearance, the Sun, the onset of oscillations in the early Universe, neutrino decoupling and N_{eff}, the CNB, supernova resonances, the fast flavour instability, the heating behind the shock; (version 3) the Majorana alternative	exact matrix exponentials and layer products; LSODA; Gauss-Legendre panels; Newton iterations; adaptive quadrature	neutrino.json	16
kohnsham.py, ks_jobs.py	Kohn-Sham (the solver library and its seven jobs): validation against the record (2 checks), free spectrum (3), interacting ground states of 8 (3) and of 592 particles (4), dark-matter quanta (3), portal responses with the three-generation sources (5), block feedback with the full adiabatic response (2)	Galerkin in the free basis, safeguarded Anderson self-consistency, Richardson in the basis size; single-threaded linear algebra (version 4)	ks/*.json (seven files)	22
electroweak.py	charges (W^{+}: $+e$, W^{-}: $-e$), anomalies of three generations, couplings, the chirality projector, $\Gamma_4 = -\mathcal{B}\gamma^{x_0}$, the chiral tip charges of Ψ_A and Ψ_B, the number of light neutrinos, the removal of the tip Z portal, the Z widths of three generations and their scheme dependence	sympy and numerics	electroweak.json	19
primordial.py	the nonlinear primordial evolution, the basin, the separatrix, the radion oscillation	DOP853, bisection	primordial.json	5
cosmology.py	TT and EE spectra with three massive neutrinos, the BAO χ^2_{BAO}, N_{eff}, helium, the variants with a thermalised fold band; (version 3) the model with the freeze-in fold bands, σ_8, the growth rate, the potential spectrum	CAMB 2.0.5	cosmology.json	8
analysis.py	the x_0 decompositions and the uncertainty relation on the cell; the gap theorem; sum rules; the twelve fermions against the uncertainty estimate; the conservation of the halves; the relic of two species with I12 alone; direct detection; the condensate; the vacuum energy; the sensitivities to the block size; the block rate today; the radion decay	mpmath, numpy, sympy	analysis.json	23

script	what it computes	method	output	checks
thermal.py	(version 3) the degrees of freedom with the x_0 masses; the relativistic thermal average; the Boltzmann solver and its validation; the relic of I12 alone; partial-wave unitarity; the unitarity-safe I12 benchmark	Gauss-Legendre ideal-gas integrals; adaptive quadrature with scaled Bessel functions; implicit Radau; Brent root finding	thermal.json	7
coupled.py	(version 3) the portal overlaps with every zero mode and their Kohn-Sham checks; the screened response; the fold band under the portals; the freeze-in of the fold bands; direct detection; I13 (the algebra of its bilinear, fold-band blindness, relic with the unitarity cap, detection, annihilation today); the $B - L$ alternative; the self-interaction channel; the mean fields in ten environments; the full adiabatic block response to every species; the proton sensitivity; the W eigenvalue with the kinetic function; the domain of validity	exact sympy matrices; self-consistent Kohn-Sham solutions with Richardson in the amplitude; Boltzmann equations; first-order perturbation theory (version 4)	coupled.json	16
series.py	(version 3) the magnitudes of the terms of the metric series: potentials, Kaluza-Klein gravitons (levels, norms, Yukawa strengths, orthogonality), the radion, order 2, the tensor bound	analytic eigenfunctions, finite elements, quadrature	series.json	4
residuals.py	(version 4) the residuals of the stored solution in 26 field equations (afresh by substitution where cheap), the sizes of the neglected terms, and 26 links in concert (section 10)	substitution (sympy, mpmath at 40 digits), independent quadratures and re-integrations	residuals.json	52

9.1 The catalogue of the 287 checks

The checks are listed by report, numbered in the order of the report; the names are those of the reports, spelled as in the code. The field-equation table of section 8 refers to them by the abbreviation of the report and the number.

alg (`um2_algebra.py`, 31 checks, all passed): 1 `tau_relations_of_the_notebook`; 2 `notebook_identities_sigma`; 3 `clifford_relations`; 4 `gammas_real`; 5 `identical_to_record_gammas`; 6 `C_is_sigma16`; 7 `C_real_symmetric_involution`; 8 `C_gamma_antisymmetric`; 9 `chirality_diag`; 10 `B_hermitian_involution`; 11 `B_signature_8_8`; 12 `B_gamma4_gamma_a_hermitian`; 13 `pin44_irreducible`; 14 `spin44_commutant_dim_2`; 15 `spin44_preserves_chiral_halves`; 16 `chiral_halves_irreducible`; 17 `chiral_halves_inequivalent`; 18 `commuting_set`; 19 `block_basis_unitary`; 20 `block_form gammaˆ(x0) = sigma3`; 21 `block_form gammaˆ(x0)gammaˆ(x1) = -i sigma2`; 22 `block_form gammaˆ(x0)gammaˆ(x4) = j sigma1`; 23 `block_form B = j s2 (number)`; 24 `block_form C = s2 sigma2`; 25 `block_form J = j`; 26 `block_form K1 = i s2`; 27 `block_form K2 = i s3`; 28 `block_form gammaˆ(x4)gammaˆ(x1) = -j sigma3`; 29 `block_form B gammaˆ(x0) = j s2 sigma3`; 30 `product_of_4D_and_internal_chirality`; 31 `chirality_squares`.

geo (`um2_geometry.py`, 25 checks, all passed): 1 `y_chart_pair_diagonal`; 2 `author_chart_pair_diagonal`; 3 `charts_agree`; 4 `einstein_from_ricci_equals_pairing_k1`; 5 `gauss_bonnet_tensor_equals_pairing_k2`; 6 `bianchi_identity_k1`; 7 `bianchi_identity_k2`; 8 `bianchi_identity_k3`; 9 `lapse_metric_pair_diagonal`; 10 `minisuperspace_variation_k1`; 11 `minisuperspace_variation_k2`; 12 `minisuperspace_variation_k3`; 13

hidden_equals_extra_times_k1; 14 hidden_equals_extra_times_k2; 15 hidden_equals_extra_times_k3; 16 record_equation constraint_x4; 17 record_equation space_x1_eq_x2_eq_x3; 18 record_equation hidden_and_extraTimes_x0_eq_x5_eq_x6_eq_x7; 19 static_block_constraint; 20 static_block_space; 21 static_block_hidden; 22 vielbein; 23 gamma_covariantly_constant; 24 gamma_mu_Omega_mu; 25 nontrivial_unless_flat.

bg (um2_background.py, 22 checks, all passed): 1 hidden_equals_extra_times_k1; 2 hidden_equals_extra_times_k2; 3 hidden_equals_extra_times_k3; 4 class_form_equals_full_tensor; 5 static_block_x4; 6 static_block_x0; 7 static_vacuum_general; 8 newton_factor_at_vacuum; 9 newton_factor_author_roots; 10 stabiliser_requirement; 11 casimir_type_static_blocks; 12 casimir_type_attractive_gravity; 13 casimir_type_positive_energy; 14 second_order_system_solvable; 15 static_point_is_a_solution; 16 block_stable_nu1; 17 block_stable_family; 18 sourced_radion_equation; 19 static_block_response; 20 internal_lovelock_densities; 21 reduction_newton_factor; 22 reduction_cosmological_term.

fld (um2_fields.py, 16 checks, all passed): 1 dirac_reduction_exact; 2 gamma4_block_form; 3 selection_operator_lorentz_invariant; 4 selected_blocks; 5 bc_lagrangian; 6 zero_mode_general_solution; 7 zero_modes_dirac16complex; 8 zero_modes_doublet; 9 zero_modes_singlet; 10 weyl_helicity_definite; 11 zero_mode_dispersion; 12 tip_mixing_self_adjoint; 13 tip_mixing_self_adjoint_three_generations; 14 gauge_x0_operator; 15 gauge_boundary_conditions; 16 tip_terms_gauge_invariant.

mass (um2_spectra.py, 14 checks, all passed): 1 unbroken_spectrum_exact; 2 fem_crosscheck_unbroken; 3 massless_zero_mode; 4 gap; 5 dirichlet_tip_bound; 6 custodial_leading_order; 7 rho_fit; 8 mZ_reproduced; 9 fem_crosscheck_W; 10 scales; 11 fermion_fit; 12 first_order_formula; 13 shooting_crosscheck; 14 localisation_lowers_mass.

flav (um2_flavour.py, 7 checks, all passed): 1 PMNS_unitary_and_Jarlskog; 2 CKM_unitary_and_Jarlskog; 3 three_generation_tip_condition_exact; 4 general_tip_matrix; 5 no_tree_level_FCNC_from_KK; 6 mu_to_e_gamma; 7 lepton_universality.

nu (um2_neutrino.py, 16 checks, all passed): 1 masses_and_cosmological_bound; 2 vacuum_oscillations; 3 MSW_resonance_13; 4 long_baseline_matter_and_CP; 5 solar_MSW; 6 early_universe_oscillation_onset; 7 plasma_integrals; 8 N_eff_method_validation; 9 N_eff_three_flavours; 10 CNB_three_mass_states; 11 supernova_MSW; 12 fast_flavour_solver_consistency; 13 fast_flavour_crossing_criterion; 14 fast_flavour_time_scale; 15 equipartition_reduces_heating; 16 N7_dirac_or_majorana.

ksv (um2_ks_jobs.py validate, 2 checks, all passed): 1 validate_N8_lamp1_a00; 2 validate_N592_lamp1_a00.

kss (um2_ks_jobs.py spectrum, 3 checks, all passed): 1 spectrum_exact_k0; 2 fold_band_massless; 3 j_mirror_k0.

ks8 (um2_ks_jobs.py ground8, 3 checks, all passed): 1 ground_basis_convergence; 2 ground_y_conservation; 3 ground_energy_equals_integrated_rho.

ks592 (um2_ks_jobs.py ground592, 4 checks, all passed): 1 ground_basis_convergence; 2 ground_y_conservation; 3 ground_energy_equals_integrated_rho; 4 fold_band_gas_radiation.

ksdm (um2_ks_jobs.py darkmatter, 3 checks, all passed): 1 dm_energy_bookkeeping; 2 dm_cold; 3 dm_hidden_pressure_identity.

ksp (um2_ks_jobs.py portals, 5 checks, all passed): 1 portal_HF I11 vector contact, electron density (v); 2 portal_HF I11 vector contact, up-quark density (v); 3 portal_HF I11 vector contact, neutrino density (v, left-chiral); 4 portal_HF I12 gluon condensate <G G> (M, uniform); 5 scalar_density_vanishes_at_tip.

ksfb (um2_ks_jobs.py feedback, 2 checks, all passed): 1 block_response; 2 friedmann_decoupled_from_block_shift.

ew (um2_electroweak.py, 19 checks, all passed): 1 electric_charges; 2 W_charges; 3 photon_coupling; 4 Z_coupling; 5 anomaly_cancellation; 6 P_plus_projector_hermitian; 7 P_plus_lorentz_invariant; 8

chiral_current_is_vector; 9 B_lorentz_invariant; 10 Gamma4_equals_minus_B_gamma0; 11 P_plus_block_form; 12 sm_zero_modes_in_psiA_blocks; 13 tip_condition_b_zero; 14 Z_charge_fold_band; 15 Z_charge_dark_matter; 16 Z_invisible_width; 17 tip_Z_portal_dropped; 18 Z_width_scheme; 19 Z_widths_three_generations.

prim (um2_primordial.py, 5 checks, all passed): 1 constraint_preserved; 2 radion_oscillation; 3 radion_dust_expansion; 4 basin_separatrix; 5 deflation_factor.

cos (um2_cosmology.py, 8 checks, all passed): 1 theta_star_matched; 2 three_massive_neutrinos; 3 cmb_spectra_close_to_lcdm; 4 bao_desi; 5 dark_radiation_variants; 6 neff_and_helium; 7 freeze_in_radiation; 8 sigma8_and_first_order_potential.

ana (um2_analysis.py, 23 checks, all passed): 1 A1_flat_form_identity; 2 A2_gap_theorem; 3 A3_sum_rules; 4 A3_uncertainty_relation_on_the_cell; 5 A3_zero_mode_localised_but_massless; 6 A3_W_mass_from_the_tip; 7 A3_heisenberg_bounds_excitations; 8 A3_fermions_far_below_heisenberg; 9 A3_sharper_localisation_lighter_fermion; 10 A4_coordinate_vs_proper; 11 A5_sensitivities; 12 B1_mass_cold_neutral; 13 B2_spin_trace; 14 B2_polarisation_sum; 15 B2_cross_section_formula; 16 B3_relic_requirement_and_unitarity; 17 B4_direct_detection; 18 B5_stability_and_neutrality; 19 C1_no_positive_energy_state_accelerates; 20 C2_condensate_not_dark_energy; 21 C3_vacuum_energy_and_tuning; 22 C4_radion_moduli_problem; 23 A6_block_frozen_today.

th (um2_thermal.py, 7 checks, all passed): 1 T1_degrees_of_freedom; 2 T2_thermal_average; 3 T3_boltzmann_solver; 4 T4_relic_numerical; 5 T5_unitarity_violated_by_effective_operator; 6 T5_unitarity_limited_relic; 7 T6_I12_benchmark.

cpl (um2_coupled.py, 16 checks, all passed): 1 C1_overlaps; 2 C1_screened_response; 3 C1_fold_band_massless; 4 C2_freeze_in_benchmark; 5 C3_direct_detection_I11; 6 C8_I13_algebra; 7 C8_I13_overlaps_and_fold_band; 8 C8_I13_relic; 9 C8_I13_direct_detection; 10 C9_B_minus_L_alternative_thermalises_nuR; 11 C10_self_interaction_channel; 12 C6_proton_sensitivity; 13 C4_mean_fields_in_environments; 14 C6_gravity_loop; 15 C5_WZ_and_photon; 16 C7_validity_domain.

ser (um2_series.py, 4 checks, all passed): 1 S2_first_order_n0; 2 S3_kaluza_klein_gravitons; 3 S4_radion; 4 S7_decreasing_magnitudes.

res (um2_residuals.py, 52 checks, all passed): 1 R_einstein_lovelock_background; 2 R_block_linear_response; 3 R_primordial_deflation_system; 4 R_four_dimensional_effective_gravity; 5 R_boson_x0_eigenvalue; 6 R_fermion_x0_tip_determinant; 7 R_dirac16complex_x0_spectrum; 8 R_kohn_sham_scf; 9 R_kohn_sham_in_gravity_block_response; 10 R_portal_overlaps_hellmann_feynman; 11 R_coupled_environment_mean_fields; 12 R_wz_x0_with_dark_matter_kinetic_function; 13 R_yang_mills_masses_and_widths; 14 R_linear_perturbations_cmb_bao; 15 R_graviton_tower; 16 R_metric_series_order2; 17 R_neutrino_flavour_evolution; 18 R_neutrino_thermal_potential; 19 R_neutrino_decoupling; 20 R_fast_flavour_dispersion; 21 R_cosmic_neutrino_background; 22 R_thermal_degrees_of_freedom; 23 R_boltzmann_relic; 24 R_freeze_in; 25 R_right_handed_neutrino_boltzmann; 26 R_self_interaction_channel; 27 L_ks_energy_density_in_block_response; 28 L_dark_matter_hidden_pressure; 29 L_z_mass_x0_vs_z_widths; 30 L_weak_angle_x0_vs_z_widths; 31 L_fermion_masses_x0_vs_thermal_history; 32 L_neutrino_masses_tip_matrices_vs_flavour_cnb_cmb; 33 L_pmns_tip_matrix_vs_flavour_evolution_and_cnb; 34 L_omega_nu_cnb_vs_camb; 35 L_neff_decoupling_plus_freeze_in_vs_camb; 36 L_relic_density_vs_cmb_omega_c; 37 L_hubble_rate_thermal_vs_four_dimensional_friedmann; 38 L_late_time_radiation_thermal_vs_cmb; 39 L_plasma_energy_density_thermal_vs_coupled_sources; 40 L_portal_responses_ks_jobs_vs_coupled_overlaps; 41 L_i12_response_0_across_scripts; 42 L_i12_benchmark_thermal_vs_coupled_mean_field; 43 L_dark_matter_mass_ks_vs_boltzmann; 44 L_relic_and_freeze_in_couplings_in_mean_fields; 45 L_w_eigenvalue_x0_vs_kinetic_function_problem; 46 L_block_response_coefficients_across_scripts; 47 L_radion_frequency_linear_vs_nonlinear; 48 L_static_block_f_star_across_scripts; 49 L_decoupling_heating_factor_F_h; 50 L_metric_series_vs_cosmology_coupled_masses; 51 L_freeze_out_temperature_relic_vs_environment; 52

`L_block_sensitivities_x0_formula_vs_block_response`.

9.2 The principal numbers and where they come from

quantity	value	script (output)
static block	$K_* = 12 + \frac{4}{3}\sqrt{51} = 21.5219$, $f_* = 0.215556$, $\mathcal{G}(K_*) = \frac{20}{3} + \sqrt{51} = 13.8081$, $\kappa\rho_{\rm stab}(f_*) = (26 + 4\sqrt{51})H^2$	background (background.json)
radion	$\omega_{\rm r}^2/H^2 = 31552/2249 - 2416\sqrt{51}/6747 = 11.4721$, $\omega_{\rm r} = 3.387H = 48.42$ TeV	background, spectra (background.json, masses.json)
unbroken x_0 spectrum ($HL = 3$)	$\tilde{m} = 0$ (flat profile); first even level $\tilde{m}_1^2 = 30.1196$; first odd level $\tilde{m}^2 = 27.0890$; gap $9H^2L^2/4 = 20.25$	spectra (masses.json)
$W^{\pm}$, Z	$\beta_{\rm W} = 0.011912$, $\beta_{\rm Z} = 0.015343$ ($\cos^2\theta_{\rm W} = 0.776368$), $\tilde{m}_{\rm W}^2 = 1.32149 \times 10^{-5}$, $\tilde{m}_{\rm Z}^2 = 1.70150 \times 10^{-5}$, $\rho_{\rm EW} - 1 = 0.00038$, model $m_{\rm Z} = 91.1879$ GeV; Dirichlet bound of the tip $\tilde{m}^2 = 0.010016$	spectra (masses.json)
scales	$m_{\rm KK} = 121.32$ TeV, $H = 14.296$ TeV, proper half cell $fL = 8.926 \times 10^{-21}$ m, x_0 period 7.227×10^{-21} m, gap $3H/(2f) = 99.48$ TeV	spectra (masses.json)
fermion localisations ($\mathcal{K} = 1$)	doublets $c_L = 5.9529$ (leptons), 3.3823 (quarks); singlets c_R: ν_1 −35.0452, ν_2 −32.8481, ν_3 −31.0702, electron −14.5539, muon −8.9809, tau −5.9529, up −15.4300, charm −8.7683, top −3.3823, down −14.6258, strange −11.5158, bottom −7.5005	spectra (masses.json)
three generations	the roots of the tip determinant equal the one-generation roots to 3.0×10^{-42}, the null vectors the columns of $\mathcal{K}$ to 1.1×10^{-16}; a general complex tip matrix: singular values to 3.5×10^{-5}	flavour (flavour.json)
mixing	PMNS ($\theta_{12} = 33.71°$, $\theta_{13} = 8.56°$, $\theta_{23} = 43.28°$, $\delta_{\rm CP} = 212°$) unitary to 1.1×10^{-16}, $J_{\rm CP} = -0.01777$; CKM $J_{\rm CP} = 3.117 \times 10^{-5}$; Kaluza-Klein gauge overlaps 0.14434 (leptons), −0.30066 (quarks), off-diagonal 1.0×10^{-17}; branching ratio of $\mu \to e\gamma$ 2.8×10^{-55}	flavour (flavour.json)
neutrino masses	$m_1 = 1.000$, $m_2 = 8.712$, $m_3 = 50.140$ meV; $\sum_i m_i = 0.0599$ eV	spectra, neutrino (masses.json, neutrino.json)
flavour conversion	1-3 MSW resonance 11.135 GeV (exact 11.133 GeV) at 2.8 g/cm^3; solar 1-2 resonance 1.93 MeV; $\mathcal{P}(\nu_\mu \to \nu_e)$: T2K 0.0517, NOvA 0.0481, DUNE 0.0701 (antineutrinos 0.0313, 0.0236, 0.0195); solar $\mathcal{P}_{ee}$ from 0.5408 (pp) to 0.3043 (^{8}B, 15 MeV)	neutrino (neutrino.json)
early Universe	oscillation onset $T = 4.99$ MeV (atmospheric), 2.39 MeV (solar); $N_{\rm eff} = 3.0476$ (separate temperatures), 3.0485 (common); adopted 3.0432	neutrino (neutrino.json)
CNB	$T_\nu = 1.9454$ K, 112.0 cm^{-3} per mass state, $\bar{v}_i = 0.435133$, 0.0604083, 0.0105352; $\Omega_\nu h_{100}^2 = 6.38217 \times 10^{-4}$ (version 4: quadrature to 3×10^{-16})	neutrino (neutrino.json, copied in cosmology.json)
supernovae	H resonance (20 MeV) at 1573 g/cm^3, adiabaticity $\gamma_{\rm ad} = 178$; $\mathcal{P}_{ee} = 0.02215$, antineutrinos 0.677; fast instability growth 0.0877 and 0.0979 in units of $\sqrt{2}G_{\rm F}n_{\nu_e}$, e-folding in 5.9×10^{-10} to 5.9×10^{-11} s; heating factors 0.884, 0.760	neutrino (neutrino.json)

quantity	value	script (output)
dirac16complex at f_*	$c_\Psi = 0.646667$; $\varepsilon_{\mathrm{DM}} = 3.102204\, m_\Psi$ ($\varpi = 1.89901$, an x_0 wave number of $2.93661\, m_\Psi$); lowest even bulk level $4.959981\, m_\Psi$; x_0-momentum fraction 0.896	ks_jobs spectrum (ks/spectrum.json)
dark matter	$M_{\mathrm{DM}} = 44.35$ TeV (both halves); $p_3/\rho = 0$; $p_0/\rho = 0.772034343 = -\mathrm{d}\ln\varepsilon_{\mathrm{DM}}/\mathrm{d}\ln f$	ks_jobs darkmatter (ks/darkmatter.json)
portal responses	the dark-matter level per unit amplitude of the mean field: I11 electron density 0.824779, up-quark density 4.328803, neutrino density 1.252406; I12 gluon condensate 0.707196 ($= \mathcal{O}_{\mathrm{DM}}$); the fold-band gap changes by 7.9×10^{-11}, 3.0×10^{-9}, -9.3×10^{-11}, -1.4×10^{-10} (zero in exact arithmetic; version 4)	ks_jobs portals (ks/portals.json)
tip Z portal (removed)	$N_\nu = 3 + 2r_{\mathrm{Z}}^2$, $r_{\mathrm{Z}}^2 < 0.0056$; at the bound decoupling at 14.7 MeV and $\Delta N_{\mathrm{eff}} = 2.02$; oriented part $\Delta N_{\mathrm{eff}} = 0.0958$ to 0.1257	electroweak (electroweak.json)
Z widths	invisible 497.7 MeV (measured 499.1), total 2422.3 MeV at tree level, 2485.2 MeV with $1 + \alpha_3/\pi$ (measured 2495.5); hadronic ratio $R_\ell = 20.83$ (measured 20.77)	electroweak (electroweak.json)
primordial era	separatrix at $K = 2.6983$, 2.5991, 2.5301 (for the initial $\mathcal{H}_f = -H/2$, $-H/3$, $-H/10$); deflation factor up to 2.9166; radion frequency $3.3868H$ measured	primordial (primordial.json)
cosmology	UM-2 (version 3, $N_{\mathrm{eff}} = 3.0442$): $\mathcal{H}_0 = 67.38$ km/s/Mpc, $\Omega_{\mathrm{m}} = 0.3150$, $r_{\mathrm{d}} = 147.104$ Mpc, $\chi^2_{\mathrm{BAO}} = 31.49$ (13 points; ΛCDM 32.55), TT and EE within 2.5×10^{-4} and 2.7×10^{-4} of ΛCDM; without the freeze-in fold bands ($N_{\mathrm{eff}} = 3.0432$): 67.37, 0.3151, 147.109 Mpc, 31.72, 2.8×10^{-4}	cosmology (cosmology.json)
relic with I12 alone (version 2)	two species, $x_{\mathrm{f}} = 30.38$, $g_\rho = 105.75$, required $\langle\sigma v_{\mathrm{rel}}\rangle = 2.01 \times 10^{-25}$ cm^3/s per species, $G_{\mathrm{I12}} = (15.5\ \mathrm{TeV})^{-3}$; the ${}^3P_0$ bound $\sqrt{\pi x_{\mathrm{f}}}/M_{\mathrm{DM}}^2 = 5.79 \times 10^{-26}$ cm^3/s exceeded 3.47 times ($v_{\mathrm{c}} = 0.263$, 79 % of the pairs above it): OPEN in version 2, resolved in version 3 by I13	analysis (analysis.json)
direct detection (gluons only)	$\sigma_{\mathrm{SI}} = 6.3 \times 10^{-53}$ cm^2 against 2.4×10^{-45} cm^2	analysis (analysis.json)
dark energy	condensate tuned to $w_{\mathrm{de},0} = -1$: $w_{\mathrm{de},a} = -6$, energy zero at $z_{\mathrm{red}} = 0.260$; natural vacuum energy 5.5×10^{64} times the observed one; $(\mathcal{H}_0/H)^2 = 1.0 \times 10^{-92}$	analysis (analysis.json)
uncertainty relation	m/m_{Heis}: first Kaluza-Klein level 7.49, first odd level 6.47, dark matter 2.29, $W^\pm$ 1.14×10^{-3}, Z 1.29×10^{-3}, top quark 3.2×10^{-3}, electron 5.5×10^{-9}, ν_1 1.1×10^{-17}; $\langle k_\xi^2\rangle\Delta\xi^2$: photon zero mode 0.497, fold zero mode of Ψ 0.097 ($\varrho_\xi(-1) = 0.489$); $\langle k_\xi^2\rangle^{1/2}\Delta\xi \geq (1 - \varrho_\xi(-1))/2$ for every tabulated state	analysis (analysis.json)
block sensitivities	logarithmic derivatives with respect to $\ln f$: Kaluza-Klein masses -1, m_{W} -0.501, fermions from -6.8 (top) to -41.0 (ν_1), m_{e} -20.5, M_{DM} -0.772, α_{em} and the gauge couplings -4, G_{N} $+2.05$; splittings Δm_{21}^2 -77.5, Δm_{31}^2 -74.0	analysis (analysis.json)
block rate today (ESTIMATE)	$\lvert\mathcal{H}_f\rvert < 3.6 \times 10^{-8}\mathcal{H}_0$ (atomic clocks, change of α_{em} below 10^{-17} per year), $< 7.1 \times 10^{-4}\mathcal{H}_0$ (lunar ranging, change of G_{N} below 10^{-13} per year)	analysis (analysis.json)

quantity	value	script (output)
radion decay (ESTIMATE)	$\Gamma_\mathrm{r} = C_\mathrm{r}\,\omega_\mathrm{r}^3/(4\pi\bar{M}_\mathrm{Pl}^2)$, $T_\mathrm{RH} = (90/(\pi^2 g_\rho))^{1/4}(\Gamma_\mathrm{r}\bar{M}_\mathrm{Pl})^{1/2}$ ($g_\rho = 10.75$): lifetime $0.43\,\mathrm{s}/C_\mathrm{r}$, $T_\mathrm{RH} = 1.85\,C_\mathrm{r}^{1/2}$ MeV (below 4.7 MeV unless $C_\mathrm{r} > 6.5$)	analysis (analysis.json)
first-order block response	$\Xi_{f3} = 0.028666$, $\Xi_{fp} = 0.037320$, $\Xi_{f\rho} = -0.009555$; adiabatic $\Delta K = (0.035852\,\kappa\Delta\rho - 0.107556\,\kappa\Delta p_3 - 0.140028\,\kappa\Delta\bar{p})/H^2$ (radiation drops out: $0.107556 = 3 \times 0.035852$); for the Kohn-Sham states $\Delta K/\kappa = 2.27 \times 10^{-6}$ (8 particles, $+\lambda_\mathrm{test}$), 0.0125 (592 particles, $+\lambda_\mathrm{test}$), 0 (free gas), in units $H = 1$	background, ks_jobs feedback (background.json, ks/feedback.json)
thermal history (version 3)	$g_\rho = 105.75$ (5 TeV), 105.73 (1.42 TeV), 104.95 (200 GeV), 86.22 (10 GeV), 10.74 (3 MeV); today $g_s = 3.9091$, $g_\rho = 3.3626$, $s_0 = 2891.3\ \mathrm{cm}^{-3}$; the relativistic thermal average exceeds the non-relativistic one by about $3/x$	thermal (thermal.json)
I12 relic and unitarity (version 3)	$G_\mathrm{I12} = (15.855\ \mathrm{TeV})^{-3}$, $x_\mathrm{f} = 31.70$, $\langle\sigma v_\mathrm{rel}\rangle(x_\mathrm{f}) = 1.891 \times 10^{-25}\ \mathrm{cm}^3/\mathrm{s}$; crossing of the ${}^3P_0$ bound at $E_\mathrm{cm} = 89510$ GeV, $v_\mathrm{rel} = 2\beta_\mathrm{cm} = 0.269$ (0.274 from the non-relativistic formula), 96 % of the rate and 75 % of the pairs above it; at the ${}^3P_0$ limit 0.0582 per species ($x_\mathrm{f} = 30.77$, $\langle\sigma v_\mathrm{rel}\rangle = 5.33 \times 10^{-26}$ against $\sqrt{\pi x}/M_\mathrm{DM}^2 = 5.84 \times 10^{-26}\ \mathrm{cm}^3/\mathrm{s}$); benchmark fraction above its crossing 5.2×10^{-4}; capped: 0.1544, 0.1081, 0.0836, 0.0649, 0.0585; the benchmark $(38.2\ \mathrm{TeV})^{-3}$ alone 8.0; $c_\mathrm{I12}/\Lambda_\star^3 = 1.106 \times 10^{-14}\ \mathrm{GeV}^{-3}$	thermal (thermal.json)
I13 relic (version 3)	$\mathcal{S}_B = 16.351$; $G_\mathrm{A} = 1.278 \times 10^{-9}\ \mathrm{GeV}^{-2}$ (capped; 1.269×10^{-9} without the cap); $x_\mathrm{f} = 31.23$; $\langle\sigma v_\mathrm{rel}\rangle(x_\mathrm{f})$: I13 9.50×10^{-26}, I12 $9.9 \times 10^{-28}\ \mathrm{cm}^3/\mathrm{s}$; bound reached above $v_\mathrm{rel} = 0.51$ (22 % of the rate; the uncapped cross-section crosses it above 0.52, 25 %); $B - L$ alternative: $G_\mathrm{A}^{B-L} = 1.072 \times 10^{-9}\ \mathrm{GeV}^{-2}$, ν_R at 0.999698 of equilibrium at $x = 40$, kinetic decoupling at $x = 24.4$, $\Delta N_\mathrm{eff} = 0.144$; with J^B the ν_R production through I11 is $r_{\nu_R} = 2.0 \times 10^{-17}$	coupled (coupled.json)
self-interaction channel (version 3)	$\tilde{I}_\lambda = 14.38$; $\Delta N_\mathrm{eff} = 1.83 \times 10^{26}\,G_\lambda^2\ \mathrm{GeV}^4$; bound $G_\lambda \le 2.3 \times 10^{-15}\ \mathrm{GeV}^{-2}$ ($1.8 \times 10^{-6} G_\mathrm{A}$); λ_test with $v_\mathrm{t} = H^{-3}$: $G_\lambda = 1.14 \times 10^{-9}\ \mathrm{GeV}^{-2}$, fold bands thermalised to 2.6 TeV, $\Delta N_\mathrm{eff} = 0.19$	coupled (coupled.json)
freeze-in and detection (version 3)	$\sum_f N_\mathrm{c}(q_f\tilde{I}_{of})^2 = 126.85$; $\Delta N_\mathrm{eff} = 5.21 \times 10^{29} G_\mathrm{V}^2\ \mathrm{GeV}^4$; $G_\mathrm{V} = 4.38 \times 10^{-17}\ \mathrm{GeV}^{-2}$; production rate per fold quantum 0.012 of $\mathcal{H}$ at 10 TeV; xenon $\sigma_\mathrm{SI} = 4.01 \times 10^{-46}\ \mathrm{cm}^2$ (0.16 of LZ); $\langle\sigma v_\mathrm{rel}\rangle$ today $9.76 \times 10^{-26}\ \mathrm{cm}^3/\mathrm{s}$	coupled (coupled.json)

quantity	value	script (output)
coupled solution (version 3)	overlaps $\tilde{I}_Q = 8.2722$, $\tilde{I}_Q^{(b)} = 2.6135$ (dark matter); Kohn-Sham checks 3.1×10^{-10}, 2.3×10^{-10}; the I13 algebra exact; $\Delta\varepsilon_{\mathrm{DM}}/M_{\mathrm{DM}} \le 1.5 \times 10^{-6}$ (at the freeze-out, $T = 1.42$ TeV; thermal condensates -8.1×10^{12}, -5.0×10^{11}, -2.6×10^{10} GeV^4; Feynman-Hellmann against the trace anomaly 10^{-16}); $\Delta \ln f \le 1.1 \times 10^{-34}$ (the hot plasma at the freeze-out; 7×10^{-39} in the electroweak era, 9×10^{-49} from nucleosynthesis on); loop gain $\le 1.1 \times 10^{-33}$; $\mathrm{d}\ln m_{\mathrm{p}}/\mathrm{d}\ln f = -46.1$; $A_{\mathrm{mf}} = 1.7 \times 10^{-16}$ at the freeze-out; $\mathrm{d}\ln \tilde{m}_{\mathrm{W}}^2/\mathrm{d}A_{\mathrm{mf}} = -0.111253$ (version 4; version 3: -0.1126); validity up to $T = 27.4$ TeV	coupled (coupled.json)
metric series and σ_8 (version 3)	$\Phi_{\mathrm{rms}} = 2.42 \times 10^{-5}$ ($k = 10^{-4}$ Mpc^{-1}) to 5.03×10^{-8} (1 Mpc^{-1}); order 2: 5.8×10^{-10}; $D_+ \mathrm{e}^{-a_4} = 0.7878$ today; graviton levels $\tilde{m}_n^2 = 30.1196$, 59.7284, 109.0764; Yukawa strengths $\mathcal{A}_n = 0.0728$, 0.1469, 0.1809 (two masses at the fold); range 1.63×10^{-21} m; radion $\le 1.1 \times 10^{-34}$; $\sigma_8 = 0.8118$ (UM-2), 0.8119 (without the freeze-in fold bands), 0.8109 (ΛCDM)	series, cosmology (series.json, cosmology.json)
Majorana alternative (version 3)	$\mathcal{K}_{\mathrm{M}}^{(i)} = 5.63 \times 10^{-13}$, 4.90×10^{-12}, 2.82×10^{-11}; $m_{\beta\beta}$ from 0.84 to 4.41 meV	neutrino (neutrino.json)
residuals (version 4)	26 field equations: residuals from 7.4×10^{-14} to 2.4×10^{-5}, each within its tolerance; 26 links: five not zero (section 10)	residuals (residuals.json)

9.3 Methods

- **Exact checks.** Symbolic identities are verified with sympy by simplifying the difference to zero; matrix identities entry by entry, with the author's matrices.
- **Roots.** The transcendental eigenvalue conditions (FE-52, FE-59, FE-60, FE-62) are solved with mpmath at 30 to 60 digits; the residual analysis measures each stored root by its implied relative error, the value of the condition divided by the eigenvalue times the derivative of the condition.
- **Richardson extrapolation.** A number computed with the discretisation size n, $2\mathsf{n}$ and $4\mathsf{n}$ is extrapolated and its observed order measured by

$$\mathsf{X}_\infty \simeq \mathsf{X}_{2\mathsf{n}} + \frac{\mathsf{X}_{2\mathsf{n}} - \mathsf{X}_{\mathsf{n}}}{\mathrm{e}^{p_{\mathrm{obs}} \ln 2} - 1}, \qquad p_{\mathrm{obs}} = \frac{1}{\ln 2} \ln \frac{\mathsf{X}_{\mathsf{n}} - \mathsf{X}_{2\mathsf{n}}}{\mathsf{X}_{2\mathsf{n}} - \mathsf{X}_{4\mathsf{n}}}$$

Symbols in this equation.

- X_{n}, X_∞: A computed number (an eigenvalue, an energy, an integral of the Kohn-Sham stress, a derivative) obtained with the discretisation size n, whose discretisation error is a constant times $\mathsf{n}^{-p_{\mathrm{obs}}}$; X_{n}, $\mathsf{X}_{2\mathsf{n}}$, $\mathsf{X}_{4\mathsf{n}}$ are its values for the sizes n, $2\mathsf{n}$, $4\mathsf{n}$, and X_∞ is the value extrapolated to an infinite size (Richardson extrapolation). Unit: the unit of the computed quantity. Status: STANDARD.
- n: The size of a discretisation that is doubled from one computation to the next: the number of free eigenfunctions per sector of the Kohn-Sham basis (60, 120, 240), the number of finite elements (100, 200, 400), or the inverse of the step of a finite difference. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- p_{obs}: The observed order of convergence of a sequence of computations with the sizes n, $2\mathsf{n}$, $4\mathsf{n}$: the exponent of the discretisation error, which behaves as a constant times $\mathsf{n}^{-p_{\mathrm{obs}}}$ (not a pressure). Unit: dimensionless. Value: 4.96 for the Kohn-Sham energy of the stress state, 2.97 for its integrated hidden pressure, 4.01 for the finite-element eigenvalue of the W. Status: DERIVED.

- **Finite elements.** As a cross-check of the boson eigenvalues, quadratic finite elements on 100, 200 and 400 elements solve the weak form of FE-50 with the tip condition of FE-51; the eigenvalue is the Rayleigh quotient of the gradient form,

$$\int_{-1}^{0} \mathrm{e}^{3HL\xi}\Big(\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big)^2 \mathrm{d}\xi + \beta\,\mathrm{e}^{-3HL}F(-1)^2 = \tilde{m}^2\int_{-1}^{0}\mathrm{e}^{3HL\xi}F^2\,\mathrm{d}\xi$$

Symbols in this equation.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- β: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.

extrapolated by Richardson (observed order 4.01; $\tilde{m}_\mathrm{W}^2$ reproduced to 3.1×10^{-14}). * **Kohn-Sham.** The orbitals of FE-63 are expanded in the exact eigenfunctions of the free problem of each sector (60, 120 or 240 functions, quadrature with ten points per function); the mean fields FE-64 are iterated to self-consistency (tolerance 10^{-13}) with a safeguarded Anderson mixing and a fixed labelling of the occupied set. Checks: the energy identity and the y-conservation law FE-68, Richardson in the basis size, the Hellmann-Feynman theorem for every portal mean field (the response of a level to a mean-field amplitude equals the expectation value of the perturbation), and the reproduction of the record's two reference states. Since version 4 the linear algebra runs single-threaded, so that the results do not depend on the number of cores (section 10). * **Background and primordial era.** The 3-space and hidden components FE-39, FE-40 are solved for the two rates $\dot{\mathcal{H}}$ and $\dot{\mathcal{H}}_f$, with $\dot{K} = -2K\mathcal{H}_f$ and the stabiliser (plus radiation where stated); the constraint FE-38 fixes the initial $\mathcal{H}$ and is monitored (residual below 6×10^{-10}); DOP853 integration, the separatrix by bisection in the initial K. * **Neutrinos.** The flavour evolution FE-104 by exact matrix exponentials (constant density) and ordered products of 20000 and 40000 layers (the Sun); the decoupling equations FE-111 to FE-114 with LSODA from 20 MeV to 5 keV (relative tolerance 10^{-11}), validated against the published values of the method; the dispersion relation FE-117 by vectorised Newton iterations from a grid of complex starting points, checked against the analytic growth of FE-119 and the eigenvalues of the discretised homogeneous equation (300 nodes); the CNB FE-120 by adaptive quadrature. * **Cosmology.** CAMB 2.0.5 solves FE-89 to FE-99 with the Planck 2018 parameters, the acoustic angle fixed (so $\mathcal{H}_0$ is an output), multipoles to 2600, lensing included. UM-2 has three massive mass states, each with the degeneracy $N_\mathrm{eff}/3 = 1.0144$; CAMB gives each the density of a Fermi-Dirac gas times the degeneracy, so the masses passed to it are the masses divided by the fourth root of the degeneracy; because CAMB recovers them by inverting the relativistic density from the neutrino density and the mass fractions, these are iterated (damped exponent 0.8, 13 iterations) until the masses agree to 7.8×10^{-12}. The comparison model is CAMB's default (one neutrino of 0.06 eV, $N_\mathrm{eff} = 3.044$). The cosmic variance of one multipole is the relative error $\sqrt{2/(2l+1)}$; the binned full-sky goodness of fit uses bins of 50 multipoles. * **Thermal history and relic.** The ideal-gas integrals FE-121 by 160-point Gauss-Legendre quadrature on 360 temperatures from $10^{-4.5}$ to $10^{4.3}$ GeV; the Boltzmann equation FE-124 for $\ln Y_\mathrm{DM}$ with the implicit Radau method (relative tolerance 10^{-10}) from $x = 5$ to 2000; the thermal average FE-125 by adaptive quadrature in the kinetic energy of the pair with exponentially scaled Bessel functions, tabulated on 200 to 260 points in x and interpolated by a cubic spline; the strengths for $\Omega_\mathrm{DM}h_{100}^2 = 0.060$ per species by Brent's method in the logarithm of the strength;

the freeze-in FE-129 and the self-interaction channel FE-136 by the trapezoid rule on 600 and 800 logarithmic temperatures. * **Determinism.** Every computation is deterministic: the Kohn-Sham scripts and the residual analysis fix the number of threads of the linear algebra to one (version 4); the document build writes no dates into the LaTeX sources, suppresses the PDF dates and trailer identifiers and fixes `SOURCE_DATE_EPOCH`.

10 The residuals of the approximate solution, and the precision fixes of version 4

10.1 The residual analysis

The order of 2026-10-18, 15:42, asks for the approximate solution to be analysed "usually in concert with other field equation". Version 4 adds the computation `um2_residuals.py` (outputs `residuals.json`, report `residuals-checks.json`, 52 checks). For every field equation (or system) that the solution uses, it records the solution (exact formula, numerical method, order of the approximation); a precisely defined residual, either taken from the stored outputs or computed afresh by substituting the stored solution into the equation; the size of the terms that the approximation drops; and the links in concert, in which the same quantity is compared in the two or more places where different field equations use it (each input is read where the script that uses it reads it). Every equation (res 1 to 26) and every link (res 27 to 52) has a named check with a tolerance fixed in the script with a one-line justification; where a residual or a link is not small it is reported as it is, and the tolerance is set by what the physics justifies. The script reads every output and writes only its own; its linear algebra is single-threaded, so it gives the same numbers on every machine. For a root of a transcendental condition the residual is the implied relative error (the value of the condition at the stored root divided by the root times the derivative of the condition); for a differential equation it is the stored solution substituted into the printed equation; for an integral it is an independent quadrature.

	field equations	residual of the stored solution	largest neglected term
res 1	FE-38 to FE-40, FE-42, FE-43	the static block in the printed equations: 3.9×10^{-13} (the printed equations equal the code's symbolic ones)	the matter sources against the stabiliser: 3.0×10^{-32} at the freeze-out, 2.4×10^{-93} today
res 2	FE-44 to FE-47	the adiabatic response in the nonlinear equations: 1.4×10^{-7} (second order, as expected)	second order and the inertia of the block at the freeze-out: 2.3×10^{-32}, 3.4×10^{-33}
res 3	FE-38 to FE-41 (primordial era)	constraint and one-step defect of the stored trajectories: 1.3×10^{-11}	radiation of zero modes: 1.0×10^{-6} of the final block
res 4	FE-48, FE-49	$\Lambda_4(K_*)$ and the K-derivative of the x_4 balance: 6.9×10^{-13}	the dark energy, $\Lambda_4/H^2 \sim 10^{-92}$ (tuned, OPEN)
res 5	FE-50 to FE-52, FE-55	implied relative error of the stored eigenvalues: 5.9×10^{-12} (W), 1.4×10^{-12} (Z)	the I12 shift of $\tilde{m}_{\mathrm{W}}^2$ at the freeze-out: 1.9×10^{-17}; Lovelock corrections to the graviton operator OPEN
res 6	FE-58 to FE-60	implied relative root error: 5.4×10^{-11} (ν_1); three generations 1.2×10^{-11}	mixing with the Kaluza-Klein modes, largest for the top quark: 2.0×10^{-6}
res 7	FE-62	the stored levels against the closed form: 8.5×10^{-13}	the shift by the Standard-Model mean fields: 1.5×10^{-6} (added in the coupled solution)
res 8	FE-63 to FE-68	a state re-solved, one more Kohn-Sham map, the energy identity: $7.4 \times 10^{-14}\, H$	the self-interaction at the assumed bound: $2.7 \times 10^{-9}\, m_\Psi$; correlation beyond the local exchange OPEN
res 9	FE-67 with FE-47	ΔK recomputed from the Kohn-Sham integrals: 1.4×10^{-12}	the block-shape source $p_0 - p_5$: NOT small, up to 4.00 times $\bar{p}$ (OPEN)
res 10	FE-70, FE-71, FE-135	the overlaps from the closed-form orbitals: 3.8×10^{-12}	screening by the interacting gas: 3.5×10^{-4} at λ_{test}, 7.2×10^{-10} at the bound

	field equations	residual of the stored solution	largest neglected term
res 11	FE-72 to FE-86	every mean field and block response recomputed: 2.0×10^{-11}	the loop gain 1.1×10^{-33}; second order in the mean fields 2.3×10^{-12}
res 12	FE-79	the first-order coefficient afresh: 3.3×10^{-12} (shooting agrees)	the physical shift at the freeze-out: 1.9×10^{-17}
res 13	FE-55, FE-102 and the Z widths	widths and m_{Z} recomputed: 2.7×10^{-12}	electroweak loop corrections 0.41 %; bottom-quark mass −0.13 %; order α_3^2 0.13 %
res 14	FE-89 to FE-99	the Poisson form against the Weyl potential: 1.5×10^{-6}; matter-era limit 4.4×10^{-5}	order ϵ^2 against order ϵ: 2.4×10^{-5}
res 15	FE-53, FE-54	the profiles in the operator: 1.2×10^{-12}; Yukawa strengths 5.2×10^{-9}	the Kaluza-Klein correction to Newton's law at 50 micrometres: zero in double precision
res 16	FE-87, FE-88	the first omitted order against the kept one: 2.4×10^{-5}	order 2 at a neutron star: 0.030 (outside the cosmological solution)
res 17	FE-103 to FE-108	unitarity, vacuum formula, accelerator and solar probabilities: 7.1×10^{-12}	non-adiabatic effects on the solar $\mathcal{P}_{ee}$: 1.3×10^{-3}
res 18	FE-109	the onset condition at the stored temperatures: 8.7×10^{-12}	the CP-asymmetric potential: 1.1×10^{-2} of the thermal one (ESTIMATE)
res 19	FE-110 to FE-115	the instantaneous limit recomputed: 2.7×10^{-12}; published values within 4.6×10^{-4}	electron mass and momentum dependence: 4.4×10^{-3} in N_{eff}
res 20	FE-116 to FE-119	moments, analytic growth, homogeneous matrix: 3.3×10^{-13}	the vacuum term: 9.0×10^{-6} of the self-interaction
res 21	FE-120	densities, energy densities, velocities, $\Omega_\nu h^2_{100}$: 2.9×10^{-12}	the non-instantaneous heating: 1.1×10^{-2} in $\Omega_\nu h^2_{100}$ (included in the CMB)
res 22	FE-121 to FE-123	g_ρ, g_s by an independent quadrature: 5.0×10^{-9}	the order-α plasma pressure: NOT small, −5.4 % at the freeze-out
res 23	FE-124 to FE-128	the relic re-solved, $\Omega_{\mathrm{DM}} h^2_{100} = 0.060000000186$ per species; late-time identity 6.1×10^{-9}	the order-α pressure: about +2.7 % in $\Omega_{\mathrm{DM}} h^2_{100}$ (ESTIMATE)
res 24	FE-129, FE-130	$\Delta N_{\mathrm{eff}}/G^2_{\mathrm{V}}$ re-integrated: 3.9×10^{-5}	inverse processes: 0.012 of the Hubble rate at $T_{\max}$
res 25	FE-131 to FE-134	the ν_R equation re-solved: 2.5×10^{-12}	the elastic rate (ESTIMATE): 9.5×10^{-6} in ΔN_{eff}
res 26	FE-135, FE-136	$\Delta N_{\mathrm{eff}}/G^2_{\lambda}$ re-integrated: 1.1×10^{-5}	inverse annihilation at the bound: 0.027 of the Hubble rate

The 26 links in concert (the quantity, the field equations that share it, the difference):

	quantity	shared by	difference
res 27	block averages of the Kohn-Sham states	FE-63 to FE-68, FE-45 to FE-47	3.4×10^{-12}
res 28	hidden pressure of the dark matter, $p_0/\rho = -\mathrm{d}\ln\varepsilon_{\mathrm{DM}}/\mathrm{d}\ln f$	FE-67, FE-62, FE-81	6.1×10^{-12}
res 29	m_{Z}	FE-55, the Z widths	2.0×10^{-12}
res 30	the weak angle and the total Z width	FE-55, FE-102, the Z widths	NOT zero: 8.3×10^{-3}

	quantity	shared by	difference
res 31	the twelve fermion masses	FE-59, FE-60, FE-121	3.1×10^{-12}
res 32	m_1, m_2, m_3 and the splittings	FE-60, FE-104, FE-120, FE-96, FE-109	5.7×10^{-12}
res 33	the PMNS matrix	FE-60, FE-104, FE-120	8.2×10^{-13}
res 34	$\Omega_\nu h^2_{100}$	FE-120, FE-96	1.9×10^{-5}
res 35	$N_{\rm eff} = 3.0432 + \Delta N_{\rm eff}$	FE-115, FE-130, FE-97	5.4×10^{-16}
res 36	$\Omega_{\rm DM} h^2_{100}$	FE-126, FE-90	3.1×10^{-9}
res 37	the Hubble rate at the freeze-out	FE-122, FE-124, FE-49, FE-129	NOT zero: 1.7×10^{-4}
res 38	g_ρ today	FE-122, FE-115, FE-97	NOT zero: 5.9×10^{-3}
res 39	the plasma energy density at the freeze-out	FE-121, FE-81	NOT zero: 1.3×10^{-3}
res 40	the portal responses of the dark-matter level	FE-70, FE-63, FE-72	3.0×10^{-12}
res 41	$\mathcal{O}_{\rm DM}$	FE-71, FE-124, FE-72, FE-79	5.7×10^{-6} (4 digits coded)
res 42	$c_{\rm I12}/\Lambda^3_\star$	FE-127, FE-72	1.6×10^{-12}
res 43	$M_{\rm DM}$	FE-62, FE-63, FE-124, FE-72	8.1×10^{-13}
res 44	$G_{\rm A}$, $G_{\rm V}$	FE-126, FE-129, FE-72	6.7×10^{-13}
res 45	$\tilde{m}^2_{\rm W}$ without the kinetic function	FE-52, FE-79	2.3×10^{-6} (finite elements)
res 46	c_ρ, c_3, c_p	FE-46, FE-47, FE-80	2.0×10^{-13}
res 47	the radion frequency	FE-44, FE-39, FE-53, FE-87	NOT zero: 7.9×10^{-5} (finite oscillation)
res 48	f_*	FE-43, FE-52, FE-59, FE-62, FE-63	2.2×10^{-12}
res 49	the heating factor $F_{\rm h}$	FE-115, FE-133	5.8×10^{-7} (4 digits coded)
res 50	$\Phi_{\rm rms}$, $\Delta \ln f$, $m_{\rm KK}$, σ_8	FE-87, FE-53, FE-98, FE-80	4.7×10^{-12}
res 51	the freeze-out temperature	FE-126, FE-86, FE-136	2.1×10^{-12}
res 52	the block sensitivities of the masses	FE-85, FE-83, FE-84	4.5×10^{-11}

Every residual lies within its tolerance and every check passes. The radion frequency (res 47) differs because the nonlinear value is measured from the zero crossings of a finite oscillation (relative amplitude 3×10^{-3}); the other links that are not zero are findings, stated next.

10.2 The findings that are not small

1. The weak angle has two values. The x_0 problem of $W^\pm$ and Z fixes the angle of the couplings with the measured electroweak parameter,

$$\rho_{\rm EW} = \frac{\tilde{m}^2_{\rm W}}{\tilde{m}^2_{\rm Z} \cos^2\theta_{\rm W}} = \rho_{\rm EW,0}, \qquad \cos^2\theta_{\rm W} = \frac{m^2_{\rm W}/m^2_{\rm Z}}{\rho_{\rm EW,0}}, \qquad m_{\rm KK} = m_{\rm W}\,\frac{\tilde{m}_1}{\tilde{m}_{\rm W}}, \qquad m_{\rm Z} = m_{\rm W}\,\frac{\tilde{m}_{\rm Z}}{\tilde{m}_{\rm W}} \qquad \text{(FE-55)}$$

Symbols in (FE-55).

- $\rho_{\rm EW}$: The electroweak parameter, $\rho_{\rm EW} = \tilde{m}^2_{\rm W}/(\tilde{m}^2_{\rm Z}\cos^2\theta_{\rm W})$. Unit: dimensionless. Status: DERIVED.
- $\tilde{m}_{\rm W}$: The dimensionless x_0 eigenvalue of the lowest $W^\pm$ mode ($\beta = \beta_{\rm W}$). Unit: dimensionless. Value: $\tilde{m}^2_{\rm W} = 1.32149 \times 10^{-5}$. Status: DERIVED.
- $\tilde{m}_{\rm Z}$: The dimensionless x_0 eigenvalue of the lowest Z mode ($\beta = \beta_{\rm Z}$). Unit: dimensionless. Value: $\tilde{m}^2_{\rm Z} = 1.70150 \times 10^{-5}$. Status: DERIVED.

- $\theta_{\rm W}$: The weak mixing angle of the couplings, $\cos\theta_{\rm W}=\hat g_2/\hat g_Z$, $\sin\theta_{\rm W}=\hat g_1/\hat g_Z$; fixed by the measured W and Z masses with $\rho_{\rm EW,0}$. Unit: dimensionless. Value: $\cos^2\theta_{\rm W}=0.776368$. Status: DERIVED.
- $\rho_{\rm EW,0}$: The measured value of the electroweak parameter (PDG fit). Unit: dimensionless. Value: 1.00038. Status: STANDARD.
- $m_{\rm W}$: The mass of W^+ and W^-. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- $m_{\rm Z}$: The mass of the Z. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- $m_n^{\rm KK}$, $m_{\rm KK}$: The mass of the n-th Kaluza-Klein graviton (and gauge) level, $m_n^{\rm KK}=\tilde m_n/(fL)$; $m_{\rm KK}=m_1^{\rm KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\rm KK}=121.32$ TeV. Status: DERIVED.
- $\tilde m$, $\tilde m_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde m=m\,fL$; $\tilde m_n$ the n-th even Kaluza-Klein level of the tower with $\beta=0$ ($\tilde m_1$ the first). Unit: dimensionless. Value: $\tilde m_1^2=30.1196$, $\tilde m_2^2=59.7284$, $\tilde m_3^2=109.0764$ ($HL=3$), i.e. $\tilde m_1=5.48813$, $\tilde m_2=7.72842$, $\tilde m_3=10.44397$. Status: DERIVED.

so that $\sin^2\theta_{\rm W}=1-0.776368=0.223632$ (on shell); this angle also enters the tip condition of the Z (FE-51), the Z part of the electroweak condensates (FE-78) and the tree-level couplings

$$\frac{G_{\rm F}}{\sqrt2}=\frac{g_2^2}{8m_{\rm W}^2},\qquad \tilde g_L^f=T_3^f-Q_{\rm em}^f\sin^2\theta_{\rm W},\qquad \tilde g_R^f=-Q_{\rm em}^f\sin^2\theta_{\rm W},\qquad \tilde g_V^f=\tilde g_L^f+\tilde g_R^f=T_3^f-2Q_{\rm em}^f\sin^2\theta_{\rm W}\tag{FE-102}$$

Symbols in (FE-102).

- $G_{\rm F}$: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- g_2: The four-dimensional SU(2) coupling ($g_2^2=4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2=0.3781$ at $2M_{\rm DM}$. Status: DERIVED.
- $m_{\rm W}$: The mass of W^+ and W^-. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- $\tilde g_L^f$, $\tilde g_R^f$, $\tilde g_V^f$: The neutral-current couplings of the fermion f to the Z (left-handed, right-handed and vector), $\tilde g_L^f=T_3^f-Q_{\rm em}^f\sin^2\theta_{\rm W}$. Unit: dimensionless. Status: STANDARD.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\rm c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- L: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L=(1+\Gamma_4)/2$. Status: STANDARD.
- T_3: The third component of the weak isospin ($+\frac12$ for the upper and $-\frac12$ for the lower member of a doublet, 0 for a singlet); T_3^f that of the fermion f. Unit: dimensionless. Status: STANDARD.
- $Q_{\rm em}$: The electric charge in units of the elementary charge, $Q_{\rm em}=T_3+Y$; $Q_{\rm em}^f$ that of the fermion f. Unit: dimensionless. Status: STANDARD.
- $\theta_{\rm W}$: The weak mixing angle of the couplings, $\cos\theta_{\rm W}=\hat g_2/\hat g_Z$, $\sin\theta_{\rm W}=\hat g_1/\hat g_Z$; fixed by the measured W and Z masses with $\rho_{\rm EW,0}$. Unit: dimensionless. Value: $\cos^2\theta_{\rm W}=0.776368$. Status: DERIVED.
- R: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R=(1-\Gamma_4)/2$. Status: STANDARD.
- V: The label V of the vector coupling of a fermion to the Z, $\tilde g_V^f=\tilde g_L^f+\tilde g_R^f$. Status: STANDARD.

while the Z widths, the running couplings (FE-77) and the couplings of the I12 benchmark use the measured $\overline{\rm MS}$ value $\sin^2\theta_{\overline{\rm MS}}=0.23122$ in the tree-level partial widths of the Z into massless fermions (with the factor $1+\alpha_3/\pi$ for the hadronic widths),

$$\Gamma_{\rm Z}^f=\frac{N_{\rm col}\,G_{\rm F}\,m_{\rm Z}^3}{6\sqrt2\,\pi}\Big[\big(T_3^f-2Q_{\rm em}^f\sin^2\theta_{\overline{\rm MS}}\big)^2+\big(T_3^f\big)^2\Big]$$

Symbols in this equation.

- $\Gamma_{\rm Z}^f$: The decay width of the Z boson; with the fermion label f it is the partial width into the pair of f and its antiparticle (tree level, massless fermion). Unit: GeV (quoted in MeV). Value: total 2485.2 MeV with $1+\alpha_3/\pi$ on the quarks (measured 2495.5 MeV). Status: DERIVED.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\rm c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.

- N_{col}: The number of colour components of a fermion multiplet: 3 for a quark, 1 for a lepton. Unit: dimensionless. Status: STANDARD.
- G_{F}: The Fermi constant. Unit: GeV^{-2}. Value: 1.1663788×10^{-5} GeV^{-2} (measured). Status: STANDARD.
- m_{Z}: The mass of the Z. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T_3: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); T_3^f that of the fermion f. Unit: dimensionless. Status: STANDARD.
- Q_{em}: The electric charge in units of the elementary charge, $Q_{\mathrm{em}} = T_3 + Y$; Q_{em}^f that of the fermion f. Unit: dimensionless. Status: STANDARD.
- $\theta_{\overline{\mathrm{MS}}}$: The weak mixing angle in the $\overline{\mathrm{MS}}$ scheme at m_{Z}. Unit: dimensionless. Value: $\sin^2\theta_{\overline{\mathrm{MS}}} = 0.23122$. Status: STANDARD.

With the on-shell angle (and the m_{Z} of the x_0 problem) the total width would be 2505.73 MeV instead of 2485.23 MeV: 0.83 % (0.82 % at tree level without the QCD factor, check `Z_width_scheme`). The difference is a scheme ambiguity of a tree-level angle, of the size of the electroweak loop corrections that UM-2 does not compute (the measured total width is 0.41 % above the computed one); the status of the widths (a tree-level comparison) is unchanged.

2. The Hubble rate of the thermal history leaves out the fold bands. The Boltzmann equation of the dark matter uses the Hubble rate of the thermal history,

$$g_\rho = \frac{30}{\pi^2 T^4}\sum_s \rho_s, \qquad g_s = \frac{45}{2\pi^2 T^3}\sum_s s_s, \qquad \mathcal{H} = \Big(\frac{8\pi^3 g_\rho}{90}\Big)^{1/2}\frac{T^2}{M_{\mathrm{Pl}}} \tag{FE-122}$$

Symbols in (FE-122).

- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- M_{Pl}: The Planck mass, $M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2}$. Unit: GeV. Value: 1.22091×10^{19} GeV. Status: STANDARD.

whose g_ρ counts the species of the Standard Model in equilibrium but not the fold bands produced by freeze-in. At the freeze-out ($T = 1.42$ TeV) the fold bands carry 3.48×10^{-4} of the energy density, so the four-dimensional Friedmann equation with all species gives a Hubble rate larger by 1.7×10^{-4}; since the relic abundance is about proportional to the Hubble rate at the freeze-out, it moves by about the same relative amount, about a factor 60 below the 1 % precision of the measured $\Omega_{\mathrm{DM}}h_{100}^2$. The dark matter itself carries 1.1×10^{-2} of the energy density at the start of the Boltzmann integration ($x = 5$), where its abundance is the equilibrium one, and 7×10^{-12} at the freeze-out.

3. The thermal history decouples the neutrinos instantaneously. Below 2 MeV the thermal history uses

$$w_{\mathrm{QGP}}(T) = \frac{1}{2}\Big[1 + \tanh\frac{T - T_{\mathrm{c}}}{T_{\mathrm{w}}}\Big], \qquad \frac{T_\nu}{T} = \Big(\frac{g_{s,e\gamma}(T)}{11/2}\Big)^{1/3}, \qquad \frac{\mathrm{d}\ln T}{\mathrm{d}x_4} = -\frac{\mathcal{H}}{1 + \frac{1}{3}\,\mathrm{d}\ln g_s/\mathrm{d}\ln T} \tag{FE-123}$$

Symbols in (FE-123).

- $w_{\mathrm{QGP}}(T)$: The weight of the quark-gluon plasma in the model of the QCD crossover (1 above, 0 below). Unit: dimensionless. Status: ASSUMED.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- T_{c}, T_{w}: The temperature and the width of the QCD crossover. Unit: GeV. Value: 155 MeV and 15 MeV. Status: ASSUMED.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $g_{s,e\gamma}$: The entropy degrees of freedom of the electrons, positrons and photons only (after neutrino decoupling). Unit: dimensionless. Value: 11/2 before and 2 after the annihilation of $e^\pm$. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- g_ρ, g_s: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.

which gives $N_{\mathrm{eff}} = 3$ and $g_\rho = 3.362644$ today, while the CMB computation uses $N_{\mathrm{eff}} = 3.0432 + 0.0010 = 3.0442$ from the decoupling equations with the precision value,

$$N_{\mathrm{eff}} = \Big(\frac{11}{4}\Big)^{4/3}\Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4 + 2\Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4\Big], \qquad \frac{\mathrm{d}T_\nu}{\mathrm{d}x_4} = -\mathcal{H}T_\nu + \frac{C_{\nu_e} + 2C_{\nu_\mu}}{3\,\mathrm{d}\rho_\nu/\mathrm{d}T_\nu} \tag{FE-115}$$

Symbols in (FE-115).

- N_{eff}: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); ΔN_{eff} an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- T_{ν_α}, T_{ν_e}, T_{ν_μ}: The temperature of the neutrinos of the flavour α (of ν_e, of ν_μ and ν_τ) during decoupling. Unit: GeV. Status: STANDARD.
- T_γ: The temperature of the photons (equal to the plasma temperature T; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- C_{ν_α}, C_{ν_e}, C_{ν_μ}: The energy transfer rate per volume into the neutrinos of the flavour α by collisions. Unit: GeV^5. Status: STANDARD.
- ρ_ν, ρ_{ν_α}: The energy density of one neutrino flavour (neutrinos and antineutrinos) at the common temperature T_ν, $\frac{7}{8}\frac{\pi^2}{15}T_\nu^4$ when massless; ρ_{ν_α} that of the flavour α at its temperature T_{ν_α}. Unit: GeV^4. Status: STANDARD.

and so $g_\rho = 3.382720$ today: 0.59 %. Below 2 MeV the thermal history enters only the entropy density today ($g_s = 43/11$, the standard value) and the environment of nucleosynthesis (effects below 10^{-58}), so the difference has no effect on a result.

4. The plasma energy density differs by 0.13 % between the thermal history and the block sources, by design. The thermal history sums the ideal gases with the x_0 masses,

$$\rho_s = \frac{d_s}{2\pi^2}\int_0^\infty \frac{k^2 E\,\mathrm{d}k}{\mathrm{e}^{E/T} \pm 1}, \qquad p_s = \frac{d_s}{6\pi^2}\int_0^\infty \frac{k^4\,\mathrm{d}k}{E\,(\mathrm{e}^{E/T} \pm 1)}, \qquad s_s = \frac{\rho_s + p_s}{T}, \qquad E = \sqrt{k^2 + m_s^2} \tag{FE-121}$$

Symbols in (FE-121).

- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, Z, quarks, leptons, nucleons, dark matter); a sum over s runs over the species present. Status: STANDARD.
- d_o, d_s: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- E: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum k in the thermal integrals. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- s, s_s: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; s_s that of the species s. Unit: GeV^3. Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.

(1.414292×10^{14} GeV^4 with the equilibrium dark matter at the freeze-out), while the sources of the block response in the coupled solution add the trace anomaly of the interacting plasma, the QCD interaction measure

$$\frac{\Theta_{\mathrm{QCD}}}{T^4} = 4\exp\Big[-\frac{\ln^2(T/0.2\,\mathrm{GeV})}{0.5}\Big] \quad (T < 1\,\mathrm{GeV}), \qquad \frac{\Theta_{\mathrm{QCD}}}{T^4} = \frac{b_3}{3}\Big(1+\frac{5N_\mathrm{f}}{12}\Big)\alpha_3^2 \quad (T > 1\,\mathrm{GeV}),$$
$$p_{\mathrm{QCD}} = \frac{8\pi^2}{45}T^4\Big[1+\frac{21N_\mathrm{f}}{32}-\frac{15}{4}\Big(1+\frac{5N_\mathrm{f}}{12}\Big)\frac{\alpha_3}{\pi}\Big], \qquad \frac{\mathrm{d}\alpha_3}{\mathrm{d}\ln T} = -\frac{b_3}{2\pi}\,\alpha_3^2, \qquad \Theta_{\mathrm{QCD}} = T^5\,\frac{\mathrm{d}}{\mathrm{d}T}\Big(\frac{p_{\mathrm{QCD}}}{T^4}\Big) \tag{FE-75}$$

Symbols in (FE-75).

- Θ_{QCD}, Θ_{G}, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: Θ_{QCD} of the quark-gluon plasma or hadron gas, Θ_{G} of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V (Θ_{W} of $W^\pm$, Θ_{Z} of Z). Unit: GeV^4. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- b_3, b_2, b_1, b_{G}: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_{\mathrm{ren}} = -b\,\alpha^2/(2\pi)$: $b_3 = 11 - \frac{2}{3}N_\mathrm{f}$ (SU(3)), $b_2 = \frac{10}{3}$ (SU(2) without a Higgs particle), $b_1 = -\frac{20}{3}$ ($\mathrm{U}(1)_Y$), b_{G} that of the group G (not related to the metric function b_4). Unit: dimensionless. Status: STANDARD.
- N_f: The number of active quark flavours (quarks lighter than the scale). Unit: dimensionless. Status: STANDARD.
- α_3: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\mathrm{Z}}) = 0.1180$; $\alpha_3(m_{\mathrm{KK}}) = 0.0606$. Status: STANDARD.

- p_{QCD}: The pressure of the quark-gluon plasma (leading orders). Unit: GeV^4. Status: ESTIMATE.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.

and the electroweak trace (1.416139×10^{14} GeV^4). The trace anomaly enters only $\rho - 3p_3$, which is what the block responds to.

5. The order-α pressure of the plasma is not in the thermal history. The interaction part of the pressure of a gauge plasma,

$$\begin{aligned} &\left\langle F^V_{\mu\nu} F^{V\,\mu\nu} \right\rangle(T) = -2\,\Theta_V \quad (\text{for } V = W^\pm, Z), \qquad \left\langle F^2_{\mathrm{geo}} \right\rangle = -4\,\frac{\partial p}{\partial\left(1/g_{\mathrm{G}}^2\right)} = 4 g_{\mathrm{G}}^2\, p_{\alpha,\mathrm{G}}, \\ &p_{\alpha,\mathrm{G}} = -\frac{g_{\mathrm{G}}^2 T^4}{144}\, d_{\mathrm{adj}} \Big[C_{\mathrm{adj}} + \frac{5}{2}\, T_{\mathrm{F}} N_{\mathrm{f}} \Big] \end{aligned} \tag{FE-76}$$

Symbols in (FE-76).

- $F^V_{\mu\nu}$: The field strength of the canonically normalised massive vector $V = W^\pm$ or Z, $F^V_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu$. Unit: GeV^2. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as μ). Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- Θ_{QCD}, Θ_{G}, Θ_V: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: Θ_{QCD} of the quark-gluon plasma or hadron gas, Θ_{G} of the gauge plasma of the group G, Θ_V of the thermal gas of the massive vector V (Θ_{W} of $W^\pm$, Θ_{Z} of Z). Unit: GeV^4. Status: STANDARD.
- F_{geo}: The field strength of the gauge group G in geometric normalisation; $\langle F^2_{\mathrm{geo}} \rangle$ is its condensate ($G^A_{\mu\nu} G^{A\mu\nu}$, $W^I_{\mu\nu} W^{I\mu\nu}$ or $B_{\mu\nu} B^{\mu\nu}$). Unit: GeV^2. Status: STANDARD.
- ∂: A partial derivative: ∂_M is the derivative with respect to the coordinate X^M, ∂_y with respect to y, ∂_{x_4} with respect to the time x_4, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- g_{G}: The four-dimensional coupling of the gauge group G (one of g_3, g_2, g_1). Unit: dimensionless. Status: STANDARD.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-α (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: GeV^4. Status: STANDARD.
- d_{adj}: The dimension of the adjoint representation of the gauge group (8, 3, 1). Unit: dimensionless. Status: STANDARD.
- C_{adj}: The quadratic Casimir of the adjoint representation (3, 2, 0). Unit: dimensionless. Status: STANDARD.
- T_{F}: The Dynkin index of the fermion representation ($\frac{1}{2}$ per Dirac fundamental; for $\mathrm{U}(1)_Y$ the sum of the squared hypercharges). Unit: dimensionless. Status: STANDARD.
- N_{f}: The number of active quark flavours (quarks lighter than the scale). Unit: dimensionless. Status: STANDARD.

is -5.4 % of the ideal-gas pressure at the freeze-out (QCD -4.7 %, SU(2) -0.7 %, $\mathrm{U}(1)_Y$ -0.10 %). The coupled solution uses it for the condensates, but the ideal-gas thermal history, and so the Hubble rate and the entropy density of the Boltzmann equation, leave it out: with it $\Omega_{\mathrm{DM}} h^2_{100}$ would rise by about 2.7 % at fixed strength, and the strength G_{A} that gives the measured relic by about 1.4 % (ESTIMATE). The physics accuracy of $G_{\mathrm{A}} = 1.278 \times 10^{-9}$ GeV^{-2} is therefore about 1 %, not the 10^{-6} of its root finding (a precision statement made in version 4; the value is unchanged).

6. The block-shape source, quantified. The order-0 metric has one block scale, so the block responds only to the hidden pressure averaged over the block,

$$\Delta S_{\rm m} = -\frac{1}{2}\int \mathrm{d}^8X\,\sqrt{|g|}\,\mathcal{T}_{MN}\,\Delta g^{MN}, \qquad \rho = -\mathcal{T}^{x_4}{}_{x_4}, \qquad p_M = \mathcal{T}^M{}_M \quad (M \neq x_4, \text{ no sum}),$$
$$\mathcal{T}^M{}_N = \mathrm{diag}\big(p_0,\ p_3,\ p_3,\ p_3,\ -\rho,\ p_5,\ p_5,\ p_5\big) \quad \text{(a diagonal source)}, \qquad \bar{p} = \frac{p_0 + 3p_5}{4} \tag{FE-2}$$

Symbols in (FE-2).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$, $\Delta S_{\rm m}$ the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- $S_{\rm m}$: The matter action: all terms of the action except $S_{\rm EL}$ (the stabiliser included). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; X^M is the coordinate with the index M, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and X^μ (four-dimensional index μ) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; g^{MN} is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{T}^M{}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- p: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, p_s the pressure of the species s; Δp the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.

through the adiabatic response

$$\Delta K = \frac{c_\rho\,\kappa\Delta\rho + c_3\,\kappa\Delta p_3 + c_p\,\kappa\Delta\bar{p}}{H^2}, \qquad \Delta\ln f = -\frac{\Delta K}{2K_*}, \qquad c_3 = -3c_\rho, \qquad \Delta\dot{\mathcal{H}} = -\frac{\kappa(\Delta\rho + \Delta p_3)}{2\mathcal{G}(K_*)} \tag{FE-47}$$

Symbols in (FE-47).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$, $\Delta S_{\rm m}$ the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.

- c_ρ: The coefficient of $\kappa\Delta\rho$ in the adiabatic block response ΔK. Unit: dimensionless. Value: 0.0358521. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- c_3: The coefficient of $\kappa\Delta p_3$ in the adiabatic block response, $c_3 = -3c_\rho$ (so radiation drops out). Unit: dimensionless. Value: -0.107556. Status: DERIVED.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- c_p: The coefficient of $\kappa\Delta\bar{p}$ in the adiabatic block response. Unit: dimensionless. Value: -0.140028. Status: DERIVED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV^8 (or GeV^4 for a four-dimensional density). Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_W and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- f: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is f^2 times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV^2. Status: DERIVED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.

while the difference $p_0 - p_5$ would drive a change of the shape of the block that the order-0 metric cannot describe (OPEN since version 3, corrections item 14). For a Kohn-Sham state

$$\begin{aligned}
&\rho = \sum_o \tfrac{1}{2}\,d_o\phi_o\,\varepsilon_o\,n_o - e_\mathrm{int}, \qquad p_3 = \sum_o \tfrac{1}{2}\,d_o\phi_o\,\frac{\mathrm{e}^{-a_4}|\vec{k}_o|}{3}\,n_o^\mathrm{t} + e_\mathrm{int}, \qquad p_5 = e_\mathrm{int},\\
&p_0 = \sum_o \tfrac{1}{2}\,d_o\phi_o\Big[(\varepsilon_o - v)\,n_o - M_\mathrm{eff}\,n_o^\mathrm{sc} - \mathrm{e}^{-a_4}|\vec{k}_o|\,n_o^\mathrm{t}\Big] + e_\mathrm{int}
\end{aligned} \tag{FE-67}$$

Symbols in (FE-67).

- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- d_o, d_s: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital o; d_s the number of internal states (spin, colour, particle and antiparticle) of the species s. Unit: dimensionless. Status: STANDARD.
- ϕ_o: The occupation of the Kohn-Sham orbital o (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- ε, ε_o: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); ε_o the Kohn-Sham level of the orbital o. Unit: GeV. Status: DERIVED.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino

mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.

- e_{int}: The interaction energy density of the contact term in the Kohn-Sham energy and energy-momentum tensor. Unit: GeV^8. Status: RECORD.
- p_3: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV^8. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: the scale factor of cosmology is e^{a_4}. Unit: dimensionless. Status: AUTHOR.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- n_o^{t}: The density of the orbital o along the momentum term ($\chi_o^\dagger \varsigma\sigma_3\chi_o$ times the weight $\mathrm{e}^{-3Hy}/\mathrm{Vol}_7$); it carries the 3-space pressure. Unit: GeV^7. Status: RECORD.
- p_5: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV^8. Status: STANDARD.
- p_0: The pressure along the hidden direction x_0, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV^8. Status: STANDARD.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along γ^{x_4}): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $M_{\mathrm{eff}}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: m_Ψ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- n^{sc}, n_o^{sc}: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume) and n_o^{sc} the contribution of the orbital o. Unit: GeV^7. Status: RECORD.

and for dark matter at rest without self-interaction ($e_{\mathrm{int}} = 0$, so $p_5 = 0$ and $\bar p = p_0/4$) the neglected source is 4.00 times the kept one; for the self-interacting test states it is 0.90 times. Its absolute size is suppressed like that of $\bar p$ ($\kappa p/\kappa\rho_{\mathrm{stab}} \le 3\times10^{-32}$ at the freeze-out), provided the shape mode is stable, which remains OPEN.

A rounding note. The response of the four-dimensional cosmological term to a shift of the block vanishes at the stabilised block,

$$\frac{\mathrm{d}}{\mathrm{d}K}\Big[6H^2K\big(1-2w_{(2)}K\big)+\kappa\rho_{\mathrm{stab}}\Big]\Big|_{K=K_*}=0$$

Symbols in this equation.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- K: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- ρ_{stab}: The energy density of the block stabiliser, a function of the block scale, $\rho_{\mathrm{stab}} \propto f^{-8}$; its value at the static block is $\rho_{\mathrm{stab}}(f_*)$. Unit: GeV^8. Value: $\kappa\rho_{\mathrm{stab}}(f_*) = (26+4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- K_*: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root K_+). Unit: dimensionless. Value: 21.5219. Status: DERIVED.

exactly (with $\kappa\rho_{\mathrm{stab}}$ proportional to K^4); the stored value -6.3×10^{-12} (`ks/feedback.json`, check `friedmann_decoupled_from_block_shift`) is the rounding of the 12-digit K_* and $\kappa\rho_{\mathrm{stab}}(f_*)$.

10.3 The precision fixes of version 4 (commit `cc2d514`)

The residual analysis found five places where a stored number was limited by the numerics, not by the physics. Each was fixed and the affected scripts rerun; every script exits with code 0 with all its checks.

1. The response of the W eigenvalue to the dark matter, by first-order perturbation theory. Through I12 the scalar density of the dark matter changes the gauge kinetic term along x_0:

$$\frac{\Delta\alpha_3}{\alpha_3} = -4G_{\mathrm{g}}\, n_{\mathrm{DM,tot}}, \qquad \frac{\Delta\alpha_2}{\alpha_2} = -4G_{\mathrm{W}}\, n_{\mathrm{DM,tot}}, \qquad \frac{\Delta\alpha_1}{\alpha_1} = -4G_{\mathrm{B}}\, n_{\mathrm{DM,tot}},$$
$$-\frac{1}{\Upsilon\, \mathrm{e}^{3HL\xi}}\frac{\mathrm{d}}{\mathrm{d}\xi}\Big(\Upsilon\, \mathrm{e}^{3HL\xi}\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big) = \tilde{m}^2 F, \qquad \Upsilon(\xi) = 1 + A_{\mathrm{mf}}\, n^{\mathrm{sc}}_{\mathrm{DM}}, \qquad n^{\mathrm{sc}}_{\mathrm{DM}} = 2a_{\mathrm{DM}}b_{\mathrm{DM}}, \tag{FE-79}$$
$$\frac{\mathrm{d}F}{\mathrm{d}\xi}(0) = 0, \qquad \Upsilon(-1)\,\frac{\mathrm{d}F}{\mathrm{d}\xi}(-1) = \beta_{\mathrm{W}} F(-1)$$

Symbols in (FE-79).

- Δ: The change (difference or small perturbation) of the quantity that follows: ΔK, $\Delta\rho$, Δp_3 are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, ΔN_{eff} an addition to N_{eff}, ΔS_{m} the change of an action under a change Δg^{MN} of the metric. Status: STANDARD.
- α_3: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\mathrm{Z}}) = 0.1180$; $\alpha_3(m_{\mathrm{KK}}) = 0.0606$. Status: STANDARD.
- G_{g}: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV^{-3}. Value: 6.09×10^{-15} GeV^{-3} (benchmark). Status: DERIVED.
- $n_{\mathrm{DM,tot}}$: The total number density of dark-matter quanta: both species (the lowest odd levels of Ψ_A and Ψ_B), particles and antiparticles, $n_{\mathrm{DM,tot}} = 2(n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}})$. Unit: GeV^3. Status: STANDARD.
- α_2: The SU(2) fine-structure constant, $\alpha_2 = g_2^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- G_{W}: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: GeV^{-3}. Value: 2.96×10^{-15} GeV^{-3}. Status: DERIVED.
- α_1: The hypercharge fine-structure constant, $\alpha_1 = g_1^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- G_{B}: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: GeV^{-3}. Value: 1.08×10^{-15} GeV^{-3}. Status: DERIVED.
- $\Upsilon(\xi)$: The factor by which the scalar density of the dark matter changes the gauge kinetic term along x_0 (I12), $\Upsilon = 1 + A_{\mathrm{mf}} n^{\mathrm{sc}}_{\mathrm{DM}}$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- A_{mf}: The amplitude of the I12 mean field in the x_0 problem of the gauge fields (a bookkeeping parameter: the result is linear in it). Unit: GeV^{-1} (with the orbital normalisation). Status: DERIVED.
- $n^{\mathrm{sc}}_{\mathrm{DM}}$: The scalar density of the dark-matter orbital, $n^{\mathrm{sc}}_{\mathrm{DM}} = 2a_{\mathrm{DM}}b_{\mathrm{DM}}$. Unit: GeV (normalised orbital). Status: DERIVED.
- a_{DM}, b_{DM}: The upper and lower components of the dark-matter orbital (block spinor $(a, \mathrm{i}b)$, normalised on one half). Unit: $\mathrm{GeV}^{1/2}$. Status: DERIVED.
- β_{W}: The Robin parameter of W^+ and W^- (set by the tip mass term of the sigma field). Unit: dimensionless. Value: 0.011912 (fitted). Status: DERIVED.

With the weight $\Upsilon\,\mathrm{e}^{3HL\xi}$ on both sides, the eigenvalue is a Rayleigh quotient whose tip term does not depend on A_{mf}, so its derivative at $A_{\mathrm{mf}} = 0$ is, exactly,

$$\left.\frac{\mathrm{d}\tilde{m}^2}{\mathrm{d}A_{\mathrm{mf}}}\right|_{A_{\mathrm{mf}}=0} = \frac{\displaystyle\int_{-1}^{0} n_{\mathrm{DM}}^{\mathrm{sc}}\,\mathrm{e}^{3HL\xi}\Big(\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big)^2\,\mathrm{d}\xi - \tilde{m}^2\int_{-1}^{0} n_{\mathrm{DM}}^{\mathrm{sc}}\,\mathrm{e}^{3HL\xi}F^2\,\mathrm{d}\xi}{\displaystyle\int_{-1}^{0}\mathrm{e}^{3HL\xi}F^2\,\mathrm{d}\xi}$$

Symbols in this equation.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless x_0 eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the n-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- A_{mf}: The amplitude of the I12 mean field in the x_0 problem of the gauge fields (a bookkeeping parameter: the result is linear in it). Unit: GeV^{-1} (with the orbital normalisation). Status: DERIVED.
- $n_{\mathrm{DM}}^{\mathrm{sc}}$: The scalar density of the dark-matter orbital, $n_{\mathrm{DM}}^{\mathrm{sc}} = 2a_{\mathrm{DM}}b_{\mathrm{DM}}$. Unit: GeV (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.

evaluated with the exact W eigenfunction of the fold condition (its tip condition holds with the stored $\tilde{m}_{\mathrm{W}}^2$ to 6×10^{-12})

$$F(\xi) = \mathrm{e}^{-3HL\xi/2}\Big(\cosh\varkappa_{\mathrm{W}}\xi + \frac{3HL}{2\varkappa_{\mathrm{W}}}\sinh\varkappa_{\mathrm{W}}\xi\Big), \qquad \frac{\mathrm{d}F}{\mathrm{d}\xi} = -\frac{\tilde{m}_{\mathrm{W}}^2}{\varkappa_{\mathrm{W}}}\,\mathrm{e}^{-3HL\xi/2}\sinh\varkappa_{\mathrm{W}}\xi,$$
$$\varkappa_{\mathrm{W}} = \Big(\frac{9H^2L^2}{4} - \tilde{m}_{\mathrm{W}}^2\Big)^{1/2}$$

Symbols in this equation.

- $F(\xi)$, F_n: The x_0 profile of a boson mode on one half of the period (a function of y or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; F_n is the profile of the n-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of m_{W} and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- L: The depth of the regular tip: each half of a period of x_0 ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- ξ: The scaled hidden coordinate, $\xi = y/L$, running from -1 (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\varkappa_{\mathrm{W}}$: The decay constant of the exact x_0 profile of W^+ and W^-, whose eigenvalue lies below the gap $9H^2L^2/4$: $\varkappa_{\mathrm{W}} = (9H^2L^2/4 - \tilde{m}_{\mathrm{W}}^2)^{1/2}$ (for such a profile the wave number ϖ of the oscillating profiles is imaginary, $\varpi = \mathrm{i}\varkappa_{\mathrm{W}}$). Unit: dimensionless. Value: 4.4999985 ($HL = 3$). Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.

- $\tilde{m}_{\mathrm{W}}$: The dimensionless x_0 eigenvalue of the lowest $W^{\pm}$ mode ($\beta = \beta_{\mathrm{W}}$). Unit: dimensionless. Value: $\tilde{m}_{\mathrm{W}}^2 = 1.32149 \times 10^{-5}$. Status: DERIVED.

and the dark-matter orbital on the 800 Gauss nodes of the Kohn-Sham cell. Result: $\mathrm{d}\ln\tilde{m}_{\mathrm{W}}^2/\mathrm{d}A_{\mathrm{mf}} = -0.111252909326$ (`coupled.json`, key `WZ_kinetic_shift/dlnLambdaW_damp`), confirmed by shooting the perturbed problem (DOP853, Richardson in the amplitude) to 3×10^{-11}. Version 3 quoted -0.1126 from finite differences of finite-element eigenvalues, which are dominated by rounding. For the photon and the gluons the flat profile $F = 1$ solves the problem for every Υ, so their eigenvalue stays exactly zero (version 3: -2.5×10^{-11}, rounding). The physical shift stays negligible: at the freeze-out, where the amplitude is largest ($A_{\mathrm{mf}} = 1.7 \times 10^{-16}$), $\tilde{m}_{\mathrm{W}}^2$ moves by 1.9×10^{-17} of itself.

2. A robust integration of the right-handed neutrinos of the $B - L$ alternative. The counterfactual of FE-131,

$$
\begin{aligned}
&G_{\mathrm{A}}^{B-L} = G_{\mathrm{A}}\Big[\frac{\mathcal{S}_B}{\mathcal{S}_{B-L}}\Big]^{1/2}, \qquad \mathcal{S}_B = \sum_f N_{\mathrm{c}}\big(q_f^B \tilde{I}_{of}^{(b)}\big)^2, \qquad \mathcal{S}_{B-L} = \sum_f N_{\mathrm{c}}\big(q_f \tilde{I}_{of}^{(b)}\big)^2, \\
&\frac{\mathrm{d}n_{\nu_R}}{\mathrm{d}x_4} + 3\mathcal{H}n_{\nu_R} = \Gamma_\nu\Big(1 - \frac{n_{\nu_R}^2}{(n_{\nu_R}^{\mathrm{eq}})^2}\Big), \qquad \Gamma_\nu = 2\, n_{\mathrm{DM,eq}}\, \bar{n}_{\mathrm{DM,eq}}\, \frac{G_\nu^2 M_{\mathrm{DM}}^2}{2\pi}, \qquad G_\nu = G_{\mathrm{A}}^{B-L}\, \tilde{I}_{\nu_R}^{(b)}, \qquad \text{(FE-131)} \\
&n_{\nu_R}^{\mathrm{eq}} = \frac{3}{4}\,\frac{\zeta(3)}{\pi^2}\,T^3, \qquad n_{\mathrm{DM,eq}} = \bar{n}_{\mathrm{DM,eq}} = 2\Big(\frac{M_{\mathrm{DM}}T}{2\pi}\Big)^{3/2}\mathrm{e}^{-x}
\end{aligned}
$$

Symbols in (FE-131).

- G_{A}^{B-L}: The strength I13 would need if it coupled to the $B - L$ current (the same relic density). Unit: GeV^{-2}. Status: DERIVED.
- G_{A}: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}}H/(2f^4 v_{\mathrm{t}}\Lambda_\star^6)$. Unit: GeV^{-2}. Value: 1.278×10^{-9} GeV^{-2} (dark-matter relic density). Status: DERIVED.
- $\mathcal{S}_B$, $\mathcal{S}_{B-L}$: The colour-weighted sums of the squared I13 overlaps of the dark matter with the baryon-number and with the $B - L$ charges. Unit: dimensionless. Value: $\mathcal{S}_B = 16.351$, $\mathcal{S}_{B-L} = 22.92$. Status: DERIVED.
- f: The label of a fermion species: a Standard-Model multiplet of one generation (its N_{c} Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over f runs over these species. Status: STANDARD.
- N_{c}: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- q_ψ^B, q_f^B: The baryon number of the multiplet ψ or of the fermion component f ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state o with the zero mode of f, $\tilde{I}_{of}^{(b)} = H^{-1}\int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}_Q^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q^{(b)} = 2.6135$. Status: DERIVED.
- o: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over o runs over the orbitals. Status: RECORD.
- q_ψ, q_f: The $B - L$ charge: q_ψ of the multiplet ψ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), q_f of the fermion component f. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_{ν_R}, $n_{\nu_R}^{\mathrm{eq}}$: The number density of the right-handed neutrinos (three generations) and its equilibrium value. Unit: GeV^3. Status: DERIVED.
- x_4: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- Γ_ν: The production rate per volume of right-handed neutrinos in the counterfactual $B - L$ variant of I13. Unit: GeV^4. Status: DERIVED.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: GeV^3. Status: STANDARD.

- G_ν: The four-dimensional coupling of the dark matter to a right-handed neutrino in the $B-L$ variant of I13. Unit: GeV^{-2}. Status: DERIVED.
- M_{DM}: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ζ: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers n. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- T: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- x: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.

is integrated for the abundance n_{ν_R}/s from zero at $x = 2$. The equilibration from zero takes only about 7×10^{-11} in x (the equilibrium abundance divided by its initial rate of growth); to resolve its start the error control of the Radau method needs steps of about 7×10^{-11} times the square root of the relative tolerance (7×10^{-15} at the version-3 tolerance 10^{-8}), at or below the smallest step it allows at $x = 2$ (ten times the spacing of the floating-point numbers, 4.4×10^{-15}), so the integration in x could stop at its first step depending on the last digits of the inputs. It now runs in the shifted variable $x - 2$ (from 0 to 38), where the steps are free, with a relative tolerance of 10^{-13}: $n_{\nu_R}/n_{\nu_R}^{\mathrm{eq}} = 0.999697681992$ at $x = 40$ (`coupled.json`, `I13_B_minus_L_alternative/Y_nuR_over_Yeq_at_x40`), stable to 3×10^{-13} when the coupling is changed by $\pm 10^{-12}$ (check `R_right_handed_neutrino_boltzmann`). The version-3 statement (0.9997 of equilibrium, $\Delta N_{\mathrm{eff}} = 0.144$) is unchanged.

3. The integrals of the cosmic neutrino background to 3×10^{-16}. The integrals of

$$f_{\mathrm{FD}}(k) = \frac{1}{\mathrm{e}^{k/T_\nu}+1}, \qquad n_i = 2\int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\, f_{\mathrm{FD}} = \frac{3\zeta(3)}{2\pi^2}\, T_\nu^3, \qquad \rho_i = 2\int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\, \sqrt{k^2+m_i^2}\, f_{\mathrm{FD}},$$
$$\bar{v}_i = \frac{\int_0^\infty k^2\, \frac{k}{\sqrt{k^2+m_i^2}}\, f_{\mathrm{FD}}\, \mathrm{d}k}{\int_0^\infty k^2\, f_{\mathrm{FD}}\, \mathrm{d}k}, \qquad \text{flavour content of } \nu_i : \ |U_{\alpha i}|^2 \tag{FE-120}$$

Symbols in (FE-120).

- f_{FD}: The Fermi-Dirac distribution of the relic neutrinos, $f_{\mathrm{FD}} = 1/(\mathrm{e}^{k/T_\nu}+1)$. Unit: dimensionless. Status: STANDARD.
- k: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T_ν: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- n, n_o, $n_{\psi i}$, n_f, n_i: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and n_o the contribution of the orbital o, $n_{\psi i}(y)$ the density of the Standard-Model fermion ψ_i (per proper 7-volume of eight dimensions); n_f the density of the fermion component f and n_i that of the neutrino mass state i (per 3-volume of four dimensions). Unit: GeV^7 (eight dimensions), GeV^3 (four dimensions). Status: STANDARD.
- i: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital o, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.

- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ζ: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers n. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- ρ: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; ρ_s of the species s, ρ_i of the neutrino mass state i); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV^8 (eight dimensions), GeV^4 (four dimensions). Status: STANDARD.
- m, m_s, m_i: A four-dimensional mass: m the mass of a mode (the eigenvalue of an x_0 problem) or of a fermion; m_s the mass of the species s; m_i the mass of the neutrino mass state i. Unit: GeV. Status: DERIVED.
- $\bar{v}_i$: The mean velocity of the relic neutrinos of the mass state i. Unit: dimensionless ($c = 1$). Value: 0.435, 0.0604, 0.0105 today. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- ν_α, ν_i: A left-handed neutrino: ν_α of the flavour $\alpha = e, \mu, \tau$ (ν_e, ν_μ, ν_τ), ν_i of the mass state i (mass m_i). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- U: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states i); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212°$ (NuFIT 6.0). Status: STANDARD.
- α: A flavour index running over e, μ, τ (the flavour states of the leptons and neutrinos). Status: STANDARD.

are of order $T_\nu^4 \approx 10^{-15}$ eV^4, far below the default absolute tolerance of the adaptive quadrature, which therefore stopped at a relative error of 5.8×10^{-7}; with a relative tolerance only they are accurate to 3×10^{-16}: $\Omega_\nu h^2_{100} = 6.38217282098 \times 10^{-4}$; mean velocities 0.435132624815, 0.0604083086104, 0.0105351736626; energy densities $9.95804523965 \times 10^{-16}$, $7.51619688306 \times 10^{-15}$, $4.31573842483 \times 10^{-14}$ eV^4 (`neutrino.json`, `CNB`, copied in `cosmology.json`). Every number quoted in version 3 is unchanged at its quoted precision.

4. Deterministic thread counts in the Kohn-Sham scripts. The number of threads of the linear algebra (BLAS, LAPACK, OpenMP) is now fixed to one in `um2_kohnsham.py`, `um2_ks_jobs.py`, `um2_coupled.py` and `um2_residuals.py`, before numpy is imported: with two threads the energy of the 8-particle state with λ_{test} had differed by $5 \times 10^{-13}\,H$, and the fresh-clone runs of versions 1 to 3 had found one Kohn-Sham integral differing by 1.1×10^{-12} relative because threaded sums are not taken in a fixed order. The rerun changed only quantities at the rounding level, among them quantities that are zero in exact arithmetic: the changes of the fold-band gap per unit amplitude, 7.9×10^{-11} (electron density), 3.0×10^{-9} (up-quark density), -9.3×10^{-11} (neutrino density), -1.4×10^{-10} (gluon condensate) and 1.2×10^{-13} (the I13 shift of the fold band); and the observed order of the Kohn-Sham energy of the stress state, now 4.956 (version 3: 4.95).

5. A stale check text. The text of the check `tip_Z_portal_dropped` of `um2_electroweak.py` still said "$c_{\mathrm{V}} = 0$ for I11" (version 2); it now states that I11 is kept with the freeze-in strength $G_{\mathrm{V}} = 4.38 \times 10^{-17}$ GeV^{-2}, at which the fold bands are produced only by freeze-in ($\Delta N_{\mathrm{eff}} = 0.001$). No number changed.

10.4 What changed in the claims

No claim of version 3 changes; the status labels are those of version 3, and the conflicts with data stand (no Higgs boson; the unitarity of W^+W^- scattering near 1.2 TeV). Numbers changed by the precision fixes: $\mathrm{d}\ln\tilde{m}_{\mathrm{W}}^2/\mathrm{d}A_{\mathrm{mf}} = -0.11125$ (was -0.1126), the photon's value exactly zero (was -2.5×10^{-11}), the observed order 4.96 (was 4.95), the rounding-level values listed in fix 4, and further digits of the CNB. Statements made precise by the residual analysis: the two values of the weak angle and their 0.83 % effect on the Z width; the physics accuracy of about 1 % of G_{A}; the block-shape source, now quantified (OPEN); the fold bands missing from the Hubble rate of the thermal history (1.7×10^{-4}); the instantaneous neutrino decoupling of the thermal history (0.59 % in g_ρ today, without effect on a result).

11 The verification record

Version 1 was rerun from fresh clones of the repository on 2026-10-17 at commit `b3c3c1a` (179 of 179 checks) and on 2026-10-18 at the commits `fa7305f`, `9285bb6` (180 of 180) and `5b73d16` (182 of 182; every computation as a separate step, the longest 247 s); version 2 on 2026-10-18 at commit `4a6421d` (205 of 205; every computation as

a separate step, 1027 s in all); version 3 on 2026-10-18 at commit `726997f` (233 of 233; every computation as a separate step, 1100 s in all, the longest the primordial integration with 269 s) and again, after the fixes of its independent review, at commit `815db9e` (235 of 235; 1116 s in all, the primordial integration 269 s). In every run every check passed and every stored number was reproduced to a relative 10^{-9} or better; in the runs from `9285bb6` on all numbers were identical except the run times, the texts of check details and one Kohn-Sham integral, which differed by 1.1×10^{-12} relative because the threaded linear algebra did not sum in a fixed order (version 4 fixes the thread count, section 10); and the LaTeX sources, the PDF documents and the figures were rebuilt byte-identical. Between the two runs of version 2, `um2_analysis.py` gained the ${}^3P_0$ unitarity bound of item 22 (new stored numbers, the same 23 checks). After the run of version 3 at `726997f` the fixes of its independent review (item 24) changed `um2_thermal.py`, `um2_coupled.py`, `um2_series.py`, `um2_cosmology.py` and the `feedback` job of `um2_ks_jobs.py`, which were rerun in dependency order before the run at `815db9e`.

Version 4: the precision fixes of commit `cc2d514` changed `um2_coupled.py`, `um2_electroweak.py`, `um2_kohnsham.py`, `um2_ks_jobs.py` and `um2_neutrino.py`; the affected computations (the seven Kohn-Sham jobs, `um2_neutrino.py`, `um2_electroweak.py`, `um2_cosmology.py`, `um2_coupled.py`) and the new `um2_residuals.py` were rerun, and every script exits with code 0 with all its checks: 287 of 287 (the reports in `universe_model_2/reports`). A rerun of version 4 from a fresh clone is not part of this record.

12 Corrections made during the work

1. The gauge-invariance test of the tip terms first used the wrong sign of the hypercharge phase; with $\psi \to \mathrm{e}^{\mathrm{i}Y\vartheta}\psi$ every term is invariant to 4×10^{-15}.
2. The finite-element cross-check of $\tilde{m}_{\mathrm{W}}^2$ was first limited by rounding; the Rayleigh quotient of the gradient form with Richardson extrapolation (observed order 4.01) agrees with the exact value to 3×10^{-14}.
3. The Kohn-Sham basis convergence and the conservation law were first checked without extrapolation; Richardson in the basis size (observed orders 4.95 for the energy, 2.97 for the hidden pressure; 4.96 for the energy since the deterministic thread count of version 4) was added.
4. The self-consistency loop did not converge for 592 particles at the record's coupling; a fixed labelling of the occupied set and a safeguarded Anderson mixing were introduced, the test coupling λ_{test} was adopted, and the record's coupling is kept as a stress test, which now converges.
5. A first statement that every primordial start above the unstable static block reaches the stable one was false in detail; the separatrix (the threshold of the initial K as a function of the initial $\mathcal{H}_f$) was computed by bisection instead.
6. The primordial figure was first sampled too coarsely (aliasing of the radion oscillation) and is now densely sampled.
7. The sum rules of the uncertainty analysis were first imprecise for the sharp fermion zero modes; exact mpmath normalisation and the tip term fixed them (10^{-8}).
8. The radion quantity was stored under a wrong sign label; it is now stored as ω_{r}^2, the determinant of the Jacobian of the block equations.
9. Portal names of the first universe model (P1 to P6) in two check texts were replaced by I9, I11, I12.
10. The gauge-kinetic portal was first written with the scale to the power 5, which is dimensionally wrong in the geometric normalisation of the gauge fields; it is $\bar{\Psi}\Psi\,\mathcal{F}/\Lambda_\star^3$ (IN-17), and the symbol of the portal scale was changed so that it no longer clashes with the Dirichlet bound of the tip.
11. Before publication every number of the documents was compared with the computed outputs; four statements were corrected (the dark-matter wave number $\varpi = 1.89901$, the finite-element agreement 5×10^{-10}, the TT and EE agreements of version 1, and the restriction $\mathcal{H} = 0$ of the sourceless static block).
12. The kinetic term of the tip sigma field was first written with the sign of the signature $(+,-,-,-)$; in the signature $(+,+,+,-)$ of $x_1, \ldots, x_4$ it is $-\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big)$, which gives mass terms $-\frac{1}{2}$ times the squared mass times the square of the vector field with a positive squared mass (the computations always used positive squared masses).
13. With the real matrices $C\gamma^a$ antisymmetric, $\bar{\Psi}\gamma^a\Psi$ is anti-Hermitian; the portal currents are now written as the Hermitian currents $\mathrm{i}\bar{\Psi}\gamma^M\Psi$ and the Standard-Model currents as $\mathrm{i}\bar{\psi}\gamma^M\psi$.
14. The block response to the hidden pressure of a Kohn-Sham state first used p_0 alone; the common block scale is driven by the block-averaged pressure $\bar{p} = (p_0 + 3p_5)/4$, and $p_0 - p_5$ drives a block-shape mode outside the order-0 metric (OPEN; quantified in version 4, section 10). The linearised sourced equations were added (`um2_background.py`, 2 checks); they first omitted the density, which enters at first order through

$3\mathcal{G}(K_*)\mathcal{H}^2 = \kappa\Delta\rho$; with it the adiabatic response reproduces the four-dimensional Friedmann acceleration equation exactly.

15. The statement that the numbers of quanta of Ψ_A and Ψ_B are separately conserved was first recorded as an asserted check; it is now computed with the author's matrices (commutators of $\mathcal{B}$ with C, γ^a, $C\gamma^a$, $\mathcal{S}^{ab}$), which also shows that extra-time momenta, outside the good sector, would mix the halves.
16. The radion lifetime was first quoted as an unverified estimate; it is now computed (ESTIMATE, with the coupling factor C_r explicit), as is the bound on the block rate today.
17. Figures: the BAO legend showed the data label twice and two labels of the uncertainty figure overlapped; the condensate figure no longer draws the vertical lines where its equation of state diverges.
18. An independent review of the five documents of version 1 against the outputs found, and the documents were corrected: the first-order fermion mass formula (it needs $\mathcal{Z}(-c_R)$ for the singlet, $c_R < 0$; the code always used the absolute value); the gap statement, which holds for the boson operators but not for Ψ (the dark matter, 44.35 TeV, lies below the boson gap of 99.5 TeV); the sensitivity of G_N to the block, which is $\mathrm{d}\ln G_\mathrm{N}/\mathrm{d}\ln f = -4 + 2K_*(\mathrm{d}\mathcal{G}/\mathrm{d}K)/\mathcal{G}$ at K_*, $= +2.05$, and not -4 (recomputed; the lunar-ranging bound becomes $7.1\times10^{-4}\mathcal{H}_0$); ΔN_eff of a thermalised fold band (recomputed); a level spacing (0.25000); a rounding (1.948); the definitions of $\bar{M}_\mathrm{Pl}$, M_Pl, the radion formulas, the spread $\Delta(fy)$ of the proper hidden length, the unitarity estimate and the cosmic-variance measure; the statement on proper and coordinate localisation; and several notational clashes (a letter then used for e^{-a_4} is now written out).
19. A second review of version 1 found, and the documents were corrected: the electroweak parameter was written with the measured masses, $m_\mathrm{W}^2/(m_\mathrm{Z}^2\cos^2\theta_\mathrm{W})$, next to the on-shell definition $\cos^2\theta_\mathrm{W} = m_\mathrm{W}^2/m_\mathrm{Z}^2$, which makes it identically 1; the computation always used $\rho_\mathrm{EW} = \tilde{m}_\mathrm{W}^2/(\tilde{m}_\mathrm{Z}^2\cos^2\theta_\mathrm{W})$. Checking the uncertainty relation for every state then showed that the bound $\langle k_\xi^2\rangle \geq 1/(4\Delta\xi^2)$ quoted for the hidden kinetic energy is not valid on the cell: the tip, where the cell ends, adds a boundary term, and the valid relation is $\langle k_\xi^2\rangle^{1/2}\Delta\xi \geq (1-\varrho_\xi(-1))/2$ (the broad fold zero mode of Ψ has $\langle k_\xi^2\rangle\Delta\xi^2 = 0.097$); a computed check of it was added. Further: the $\overline{\mathrm{MS}}$ angle in the Z widths, the shooting accuracy, the normalisation volume Vol_7 (not the proper seven-volume), the occupations ϕ_o, the tip coupling matrices, and the notation (with a list of the symbols with more than one use in each document; version 4 replaces these lists by one meaning per symbol).
20. A third review of version 1 found, and the computations and documents were corrected: (a) the $W^\pm$, Z fit set the angle of the gauge couplings equal to the on-shell angle of the measured masses and fitted the tree-level $\rho_\mathrm{EW} - 1$ to $\rho_{\mathrm{EW},0} - 1$, which put the model's $m_\mathrm{Z} = 91.1706$ GeV 17 MeV below the measured value; since $\rho_{\mathrm{EW},0}$ measures the custodial violation beyond the Standard-Model loops, the angle of the couplings is now fixed by the measured masses, $\cos^2\theta_\mathrm{W} = (m_\mathrm{W}^2/m_\mathrm{Z}^2)/\rho_{\mathrm{EW},0} = 0.776368$, and the model reproduces both masses (check `mZ_reproduced`); $\beta_\mathrm{W} = 0.011912$ and every scale in GeV moved by $+8.50\times10^{-4}$ ($m_\mathrm{KK} = 121.32$ TeV, $H = 14.296$ TeV, gap 99.48 TeV, radion 48.42 TeV, $M_\mathrm{DM} = 44.35$ TeV). The tree-level Z widths with the angle of the couplings were added (check `Z_width_scheme`). (b) The familiar bound $\langle k_\xi^2\rangle \geq 1/(4\Delta\xi^2)$ is guaranteed only for profiles negligible at the tip. (c) Symbol lists and wording.
21. **Version 2 (2026-10-18): three generations and three neutrino flavours.** By the author's orders of 2026-10-18 (option A): (a) three generations of quarks and leptons with one doublet bulk mass per sector, the twelve masses as exact x_0 eigenvalues and the PMNS and CKM matrices as the tip matrices, proved exact for three generations (`um2_flavour.py`, new; `um2_spectra.py`, `um2_fields.py` extended); (b) the flavour conversion of the three neutrinos (`um2_neutrino.py`, new): vacuum, matter, accelerators, the Sun, the early Universe and N_eff, the CNB of three mass states, supernovae; (c) the chiral tip Z portal I9 is removed: with three neutrinos it has no role left, its even part is bounded by the invisible width ($N_\nu = 3 + 2r_\mathrm{Z}^2$, $r_\mathrm{Z}^2 < 0.0056$) and would still keep the fold band in equilibrium to 14.7 MeV ($\Delta N_\mathrm{eff} = 2.02$), and its oriented part would thermalise the fold band through the decays of the odd Ψ_A states ($\Delta N_\mathrm{eff} = 0.0958$ to 0.1257, 2.1 to 2.4 standard deviations above ACT DR6); consequently both halves of Ψ keep an exact fold parity and the dark matter consists of the lightest odd states of both halves (two species, relic recomputed: $x_\mathrm{f} = 30.38$, $G_\mathrm{I12} = (15.5\ \mathrm{TeV})^{-3}$); (d) the invisible width uses the PDG 2025 value $N_\nu = 2.9963 \pm 0.0074$ instead of the LEP 2006 value; (e) N_eff is computed for three flavours (3.0476 to 3.0485) and the precision value 3.0432 (Cielo et al.) replaces 3.044; (f) the cosmology has three massive neutrinos ($\sum_i m_i = 0.0599$ eV) instead of one of 0.05 eV and no dark radiation; (g) the Z widths of three generations (2485.2 MeV with the QCD factor) remove the one-generation conflict (1232 MeV); (h) the Kohn-Sham portal sources use the three-generation localisations. Errors found and corrected during the version-2 work: the 1-3 MSW resonance energy was first evaluated with a spurious factor $1/c_{13}^2$ (11.49 GeV); it is $E = \Delta m_{ee}^2\cos 2\theta_{13}/(2V_\mathrm{CC}) = 11.135$ GeV with $\Delta m_{ee}^2 = c_{12}^2\Delta m_{31}^2 + s_{12}^2\Delta m_{32}^2$, confirmed by the exact three-flavour root 11.133 GeV (now a check);

CAMB's built-in normal-hierarchy option put the mass in one mass state, so the three mass states are now set up explicitly with the iteration of section 9; a check of the analysis (fermions below the uncertainty estimate) first used a threshold that the muon and the strange quark exceed and now tests the documented statement ($m/m_{\mathrm{Heis}} < 0.01$ for every fermion); in the documents the muon eigenvalue ($\tilde{\varepsilon} = 4.7795 \times 10^{-6}$), the Landau-Zener bound ($\mathcal{P}_{\mathrm{LZ}} \leq 2.6 \times 10^{-77}$) and the explanation of $\Omega_\nu h^2_{100}$ (the non-relativistic coefficient 94.06 eV at the instantaneous-decoupling temperature, 93.14 eV with the heating) were corrected; the published values of the decoupling method were read from the source (Escudero 2020, Table 1) and the separate-temperature value 3.048 added to the validation.

22. **Independent review of version 2 (2026-10-18).** A review of the five documents of version 2 against the outputs found every computed number in agreement except three roundings, and found the following, all corrected in the documents: (a) the dark-matter relic: the operator $\bar{\Psi}_{\mathrm{DM}}\Psi_{\mathrm{DM}}$ times a squared field strength annihilates a pair only from the state ${}^3P_0$, so the unitarity bound on the spin-averaged cross-section is $\sigma \leq \pi/(4k^2_{\mathrm{cm}})$ and $\langle \sigma v_{\mathrm{rel}} \rangle \leq \sqrt{\pi x_{\mathrm{f}}}/M^2_{\mathrm{DM}} = 5.79 \times 10^{-26}$ cm^3/s; the freeze-out requirement 2.01×10^{-25} cm^3/s exceeds it 3.47 times (the documents had compared it with $(4\pi/M^2_{\mathrm{DM}})\sqrt{x_{\mathrm{f}}/\pi} = 2.32 \times 10^{-25}$ cm^3/s, the bound of one helicity channel without the spin average, and called the requirement 87 % of it); the relic abundance of the dark matter is therefore OPEN in version 2, and the check `B3_relic_requirement_and_unitarity` of `um2_analysis.py` now computes the bound, its numerical thermal average, the crossing velocity $v_{\mathrm{c}} = 0.263$ and the fraction (79 %) of pairs above it; (b) "no tip operator reaches Ψ_B" holds only for chiral tip currents with P_L and for operators built on $\bar{\Psi}\Psi$ (a non-chiral tip current would reach Ψ_B); (c) the fast-flavour dispersion relation is written with the shifts by the moments of the electron-lepton-number distribution (now $\mathcal{I}_0$, $\mathcal{I}_1$) and the stated metric diag$(1, -1)$ on (x_4, x_3); (d) $c_{\mathrm{Z}} = \tilde{c}_{\mathrm{Z}} = 0$ is a modelling choice (ASSUMED): the invisible width bounds the even part and N_{eff} disfavours, but does not exclude, the oriented part; (e) the coordinate rule: the flavour, decoupling and fast-flavour equations had written a letter t for x_4 and z for the symmetry axis x_3; (f) roundings: the adiabatic solar $\mathcal{P}_{ee}$ at 15 MeV (0.306), the bulk mass of the tau singlet ($-9.205H$), the Kohn-Sham accuracy (2.2×10^{-9} relative); (g) minor items: the largest possible J_{CP} computed from the best-fit angles (0.0335) against NuFIT's marginalised value (0.0333 ± 0.0007); the conventions of the supernova resonances (two-flavour conditions); the tip matrices equal the mixing matrices only with $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$ (ASSUMED); the three roles of the measured values; the sum rules of the uncertainty analysis (absolute 1.0×10^{-8}, relative 7.8×10^{-4} and 6.0×10^{-4} for $W^{\pm}$ and Z); the 50-digit roots against the stored masses (4.5×10^{-11}); the labels ESTIMATE and STANDARD; the strength required for a single species ((17.6 TeV)$^{-3}$ with $g_\rho = 105.75$, not a version-1 value); further notational clashes added to the symbol lists; (h) self-containment: equations and inputs used but not written in a document (the relic and ΔN_{eff} formulas, the Einstein-Boltzmann equations, the BAO goodness of fit and data, the nonlinear background equations, the fermion equations with their boundary conditions, the decoupling equations, the CNB, Landau-Zener, adiabatic solar, condensate and pairing formulas, the Kohn-Sham equations behind the quoted equations of state) are now written out in every document that uses them.

23. **Version 3 (2026-10-18).** By the author's order for the coupled solution: (a) the computations `um2_thermal.py`, `um2_coupled.py` and `um2_series.py` (new), the matter power spectrum and the version-3 model in `um2_cosmology.py`, the Majorana alternative in `um2_neutrino.py`, and an optional potential on the lower component in the Kohn-Sham solver `um2_kohnsham.py` (the record's reference states reproduced again to 10^{-12}); (b) the portal I13 was first coupled to the $B - L$ current; the computation C9 then showed that the annihilation of the dark matter into right-handed neutrinos and its inverse would thermalise them and add $\Delta N_{\mathrm{eff}} = 0.144$ (2.5 standard deviations above ACT DR6), so I13 couples to the baryon-number current (G_{A}: $1.072 \to 1.278 \times 10^{-9}$ GeV^{-2}; xenon: 0.12 to 0.16 of LZ); (c) the first version of the C9 rates compared the production rate of three flavours with the equilibrium density of one; corrected (per flavour) before the result was recorded; (d) `um2_thermal.py` read a key of the analysis output that item 22 had renamed; corrected; (e) the early-Universe environments first put the whole net baryon density into protons; it is now shared by weak equilibrium, $n_{\mathrm{n}}/n_{\mathrm{p}} = \mathrm{e}^{-(m_{\mathrm{n}} - m_{\mathrm{p}})/T}$, and the QCD interaction measure had the wrong flavour thresholds in its perturbative branch and a special case at exactly 200 MeV (neither affected a stored number); (f) the text of the validity check C7 now refers to the benchmark strengths; (g) the thermal numerical result refines the version-2 statement on I12: a cross-section at the ${}^3P_0$ limit at every energy leaves $\Omega_{\mathrm{DM}} h^2_{100} = 0.058$ per species, so I12 alone could give the relic only as a strongly coupled channel at the unitarity limit; (h) in MODEL a cross-reference to the neutrino decoupling pointed to section 9.4 instead of 10.4 (since version 2).

24. **Independent review of version 3 (2026-10-18).** A review of the five documents of version 3 at commit `726997f` against the outputs (2055 s, 74 tool calls) found every version-3 number in agreement with the stored

outputs except the items below, and the following errors and omissions, all corrected in the computations and the documents (the affected scripts rerun; 235 checks, all passing): (a) the I13 bilinear $\mathrm{i}\bar{\Psi}\gamma^M P_b\Psi$ is Hermitian only for $M \neq x_0$, since the Hermitian conjugate of $\mathrm{i}C\gamma^{x_0}P_b$ is $\mathrm{i}C\gamma^{x_0}P_a$; the action now sums over $M = x_1, \ldots, x_7$, and the new check `C8_I13_algebra` verifies with the author's matrices the Hermiticity, its failure for x_0, $CP_bC^{-1} = P_a$, the Lorentz property and the structure under $\mathcal{B}$ ($C\gamma^M P_b$ commutes with $\mathcal{B}$ for $x_1, \ldots, x_4$ and anticommutes with it for x_5, x_6, x_7, whose currents vanish for spinors in one half, so the Standard-Model zero modes, all in the half $\mathcal{B} = -1$, are not mixed); no number changes, since the zero-mode reduction uses the four-dimensional components only; (b) the block response of the coupled solution kept only the hidden pressure of non-relativistic matter and only the $\Delta\bar{p}$ term of the adiabatic response; it now uses the full response (c_ρ, $c_3 = -3c_\rho$, c_p) and the hidden pressure $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$ of every species, the hot plasma included (ideal gases with the x_0 masses, the QCD interaction measure with the block dependence of the QCD scale, the order-α electroweak pressure with the block dependence of the couplings): $\Delta \ln f$ at the freeze-out $5.6 \times 10^{-45} \to 1.1 \times 10^{-34}$ (7×10^{-39} in the electroweak era, at most 9×10^{-49} from nucleosynthesis on), the loop gain $1.3 \times 10^{-44} \to 1.1 \times 10^{-33}$, the radion term of the metric series accordingly; the Kohn-Sham block feedback also reports the full response; the conclusion, a negligible back-reaction, is unchanged; (c) the thermal W, Z condensates had only the free massive-vector part; the order-α interaction part (FE-76; Feynman-Hellmann; for QCD it reproduces the trace-anomaly condensate to 10^{-16}, now part of check C4) dominates above m_W: the W condensate at the freeze-out is -5.0×10^{11} GeV^4; (d) the freeze-out environment used the version-2 $x_\mathrm{f} = 30.38$ ($T = 1.46$ TeV); it uses the version-3 $x_\mathrm{f} = 31.23$ ($T = 1.42$ TeV), and $\Delta\varepsilon_\mathrm{DM}/M_\mathrm{DM}$ there is -1.5×10^{-6} (was -1.6×10^{-6}); the degrees-of-freedom table lists 1.42 TeV instead of 1.46 TeV; (e) the thermal history did not state λ: the self-interaction contains the channel in which two dark-matter particles become two fold quanta through the transition scalar density $n^\mathrm{sc}_\mathrm{DF} = a_\mathrm{DM}b_\mathrm{fold} + b_\mathrm{DM}a_\mathrm{fold}$; the new computation C10 gives $\Delta N_\mathrm{eff} = 1.83 \times 10^{26} G_\lambda^2$ GeV^4 and the bound $G_\lambda \leq 2.3 \times 10^{-15}$ GeV^{-2}, now ASSUMED (the Kohn-Sham test value λ_test with $v_\mathrm{t} = H^{-3}$ would thermalise the fold bands and add 0.19); (f) the elastic ν_R rate of the $B-L$ alternative used $\sigma_\mathrm{el} = G_\nu^2 E^2/\pi$ instead of $G_\nu^2 E^2/(2\pi)$ for a Weyl ν_R on a heavy Dirac fermion: kinetic decoupling at $x = 25.1 \to 24.4$ ($T = 1769 \to 1816$ GeV), $\Delta N_\mathrm{eff} = 0.144$ unchanged; (g) the I11 production of ν_R, quoted as 10^{-14} of the I13 rate without a source, is now computed: 2.0×10^{-17}; (h) the fractions of the annihilation rate above the unitarity crossings were integrated inaccurately (one of them exceeded 1): they are now computed by adaptive quadrature with the break points of the thermal weight (I12 at the relic strength: 96 %; the I12 benchmark $5.3 \to 5.2 \times 10^{-4}$; I13 uncapped $24.5 \to 24.8$ %, capped $19.0 \to 21.7$ %); the exact I12 crossing at $E_\mathrm{cm} = 89510$ GeV is quoted with its relative velocity $v_\mathrm{rel} = 2\beta_\mathrm{cm} = 0.269$ (0.274 is the non-relativistic formula); the relic at the ${}^3P_0$ limit compares the thermal average and $\sqrt{\pi x}/M_\mathrm{DM}^2$ at the same x (5.33 against 5.84×10^{-26} cm^3/s); (i) stale or mixed statements: "$c_\mathrm{V} = 0$" and "no dark radiation" or "no extra radiation" of version 2; the cosmology labels (UM-2 version 3 is the model with the freeze-in fold bands: $N_\mathrm{eff} = 3.0442$, $\sigma_8 = 0.8118$, $\chi^2_\mathrm{BAO} = 31.49$, $\mathcal{H}_0 = 67.38$ km/s/Mpc; the model without them is the neutrino content of version 2); "relic OPEN" qualified as the result of version 2; the check counts of `um2_neutrino.py` (16) and `um2_cosmology.py` (8); the amplitude of the kinetic-function shift at the freeze-out (1.7×10^{-16}, not below 10^{-50}); "perturbative" made precise (22 % of the freeze-out rate at the bound of total angular momentum 1; $G_f^{-1/2} = 30$ TeV for the quark doublet, below $E_\mathrm{cm} \approx 89$ TeV); the ν_R couplings (the I11 coupling as well as the tip mixing); the dark matter is not counted in g_ρ, g_s (as in the code); roundings ($\tilde{I}^{(b)}_{\nu_R} = 0.301$ to 0.308, $g_\rho = 104.95$ at 200 GeV, $D_+\mathrm{e}^{-a_4} = 0.7878$ today, $\Phi_\mathrm{rms} = 5.03 \times 10^{-8}$ at 1 Mpc^{-1}, the Poisson agreement 1.5×10^{-6}); the signature convention of the trace anomaly and of the Proca Lagrangian, written for $(+,-,-,-)$; the $\overline{\mathrm{MS}}$ angle in the running couplings; the Yukawa strengths $\mathcal{A}_n$ for two different profiles; the order-2 magnitude $(2.4 \times 10^{-5})^2 = 5.8 \times 10^{-10}$; a heading labelled DERIVED over estimates; (j) self-containment: the quark sensitivities to the block scale, the scale $2\pi T$ of α_3, the densities of the environments, the hadron list, Ω_m, the symbols N_c, N_fold, G_A^{B-L}, the factor 2 of the two dark-matter species, $g_{s,e\gamma}$, K_1, K_2, the squared centre-of-mass energy, β_cm, the total angular momentum of a partial wave and the version-3 symbols are now stated in every document that uses them, and the variable $E^2_\mathrm{cm}/(4M^2_\mathrm{DM}) - 1$ is no longer written with the letter z (in version 4 z is only $6Hx_0$ and the redshift is z_red).

25. **Version 4 (2026-10-18): the organisation and presentation ordered on 2026-10-18 at 15:42.** (a) *The presentation rules* S1 to S6 (section 2) for MODEL, FIELD_EQUATIONS, SOLUTION, ANALYSIS and PROVENANCE: every document starts with "How to read this document"; every display equation is followed by the complete list of its symbols; every field equation or interaction term that a document uses is written out there with its tag FE-n or IN-n and its symbols; a solution is presented as the field equation, the approximation (the small parameter and the size of what is neglected), the solution, its check in its own

equation and in concert with the other field equations, and the status labels; the reader is never sent to another file; every change of a claim is stated here (commit `9dbc424`). (b) *The notation.* Version 3 used many letters for several things (k, s, M, H, p, a, κ, Λ, δ, n, μ, e, J, u, G, β, w, f, c and others); version 4 gives every meaning its own symbol, among them the block variable $K = 1/f^2$ (was k, now only a wave number); the Hubble rate $\mathcal{H}$ and the rate $\mathcal{H}_f$ of the block scale (were H or A, and h; H is only the author's constant); the Lovelock order as the index p in brackets, $\alpha_{(p)}$, $w_{(p)}$, $\mathcal{L}_{(p)}$, $\mathcal{E}_{(p)}$; the energy-momentum tensor $\mathcal{T}^M{}_N$ (was T, now only the temperature); the pressure p_5 of the extra times and the stabiliser density ρ_{stab}; the gauge couplings g_3, g_2, g_1 and $\hat{g}_3$, $\hat{g}_2$, $\hat{g}_1$ (were g_s, g, g'), so that g is only the metric; the tip mixing matrices $\mathcal{K}$ and the tip Yukawa matrices $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$; the scalar density n^{sc} of dirac16complex; the mass parameter m_Ψ and the record's parameter $\mathcal{M}$; the four-dimensional dark-matter and fold-band fields Ψ_{DM}, Ψ_{fold} (χ is now only the block spinor); the dimensionless boson eigenvalue $\tilde{m}^2$, its wave number ϖ and the profiles F; the Newtonian-gauge potentials Φ and Ψ_{N}; the gauge fields in the geometric normalisation (G, W, B) distinguished from the canonically normalised $W^\pm$, Z, A; the overlaps $\tilde{I}_{of}$ made dimensionless by an explicit factor $1/H$; Euler's number, the imaginary unit and the differential upright. The table of the decisions (the version-3 symbol and meaning, the version-4 symbol) is `docs_src/NOTATION.md` (commit `fd749e8`); this document applies it in every display, inline formula and sentence. (c) *The symbol registry and its check.* Every symbol of the documents has one entry (meaning in complete sentences, unit, value and status) in the registry `docs_src/symbols/*.json`: the core file with the symbols of the field equations (488 entries when the inventory was made, commit `fd749e8`) and one file per document (`provenance.json` for this document). `tools/symbols_tool.py` (commit `9dbc424`; extended in `fd749e8`: exponents of e and of brackets, evaluation bars and integral limits are expressions, not indices) writes the symbol list under every display equation from the registry and checks that every atom of every display (symbols, name subscripts, index letters, primes and daggers) is defined by a listed key, that every display has its list, and that no atom has two entries; since commit `1e264d0` it also checks that every display tagged FE-n or IN-n is written exactly as in the inventory, with the same keys. (d) *The inventory.* `docs_src/equations.json` lists the 136 field equations FE-1 to FE-136 and the 18 interaction terms IN-1 to IN-18 in the version-4 notation, each with its title, status, fields, the place where it is solved and the registry keys of all its symbols; a test document of all 154 equations passed the check and built with 0 overfull boxes (commit `fd749e8`). Commit `1e264d0` aligned it with the documents: FE-40, FE-42 and FE-43 now write the x_0 component (with p_0) and the extra-time components x_5, x_6, x_7 (with p_5) separately, next to the block-isotropic form with $\bar{p}$; FE-52 and FE-53 quote the squared eigenvalues $\tilde{m}_1^2$, $\tilde{m}_n^2$; FE-79 records the first-order result of fix 1 of section 10. (e) *The residual analysis.* The new computation `um2_residuals.py` (52 checks; 287 in all) measures the residual of the stored solution in 26 field equations and 26 links in concert; its findings are section 10 (commit `cc2d514`). (f) *The precision fixes:* the first-order perturbation theory for $\mathrm{d}\ln\tilde{m}_{\mathrm{W}}^2/\mathrm{d}A_{\mathrm{mf}}$, the robust integration of the right-handed neutrinos, the CNB quadrature, the deterministic thread counts in the Kohn-Sham scripts, and a stale check text (section 10; commit `cc2d514`). (g) *Claims.* No claim of version 3 changes; the changed numbers and the statements made precise are listed at the end of section 10. (h) The documents of version 4: SOLUTION and ANALYSIS (commit `1e264d0`), MODEL, FIELD_EQUATIONS and PROVENANCE (this version).

13 What is not in the repository, and why

By the author's instruction the following private inputs are never committed (they are git-ignored): the prompt files, the author's private PDF with the supernova dark-energy fits, the folders `dirac-main/` and `vendor/`, the archive `dirac-main_2.zip`, the files `Generalized_Kronecker_Delta.*`, and the notebook `WaveFunctionUniverse-4+4-Einstein-Lovelock.nb` with its `.mx` files. The paper "Neutrino Flavor Conversion.pdf" (Gogilashvili and Tamborra, arXiv:2605.16504) is a third-party publication and is not committed either; UM-2 cites its results. UM-2 uses from these inputs only what is written out in this document (the notebook formulas of the metric and of the gamma matrices, the two supernova fits, the explosion statistics).

14 Reproduction

From the repository root, with Python 3 and the packages sympy, numpy, scipy, mpmath, matplotlib and camb installed (`pip install sympy numpy scipy mpmath matplotlib camb`), the commands

```
python universe_model_2/code/um2_algebra.py
python universe_model_2/code/um2_geometry.py
python universe_model_2/code/um2_background.py
python universe_model_2/code/um2_fields.py
python universe_model_2/code/um2_spectra.py
python universe_model_2/code/um2_flavour.py
python universe_model_2/code/um2_neutrino.py
python universe_model_2/code/um2_ks_jobs.py validate
python universe_model_2/code/um2_ks_jobs.py spectrum
python universe_model_2/code/um2_ks_jobs.py ground8
python universe_model_2/code/um2_ks_jobs.py ground592
python universe_model_2/code/um2_ks_jobs.py darkmatter
python universe_model_2/code/um2_ks_jobs.py portals
python universe_model_2/code/um2_ks_jobs.py feedback
python universe_model_2/code/um2_electroweak.py
python universe_model_2/code/um2_primordial.py
python universe_model_2/code/um2_cosmology.py
python universe_model_2/code/um2_analysis.py
python universe_model_2/code/um2_thermal.py
python universe_model_2/code/um2_coupled.py
python universe_model_2/code/um2_series.py
python universe_model_2/code/um2_residuals.py
python universe_model_2/tools/symbols_tool.py check
python universe_model_2/tools/build_docs.py
python universe_model_2/tools/compare_outputs.py
```

rerun every computation in dependency order (each exits with code 0 if and only if all its checks pass; the residual analysis reads the outputs of all the others and must run last), check the symbol lists of the five documents against the registry (the check must end with `symbols_check=OK`), rebuild the LaTeX and PDF documents from the Markdown sources (pandoc and a TeX Live installation with the `lmodern` package are needed; every document must build with 0 overfull boxes), and compare the regenerated files with the committed ones (numbers to a relative 10^{-9}, run times and check texts listed separately). The document build is deterministic: no dates in the LaTeX sources, PDF dates and trailer identifiers suppressed, a fixed `SOURCE_DATE_EPOCH`.

Software used: Python 3.13.16, sympy 1.14.0, numpy 2.5.3, scipy 1.18.1, mpmath 1.3.0, matplotlib 3.11.2, camb 2.0.5, pandoc 3.1.3, pdfTeX 1.40.25 (TeX Live 2023).

15 History in the repository

UM-2 was started on top of commit `0cb4916`. Version 1: its computations were committed in the order algebra and geometry (`ca523d9`), background gravity (`0b3e10d`), field reductions (`6fc35a0`), mass spectra (`dcfd528`), Kohn-Sham (`2f3018e`), electroweak sector (`2e4c48d`), primordial era (`a502e62`), cosmology (`717b46e`), analysis (`a06b55e`), followed by the corrections listed above and the documents MODEL, PROVENANCE, FIELD_EQUATIONS, SOLUTION and ANALYSIS (Markdown, LaTeX and PDF each); the fixes of the three independent reviews are in `b386446`, `fa7305f`, `b0e686e`, `9285bb6`, `d92a61b`, `c98de95`, `1561566`, `df358db` and `5b73d16`; a checklist for manual reruns and the comparison tool are in `7b86239`. Version 2: three generations and the masses and mixing (`94621af`), the neutrino flavour conversion (`a0d750a`), the electroweak sector of three generations (`cf824d5`), the removal of the tip Z portal and the cosmology with three massive neutrinos (`53c5139`), the dark matter of both halves, the twelve-fermion tables and the three-generation tip and gauge checks (`b755990`), the corrected 1-3 resonance (`b999f57`), the documents MODEL (`9a5b92b`), FIELD_EQUATIONS (`79d9d31`), SOLUTION (`a3dcc6a`), ANALYSIS (`df6d94a`) and PROVENANCE (`4a6421d`), and the fixes of the independent review of version 2 (item 22; `ecca926`). Version 3: the thermal history, the coupled solution, I13, the metric series and σ_8 (`15c2aab`), I13 on the baryon-number current (`3fa1b49`), the documents MODEL (`765f0b8`), FIELD_EQUATIONS (`41e921a`), SOLUTION (`df23013`), ANALYSIS (`19d2c06`) and PROVENANCE (`726997f`), the fixes of the independent review of version 3 with the record of the fresh-clone verification of `726997f` (item 24; `815db9e`), and the record of the fresh-clone verification of `815db9e` (`93e5c03`). Version 4: the presentation rules and the symbol tool (`9dbc424`), the residual analysis

and the precision fixes (`cc2d514`), the notation, the inventory and the core registry (`fd749e8`), the documents SOLUTION and ANALYSIS of version 4 with the inventory aligned and the extended symbol check (`1e264d0`), and the documents MODEL, FIELD_EQUATIONS and PROVENANCE of version 4 (this version).